

Database Design Tuning



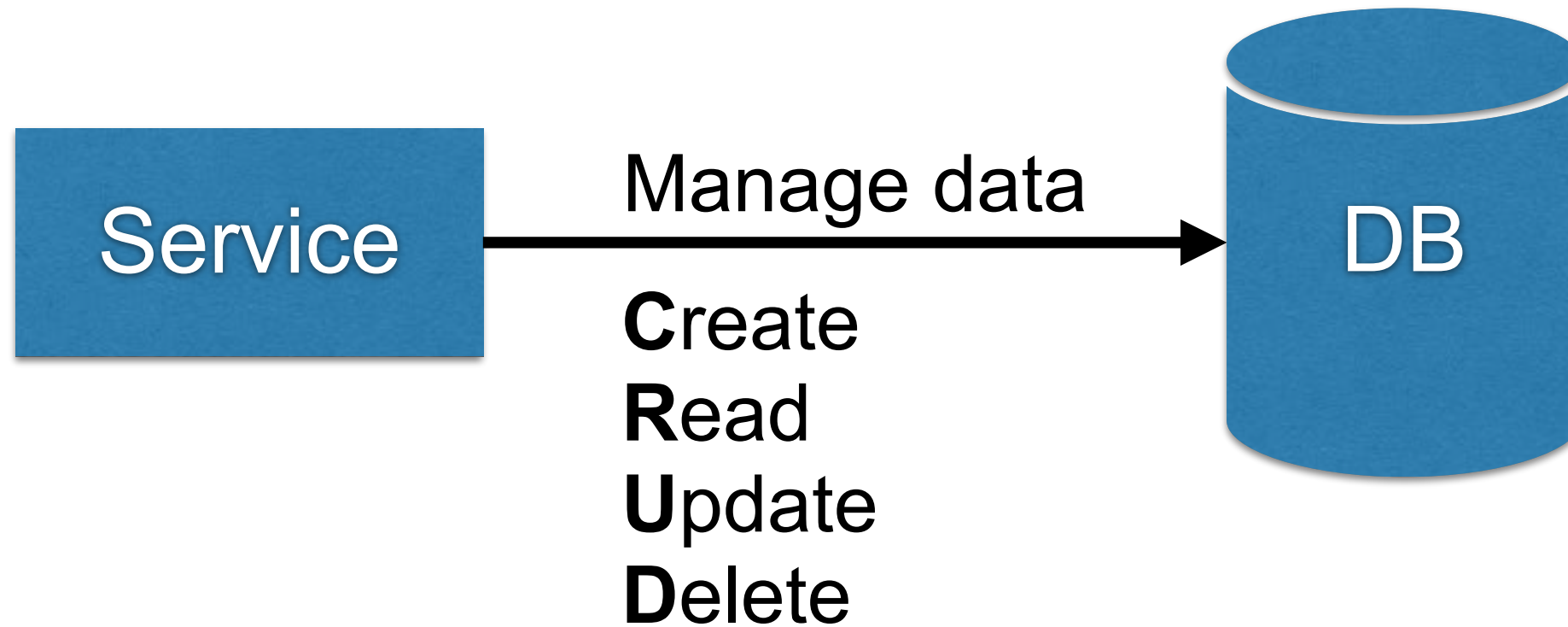
**[https://github.com/up1/
workshop-database](https://github.com/up1/workshop-database)**



Database

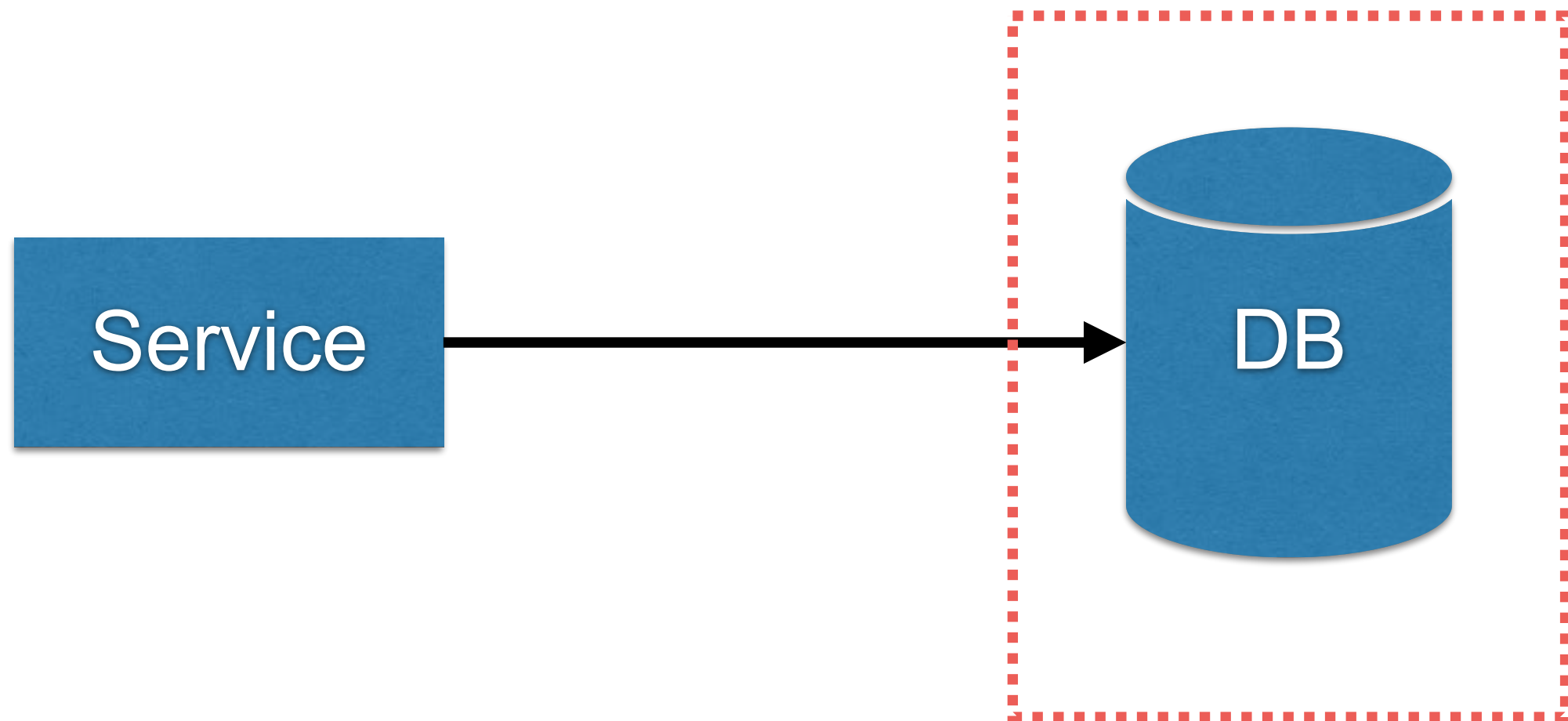


Database



Working with Database

Directly to database

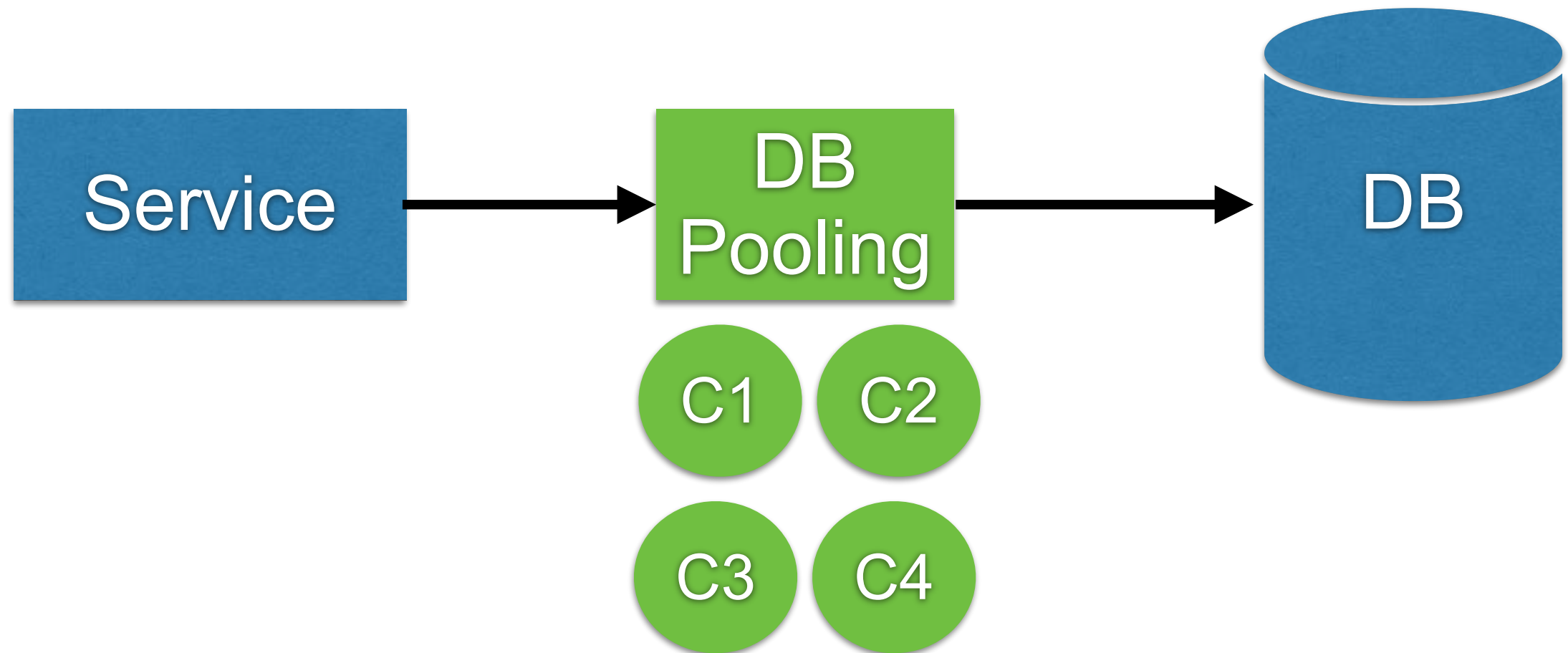


Data modeling
Database architecture



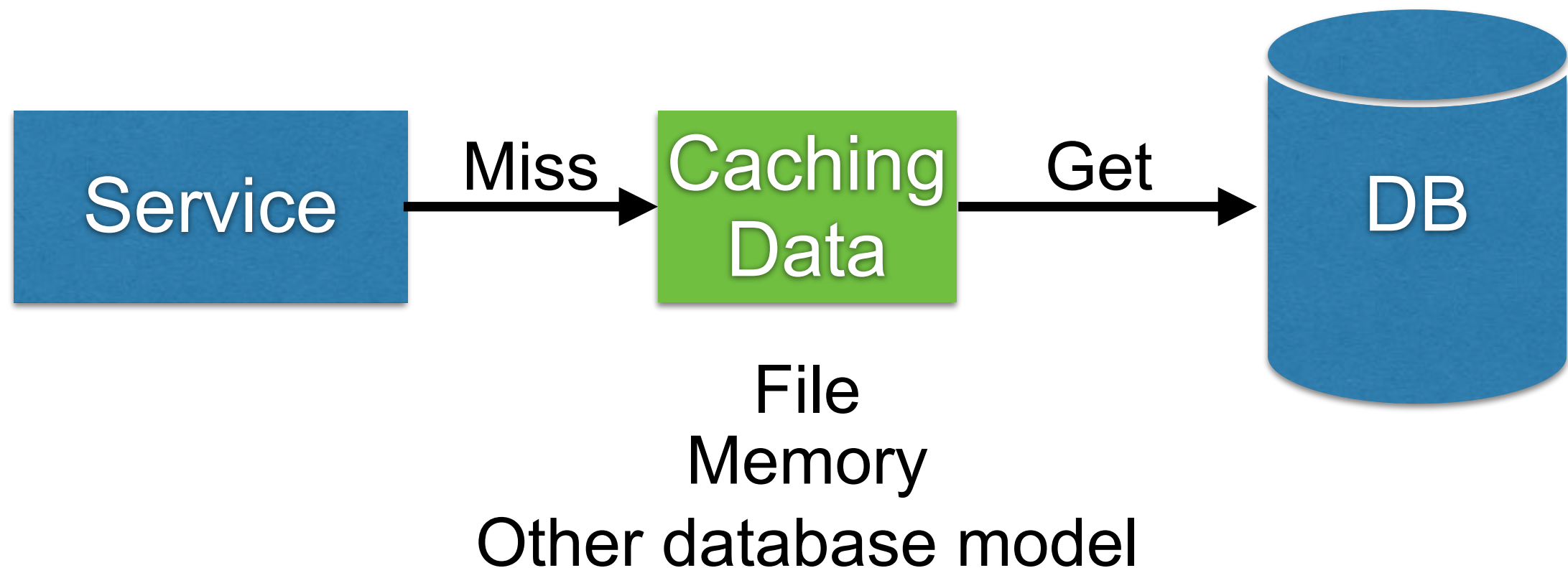
Working with Database

Connection pool



Working with Database

Caching data layer



Scalability metrics

Throughput (transactions per second)

Latency (response time)

Availability (uptime)

Durability (data safety)



Database ?

Structured data collection

Records

Relationships



Database Management System ?

Software systems for storing and querying databases



User's perspective

Data modeling

Querying data

Store data

Read

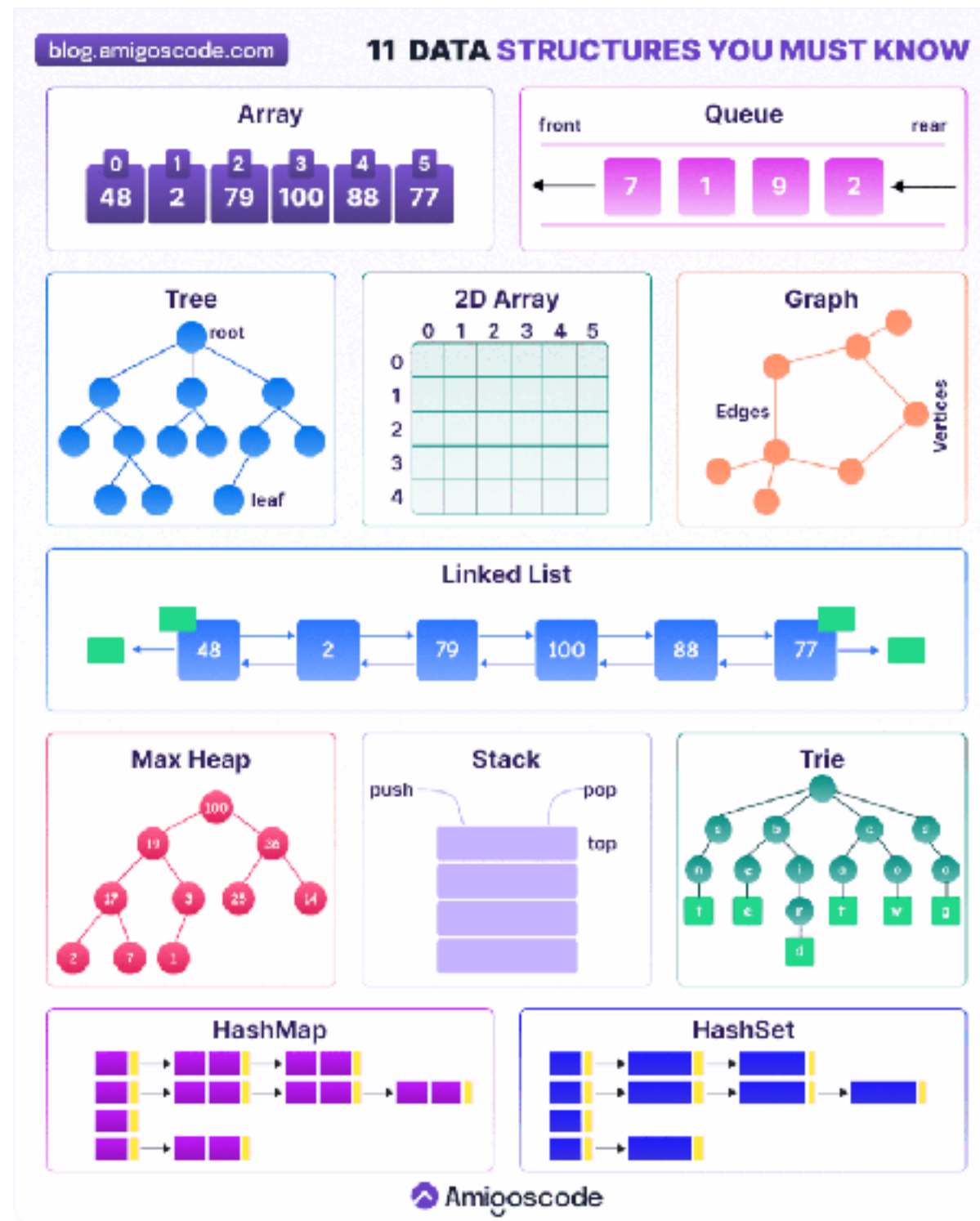
Write



Database Models ?



Data Structure



<https://blog.amigoscode.com/p/11-data-structures-every-developer>



Database Models ?

424 systems in ranking, July 2025

Rank			DBMS	Database Model	Score		
Jul 2025	Jun 2025	Jul 2024			Jul 2025	Jun 2025	Jul 2024
1.	1.	1.	Oracle	Relational, Multi-model 	1217.05	-13.33	-23.31
2.	2.	2.	MySQL	Relational, Multi-model 	940.73	-12.85	-98.73
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model 	771.14	-5.61	-36.51
4.	4.	4.	PostgreSQL	Relational, Multi-model 	680.89	+0.23	+41.98
5.	5.	5.	MongoDB 	Document, Multi-model 	403.83	+0.99	-25.99
6.	6.	 7.	Snowflake	Relational	176.17	+1.68	+39.64
7.	7.	 6.	Redis	Key-value, Multi-model 	149.72	-2.01	-7.05
8.	8.	 9.	IBM Db2	Relational, Multi-model 	127.51	+2.38	+3.11
9.	9.	 8.	Elasticsearch	Multi-model 	118.83	-2.45	-12.00
10.	10.	10.	SQLite	Relational	115.44	-1.60	+5.49
11.	11.	 12.	Apache Cassandra	Wide column, Multi-model 	108.76	+0.49	+9.63
12.	12.	 15.	Databricks	Multi-model 	108.04	+3.36	+24.74
13.	13.	 14.	MariaDB 	Relational, Multi-model 	95.44	+0.91	+4.85
14.	14.	 11.	Microsoft Access	Relational	90.47	+2.19	-10.17
15.	15.	 17.	Amazon DynamoDB	Multi-model 	84.15	+0.81	+13.20

<https://db-engines.com/en/ranking>



Database Model

Relational

Document

Key-value

Search engine

Wide column

Graph

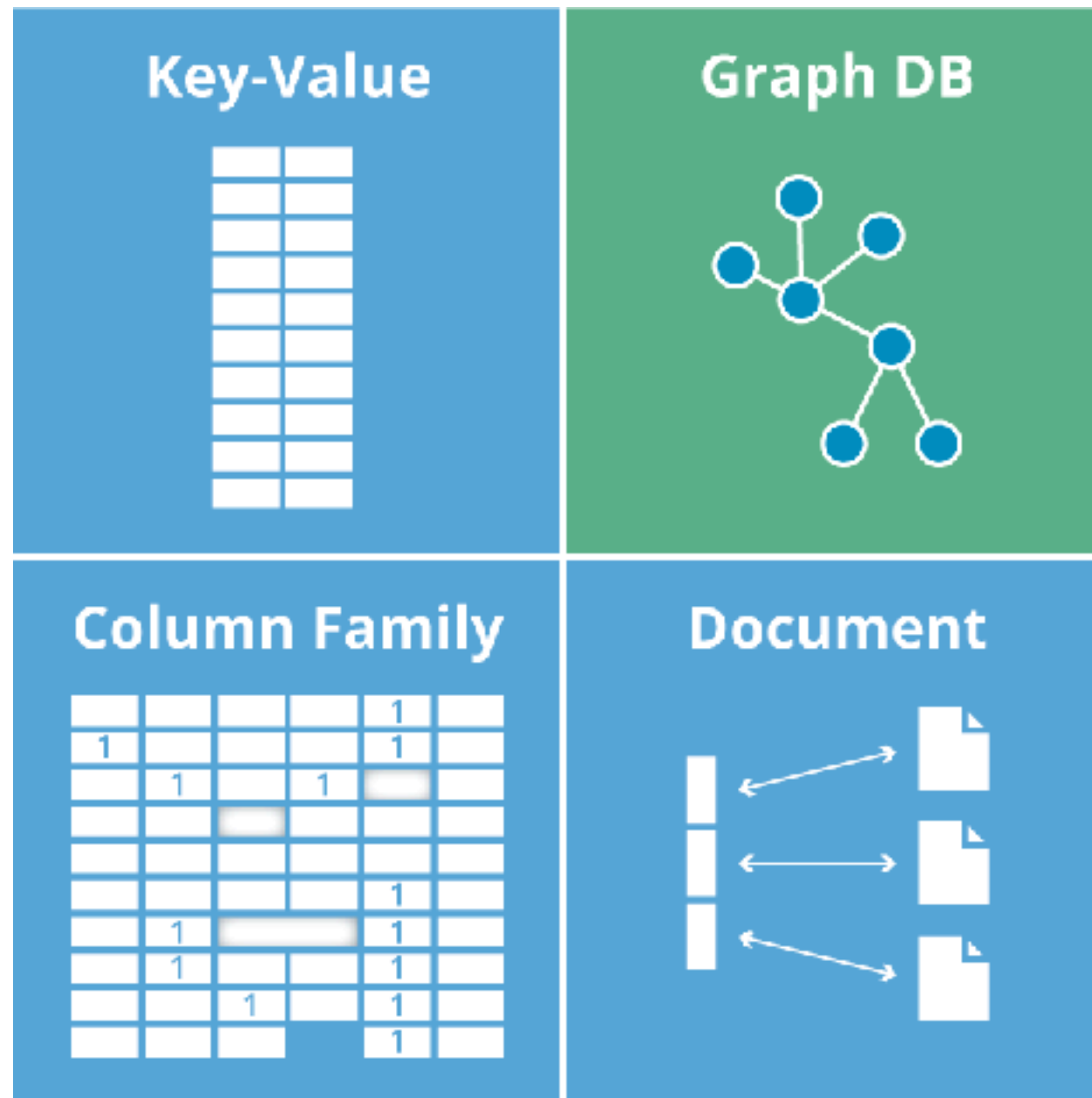
Time series

Vector

Non-Relational
NoSQL



NoSQL



Relational Database

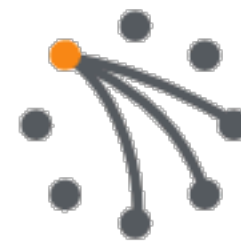


PostgreSQL

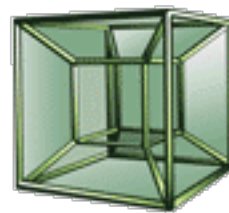


NoSQL

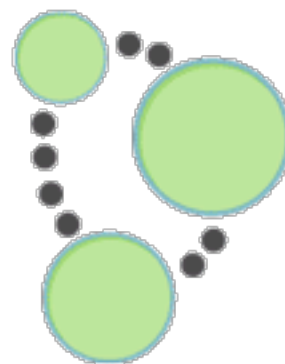
APACHE
HBASE



riak



HYPERTABLE INC

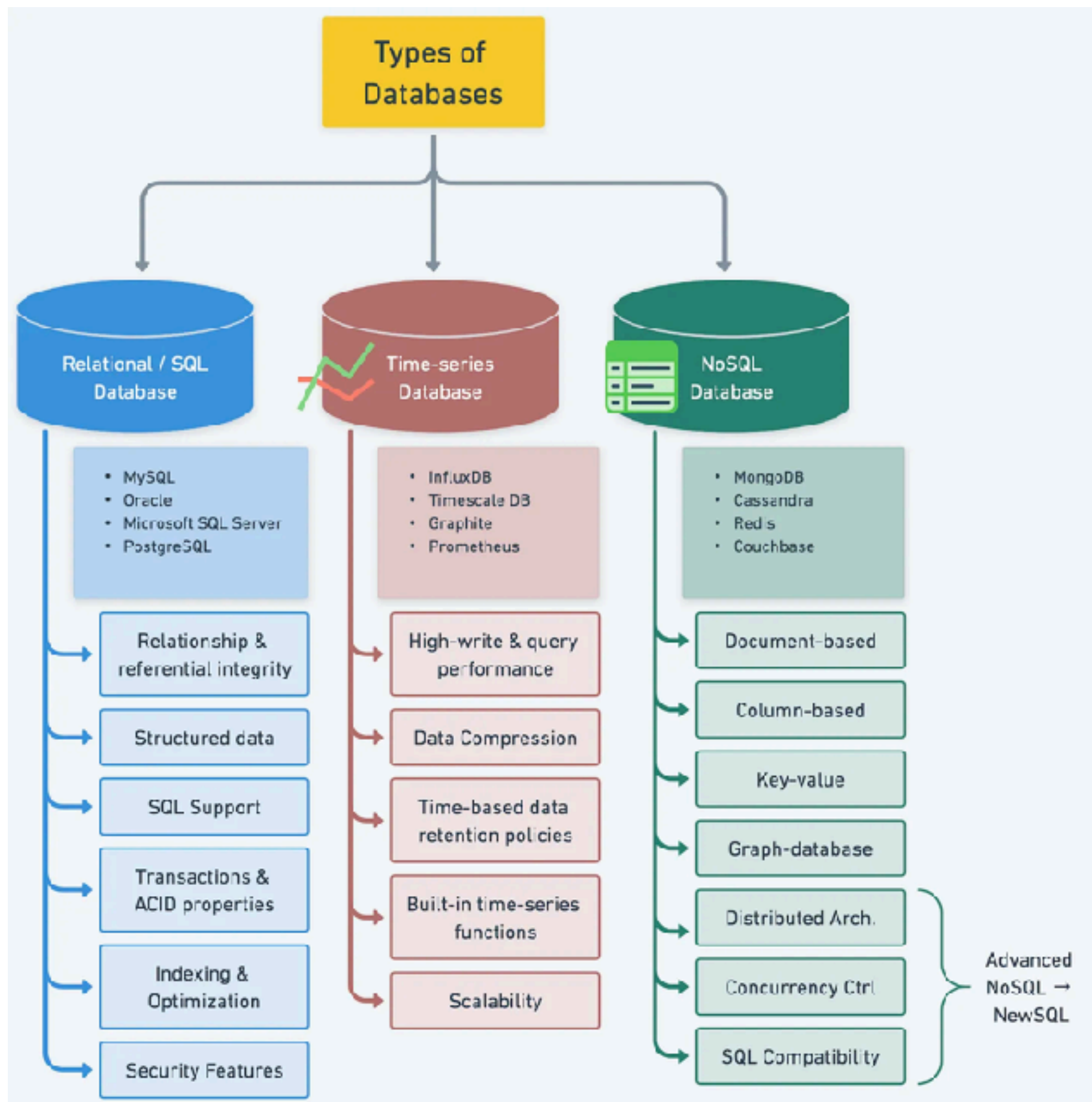


Neo4j



redis





<https://blog.bytebytego.com/p/understanding-database-types>



Relational Database ?



Relational Database ?

Relational DataBase Managment System

Maintain data in **tables**

Pre-define structure

Each table has **Primary key**

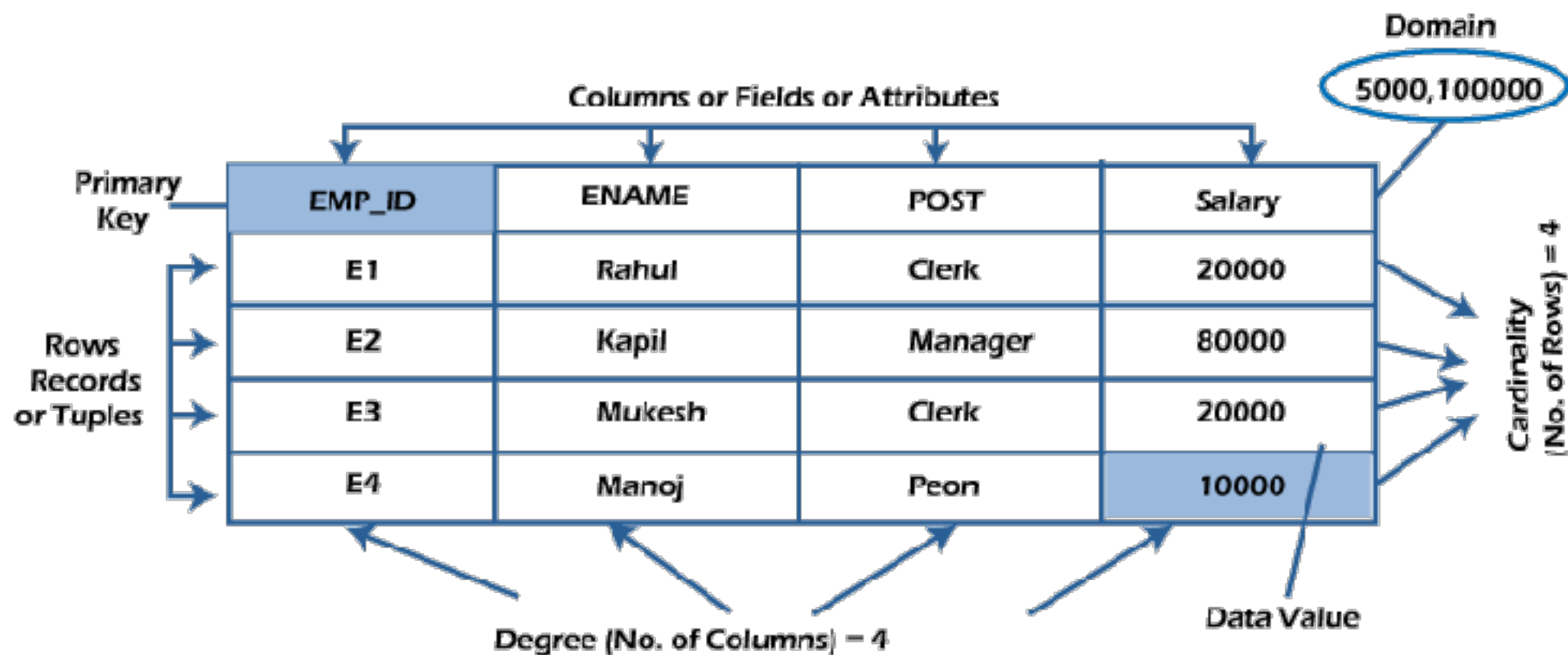
Many **columns**

Data is represented in term of **row**



Relational Database ?

Table = Employee

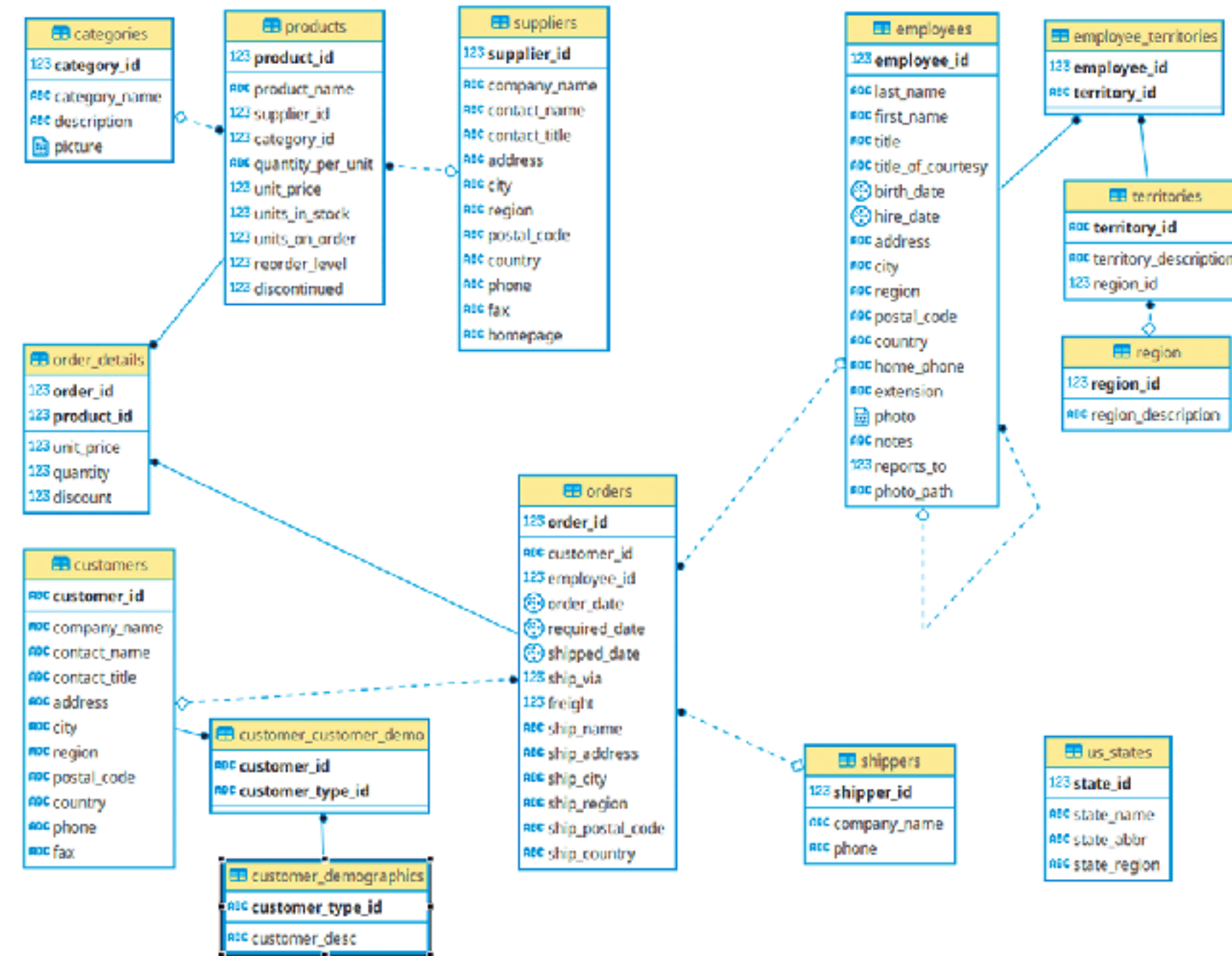


Relational Database

Database	Ease of use	Support complex query	Community support	Licence
MySQL	High	High	Strong	Open source
PostgreSQL	Medium	Very high	Strong	Open source
MSSQL Server	Medium	Very high	Strong	Proprietary
Oracle	Low	Very high	Moderate	Proprietary



Real Design of RDBMS ?



Questions ?

How to write data ?

How to read data ?

Aggregate data ?

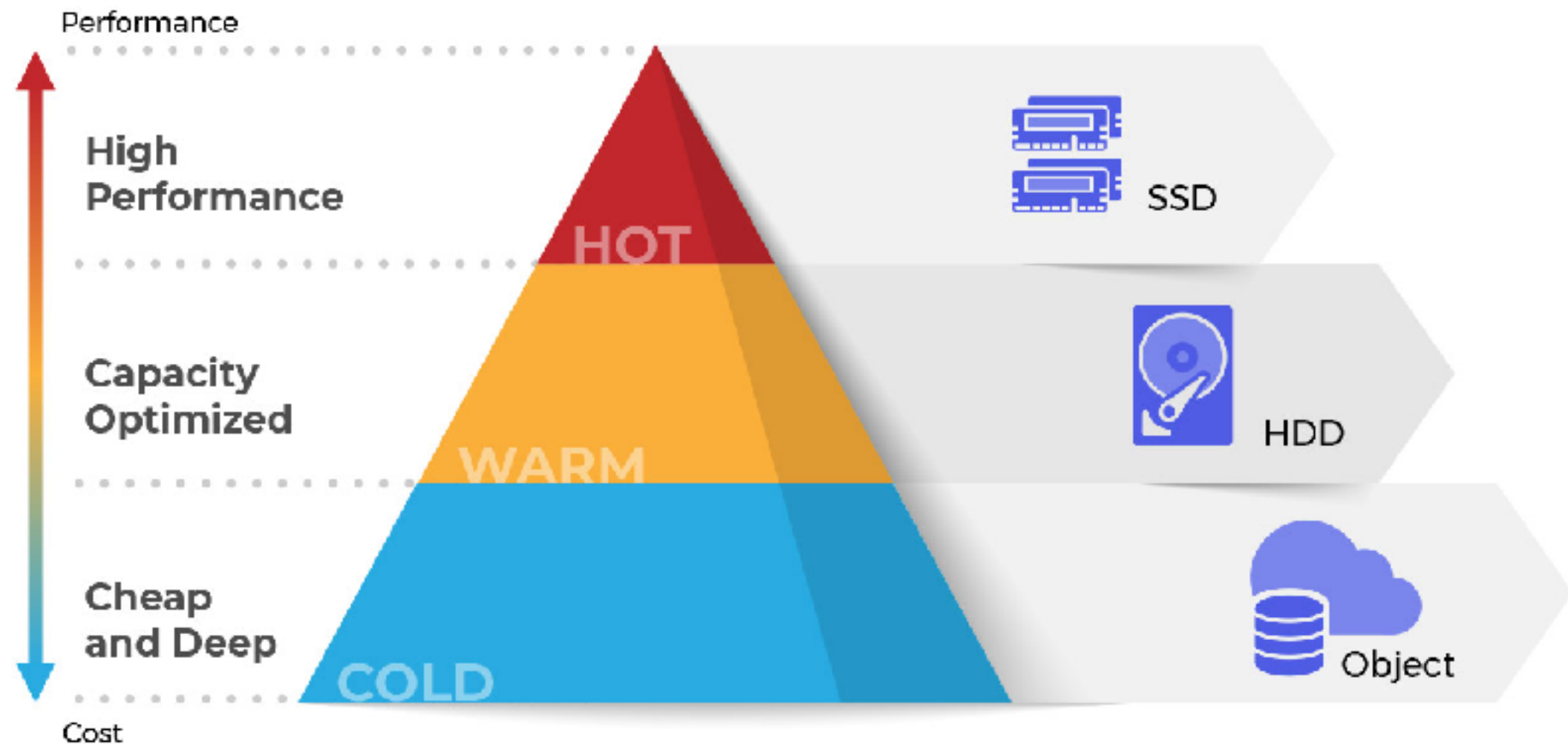
Reporting

Data retention ?



Questions ?

Hot vs Warm vs Cold data ?

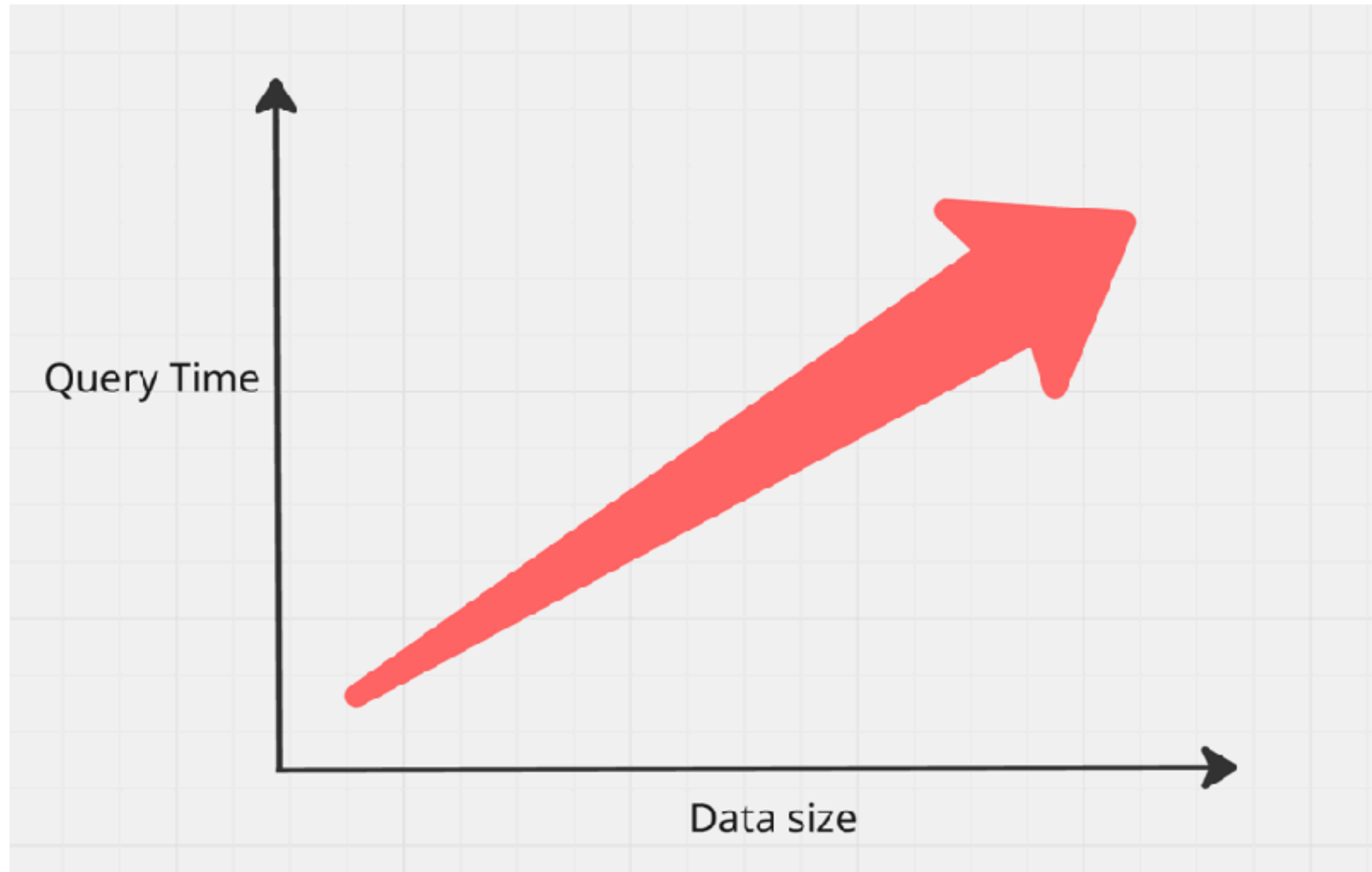


Complex Query ?

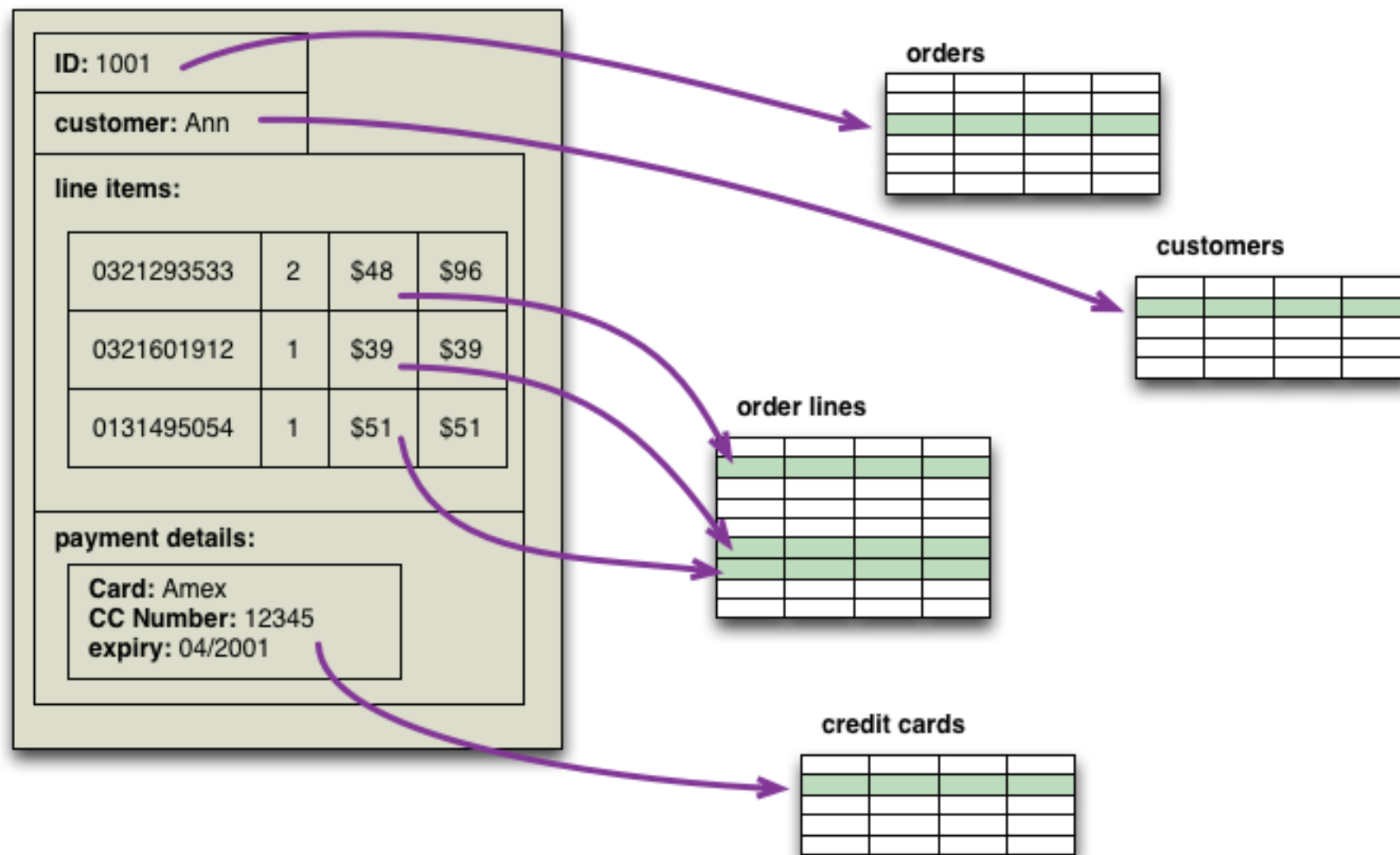
```
1  WITH tmp_1 AS
2  (
3  SELECT Calc1 =
4  ( (SELECT TOP 1 DataValue
5  FROM (
6  SELECT TOP 50 PERCENT DataValue
7  FROM SOMEDATA
8  WHERE DataValue IS NOT NULL
9  ORDER BY DataValue
10 ) AS A
11 ORDER BY DataValue DESC
12 ) + (SELECT TOP 1 DataValue
13 FROM (
14 SELECT TOP 50 PERCENT DataValue
15 FROM SOMEDATA
16 WHERE DataValue Is NOT NULL
17 ORDER BY DataValue DESC
18 )AS A
19 ORDER BY DataValue ASC))/2
20 ),tmp_2 AS
21 (SELECT AVG(DataValue) Mean, MAX(DataValue) - MIN(DataValue) AS MaxMinRange
22 FROM SOMEDATA
23 ),tmp_3 AS
24 (
25 SELECT TOP 1 DataValue AS Calc2, COUNT(*) AS Calc2Count
26 FROM SOMEDATA
27 GROUP BY DataValue
28 ORDER BY Calc2Count DESC
29 )
30 SELECT Mean, Calc1, Calc2 , MaxMinRange AS [Range]
31 FROM tmp_1 CROSS JOIN tmp_2 CROSS JOIN tmp_3;
```



Problems ?



NoSQL

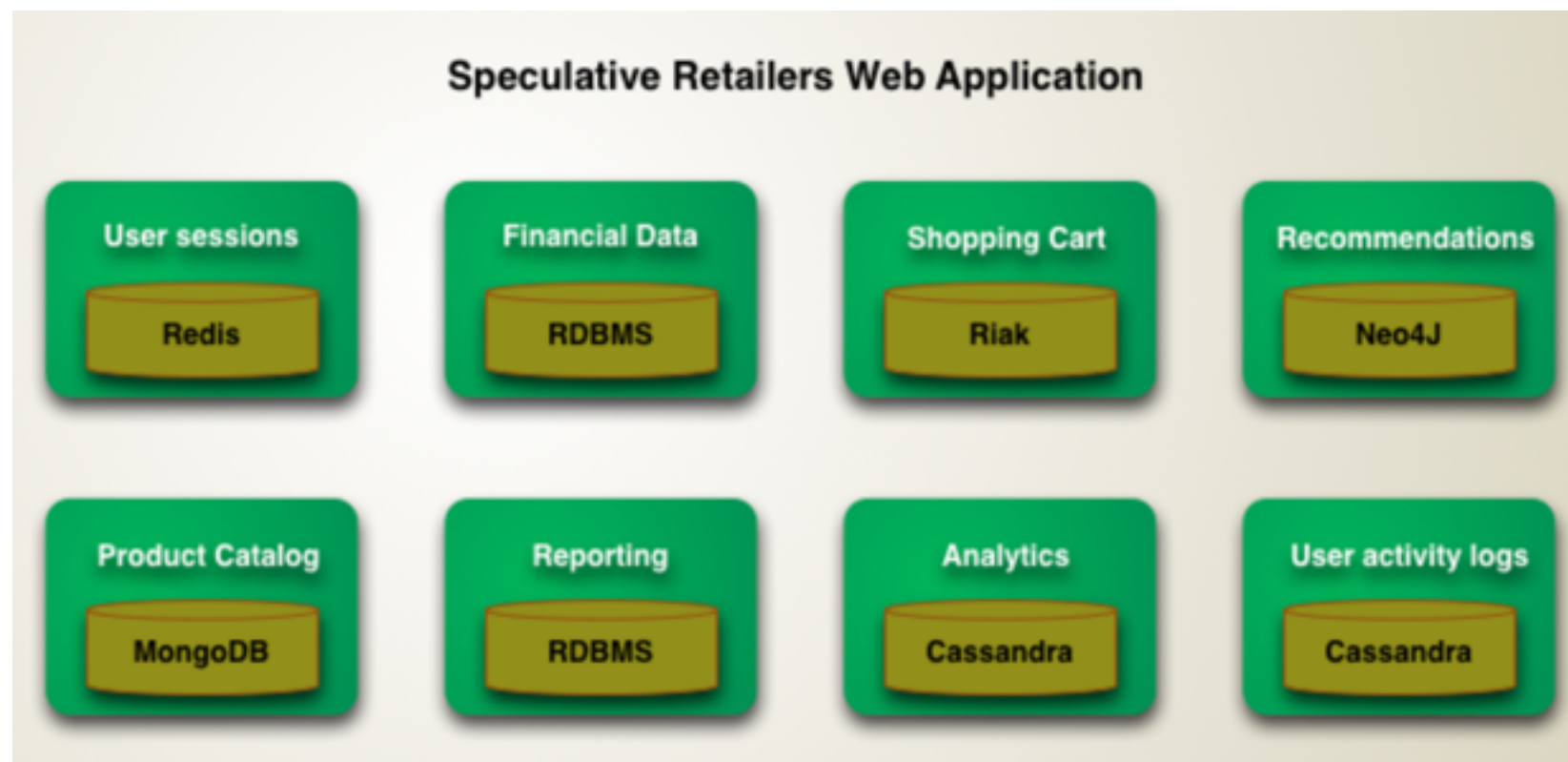


<https://martinfowler.com/bliki/AggregateOrientedDatabase.html>



Polyglot Persistence

Right tool for the right job



<https://martinfowler.com/bliki/PolyglotPersistence.html>



Data Modeling

Read

Write



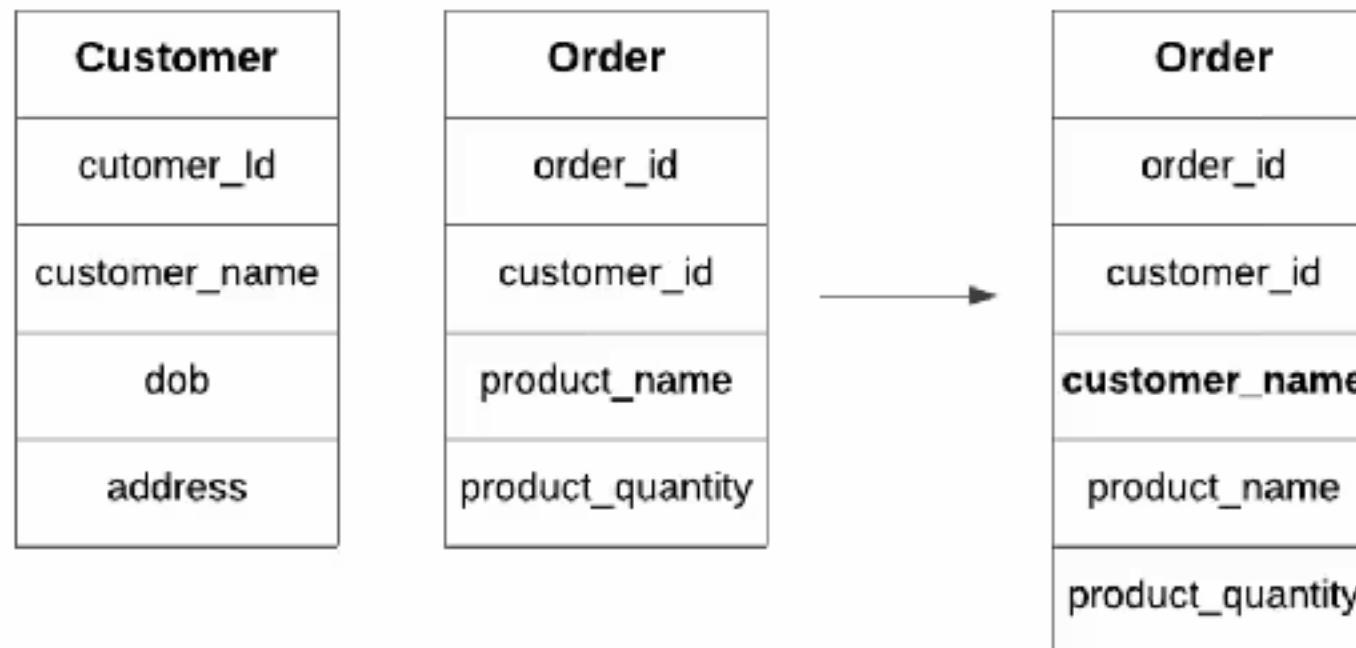
Data Modeling

Normalization

Denormalization

Pre-joins, pre-aggregation

Caching strategies



Normalization ?

Cleansing data

Reorganized data

Remove redundant data

Improve efficient for store data

Organize data in consistent way



Normalization

Divide in to several tables
Link tables with **relationships !!**

Design for write or read ?

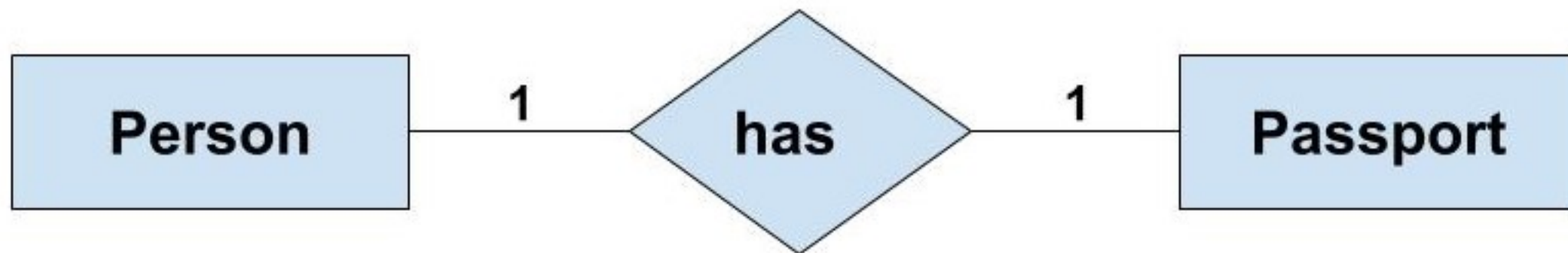


Relationships



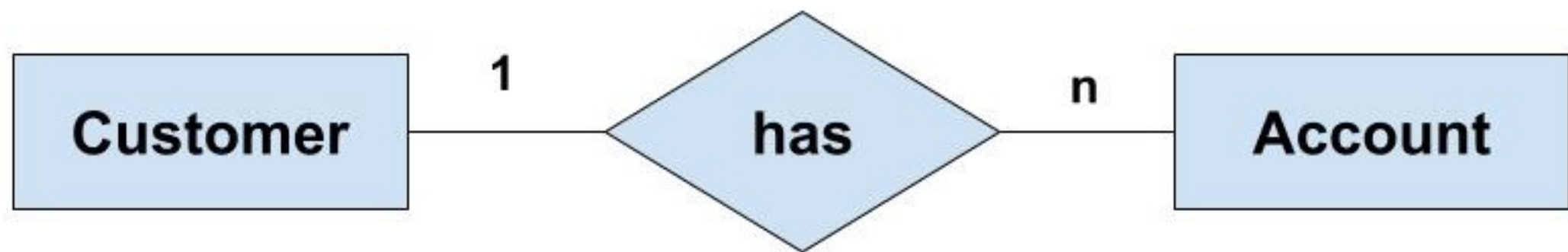
Relationship between table

One-to-one
One-to-many
Many-to-many



Relationship between table

One-to-one
One-to-many
Many-to-many



Relationship between table

One-to-one
One-to-many
Many-to-many

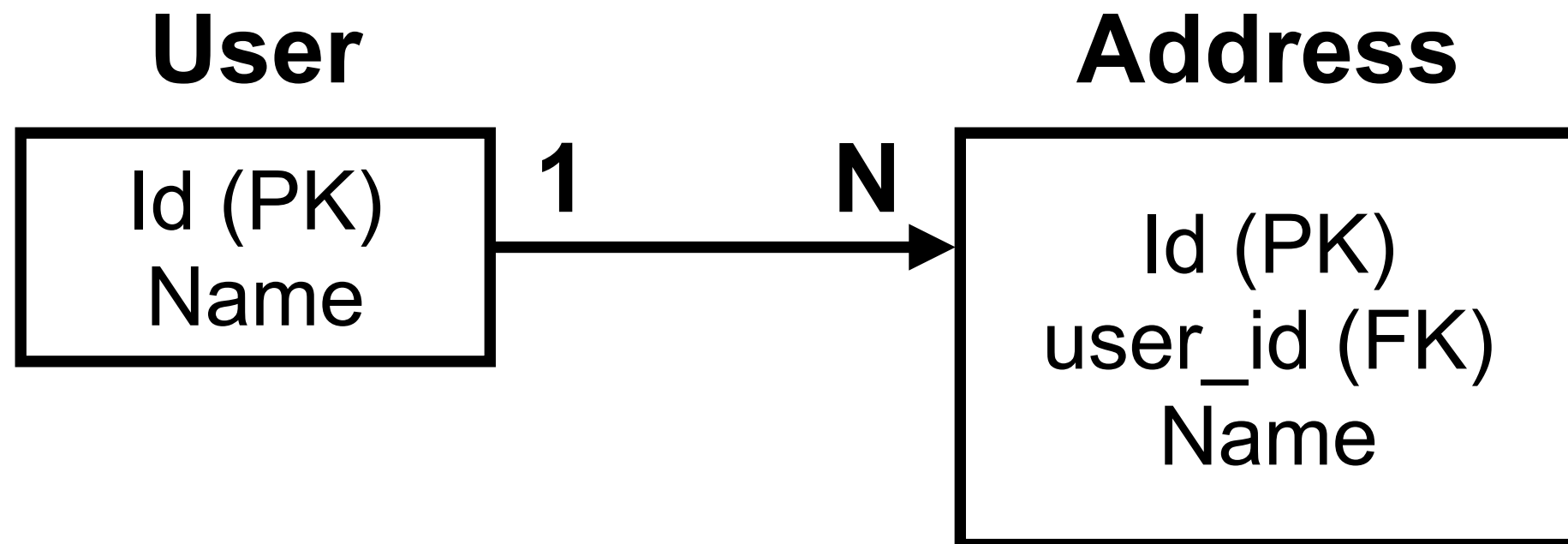


Relationships

Primary key (PK)

Composite key

Foreign key (FK)



Normalization forms

First Normal form (1NF)

2NF

3NF

4NF

5NF

6NF



First Normal Form (1NF)

No repeat data in table

Every column should only have single value

Every row should be unique



Sample data in 1NF

Name	Address	Gender	T-shirt Size
User 1	Address 1	Male	Large
User 2	Address 2	Female	Small
User 2	Address 2	Female	Medium
User 3	Address 3	Male	Medium



Second Normal Form (2NF)

Apply 1NF

Have one primary key



Problem in 1NF

Name	Address	Gender	T-shirt Size
User 1	Address 1	Male	Large
User 2	Address 2	Female	Small
User 2	Address 2	Female	Medium
User 3	Address 3	Male	Medium



Sample data in 2NF

User ID	Name	Address	Gender
1	User 1	Address 1	Male
2	User 2	Address 2	Female
3	User 3	Address 3	Male



Sample data in 2NF

User ID	T-shirt Size
1	Large
2	Small
2	Medium
3	Medium



Third Normal Form (3NF)

Apply 2NF

It should only be dependent on the primary key

if any changes to the primary key are made,
all data that is impacted must be put into a
new table



Problem in 2NF

User ID	Name	Address	Gender
1	User 1	Address 1	Male
2	User 2	Address 2	Female
3	User 3	Address 3	Male



Sample data in 3NF

User ID	Name	Address	Gender
1	User 1	Address 1	1
2	User 2	Address 2	2
3	User 3	Address 3	1



Sample data in 3NF

Gender ID	Gender name
1	Male
2	Female



Sample data in 3NF

User ID	T-shirt Size
1	Large
2	Small
2	Medium
3	Medium



3NF is normal form ...



Benefits of data normalization

Freeing up space
Improve query response time
Reduce data inconsistency
Easier to use and analyze



Challenges of data normalization

Slower query response rates

Accurate knowledge is required

Add complexity for teams

Denormalization as an alternative !!



Primary Key (PK)



Foreign Key (FK)



Other Constraints

Ensure data quality

Prevent inconsistency at the database level

Not Null

Unique

Check

Default



Workshop with PostgreSQL



PostgreSQL



PostgreSQL Database

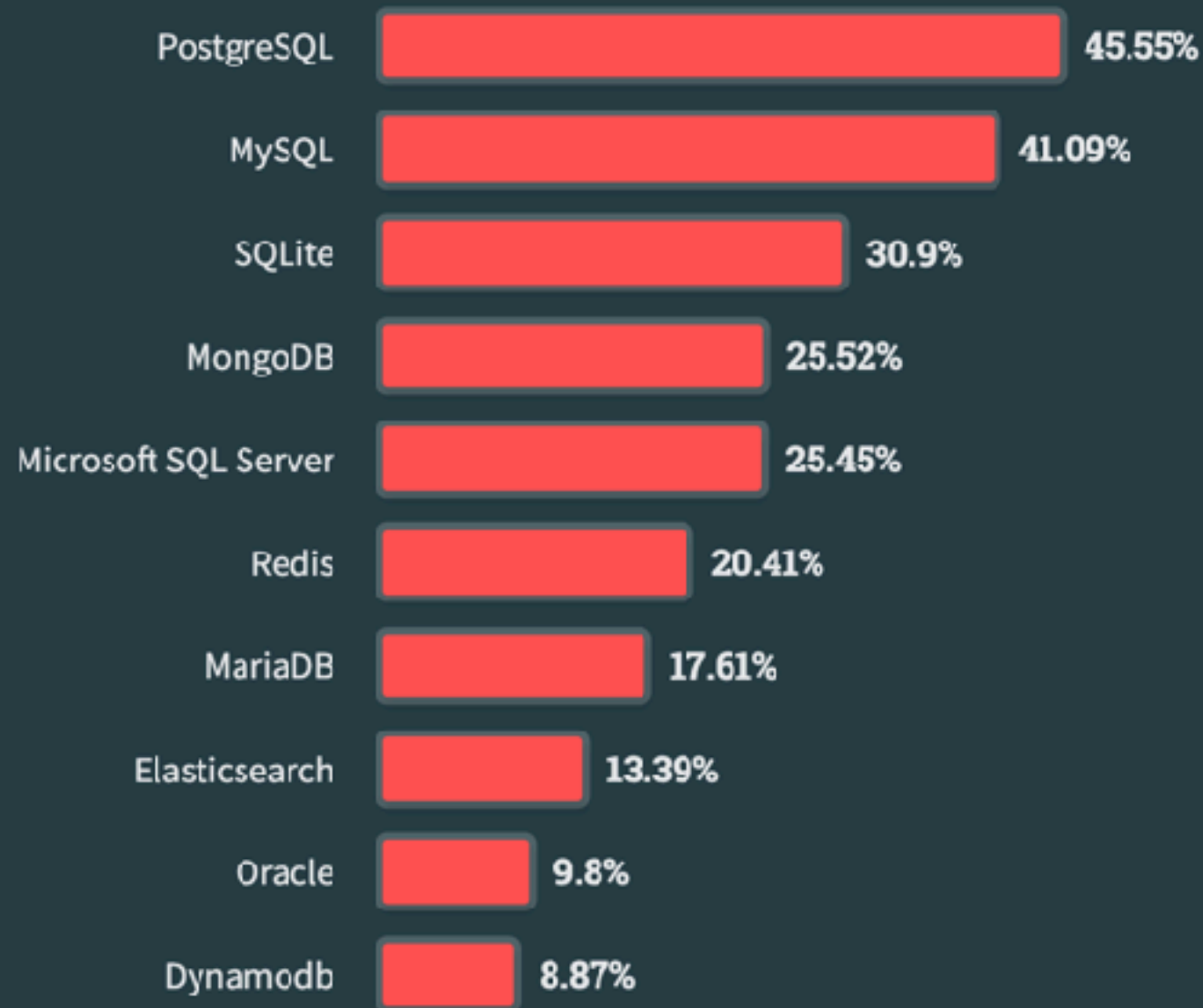
All Respondents

Professional Developers

Learning to Code

Other Coders

76,634 responses



<https://survey.stackoverflow.co/2023/#most-popular-technologies-database>



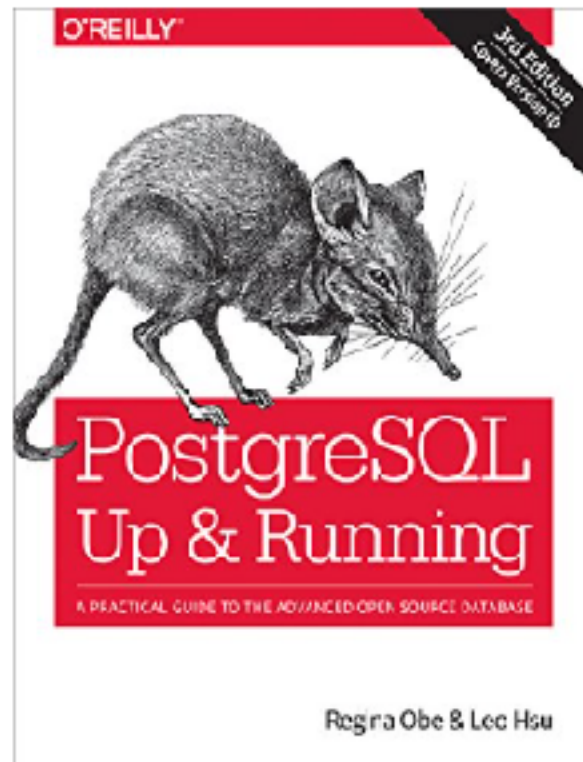
Book Store

QuickStart
Guides

Master SQL and Manage, Analyze and Manipulate Data

\$25¹⁹

Kindle Store › Kindle eBooks › Computers & Technology



Read sample

Follow the Author



Regina Obe

Follow

PostgreSQL: Up and Running: A Practical Guide to the Advanced Open Source Database 3rd Edition, Kindle Edition

by Regina O. Obe (Author), Leo S. Hsu (Author) | Format: Kindle Edition

4.4 ★★★★★ 99 ratings

[See all formats and editions](#)

Kindle
\$10.09 - \$28.99

Read with Our Free App

Paperback
\$44.99

6 Used from \$36.00
20 New from \$34.18

Thinking of migrating to PostgreSQL? This clear, fast-paced introduction helps you understand and use this open source database system. Not only will you learn about the enterprise class features in versions 9.5 to 10, you'll also discover that PostgreSQL is more than a database system—it's an impressive application platform as well.

With examples throughout, this book shows you how to achieve tasks that are difficult or impossible in other databases. This third edition covers new features, such as ANSI-SQL constructs found only in proprietary databases until now: foreign data wrapper (FDW) enhancements, new full-text search and partitioning introduced in version 9.5, and more. [Read more](#)

ISBN-13



978-1491963418

Edition



3rd

Sticky notes



[On Kindle Scribe](#)

Publisher



O'Reilly Media

Publication date



October 10, 2017

Language



English



Database design

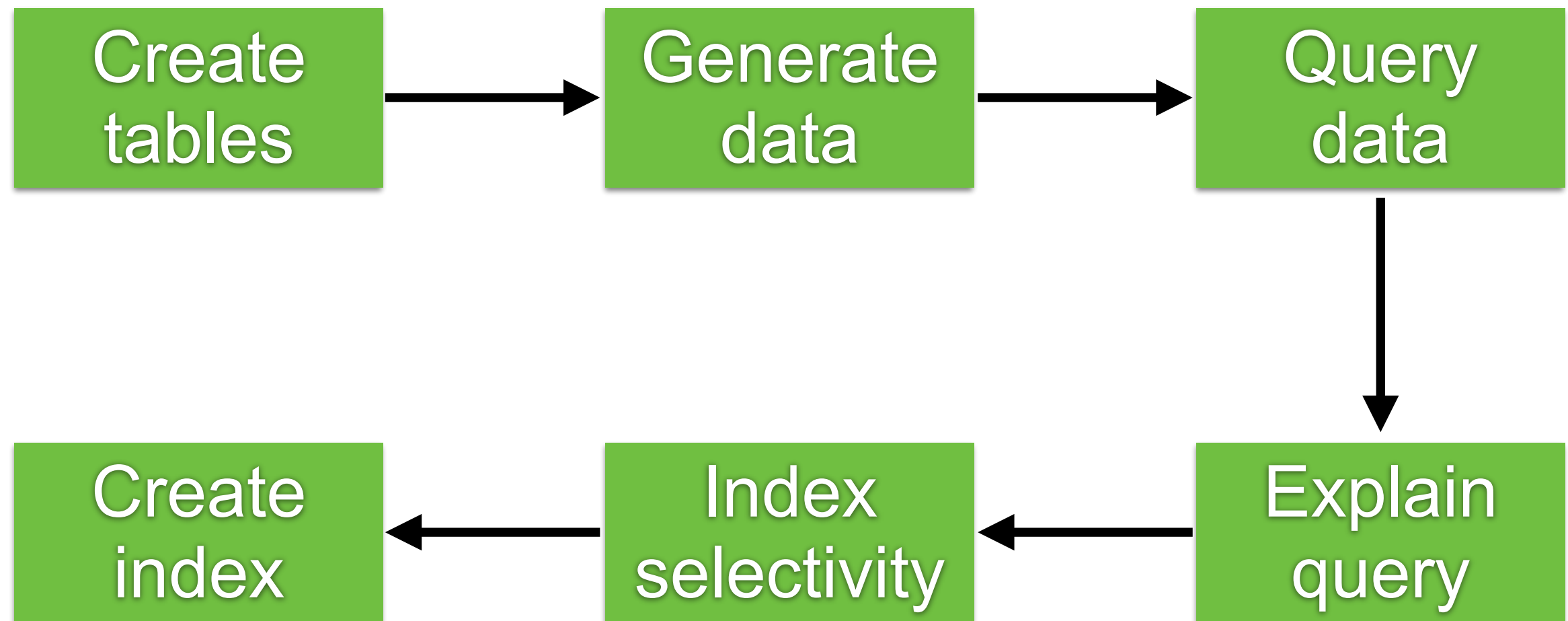
© 2020 - 2025 Siam Chamnankit Company Limited. All rights reserved.

60

Book Store



Step in workshop



<https://github.com/up1/workshop-database/tree/main/workshop/basic-workshop>

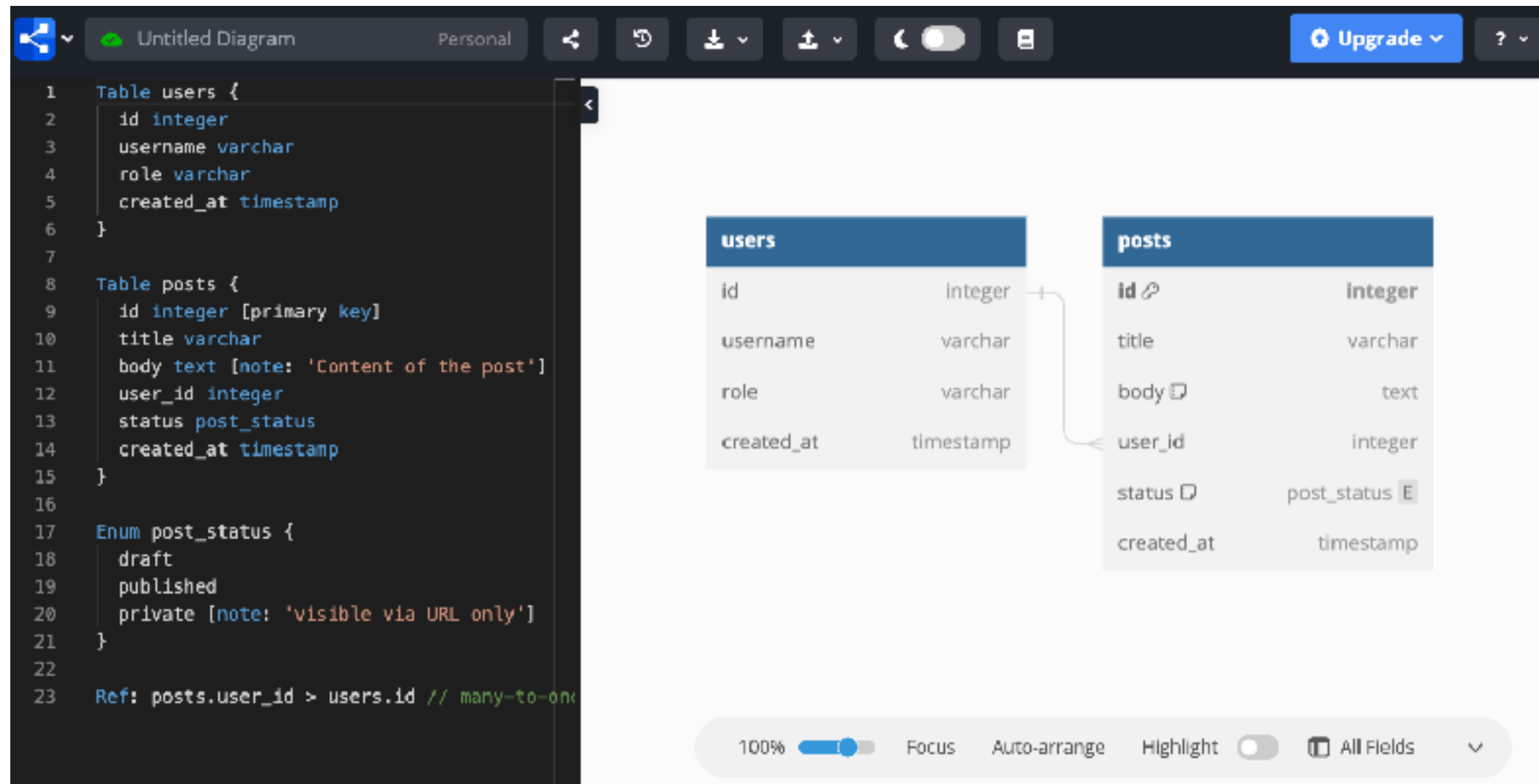


Design



Design table with DBML

Database Markup Language

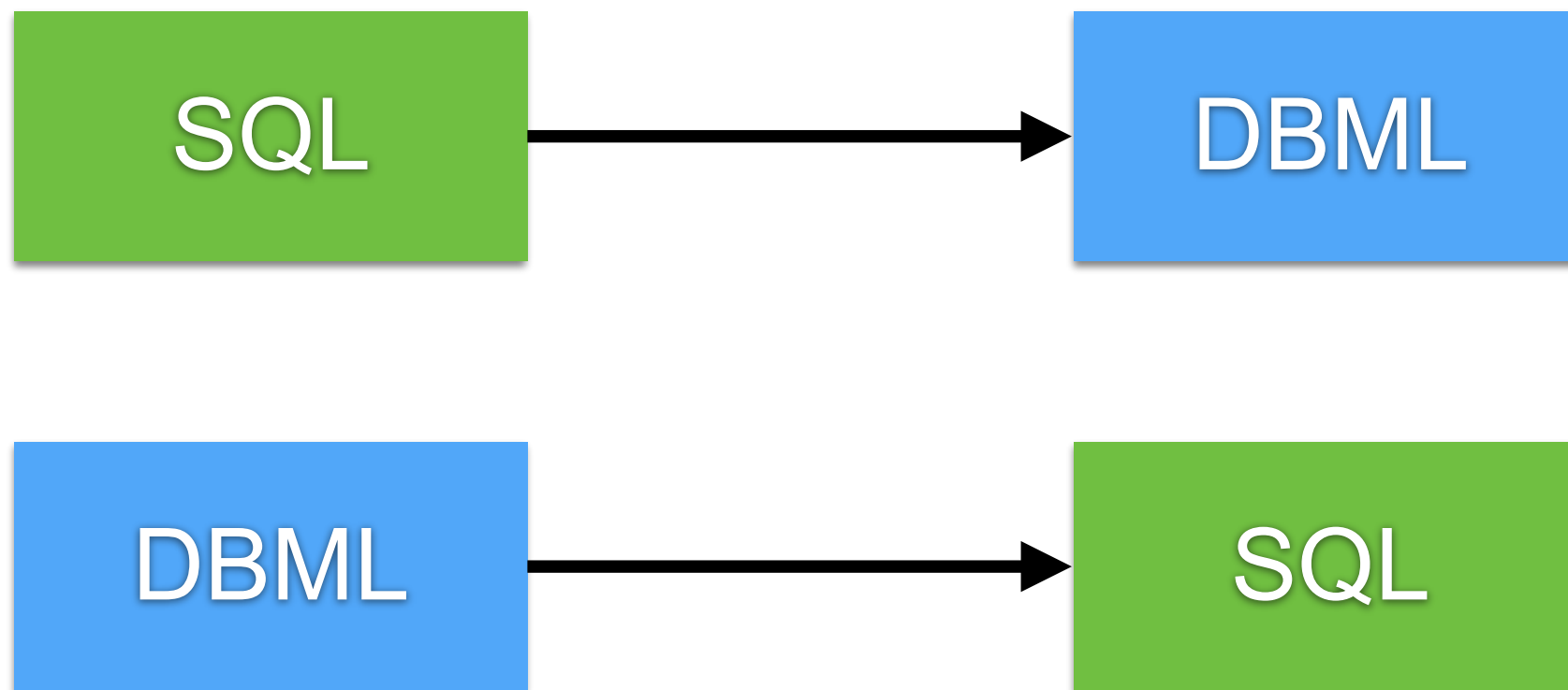


<https://dbml.dbdiagram.io/home/>



Convert SQL to DBML

```
$npm install -g @dbml/cli
```

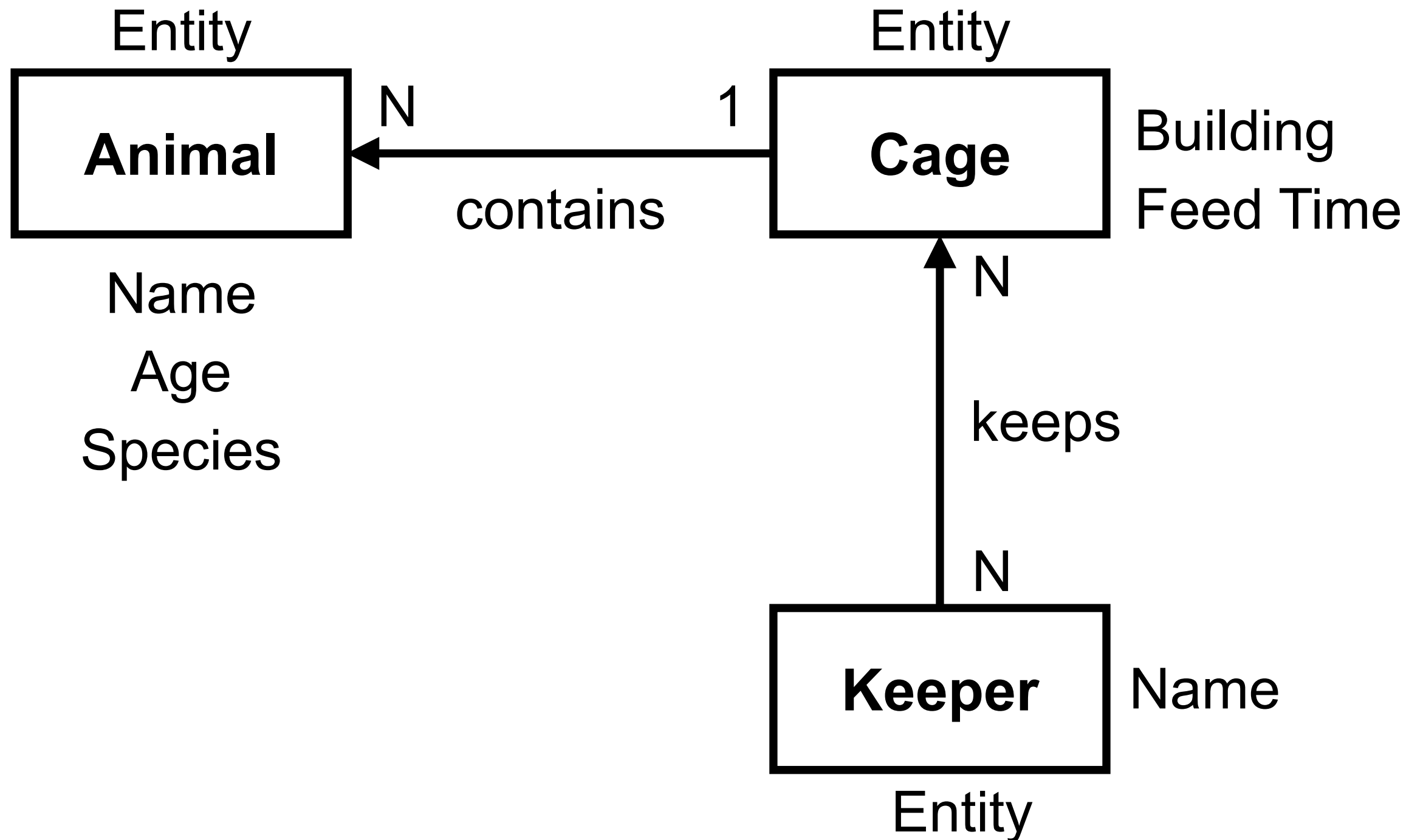


<https://dbml.dbdiagram.io/cli/#convert-a-sql-file-to-dbml>



Design with entity relation diagram





How to manage RDBMS ?



Database Language

Data Definition Language (DDL)
Data Manipulation Language (DML)
Data Control Language (DCL)
Transaction Control Language (TCL)



Data Definition Language (DDL)

Create
Alter
Drop
Truncate



Data Manipulation Language (DML)

Select
Insert
Update
Delete



Delete vs Truncate ?



Delete vs Truncate

Delete

Row-level operation

Mark every row matching the WHERE-clause as deleted

Slow operation

Truncate

Table operation

Empty the table

Start a new data file



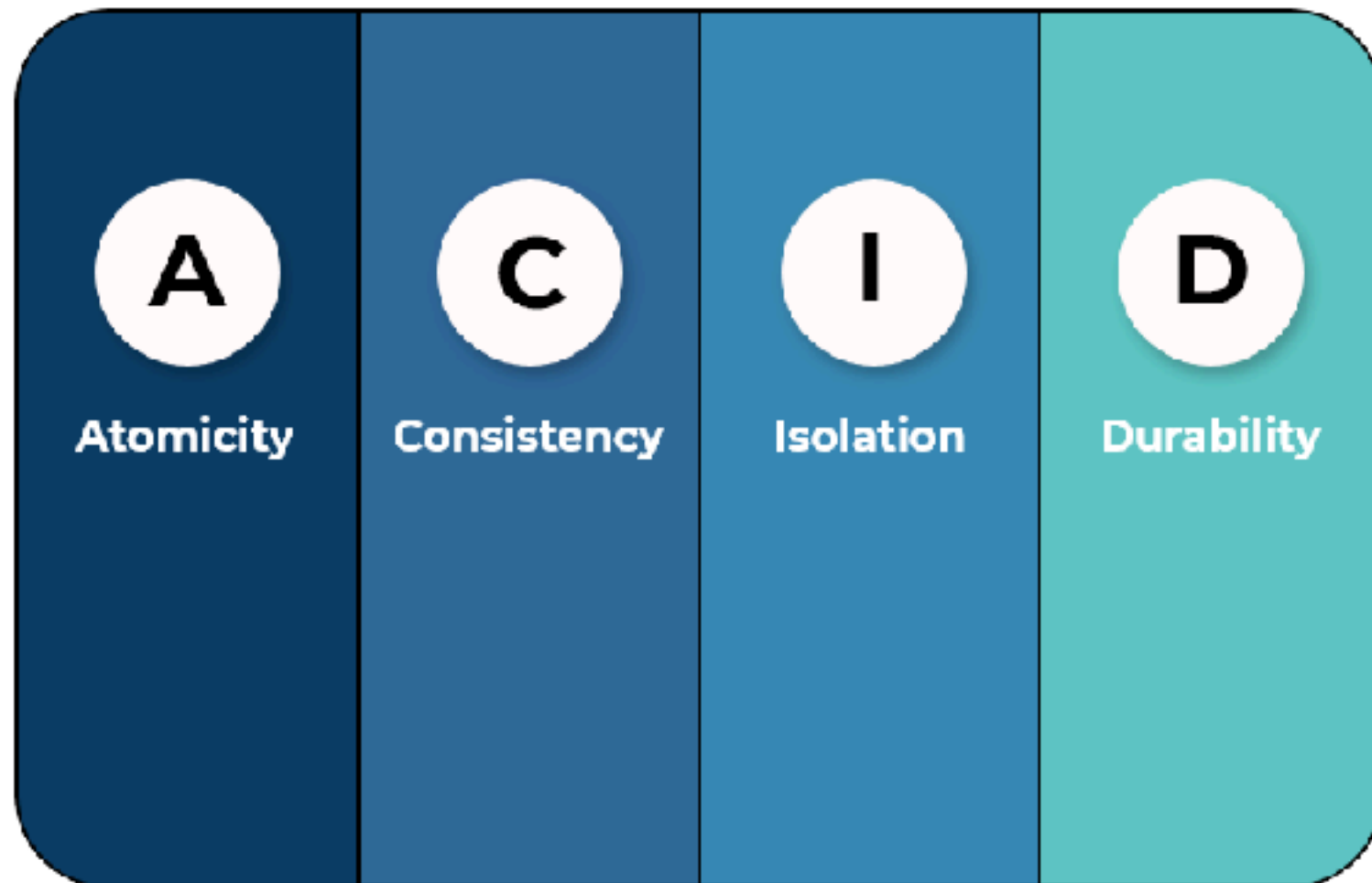
ACID in RDBMS

<https://en.wikipedia.org/wiki/ACID>

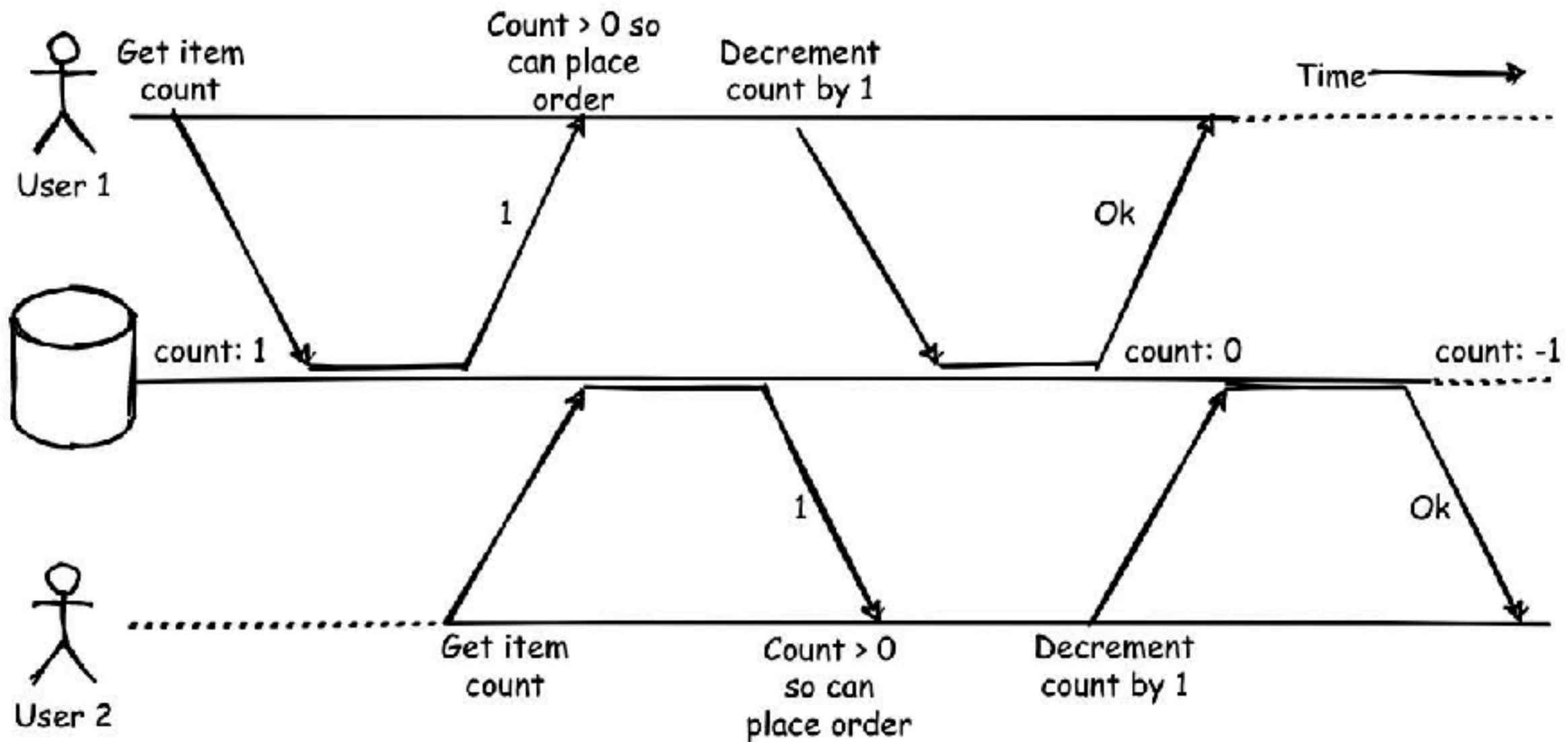


ACID in RDBMS

Ensure database in valid state even in the event of unexpected errors



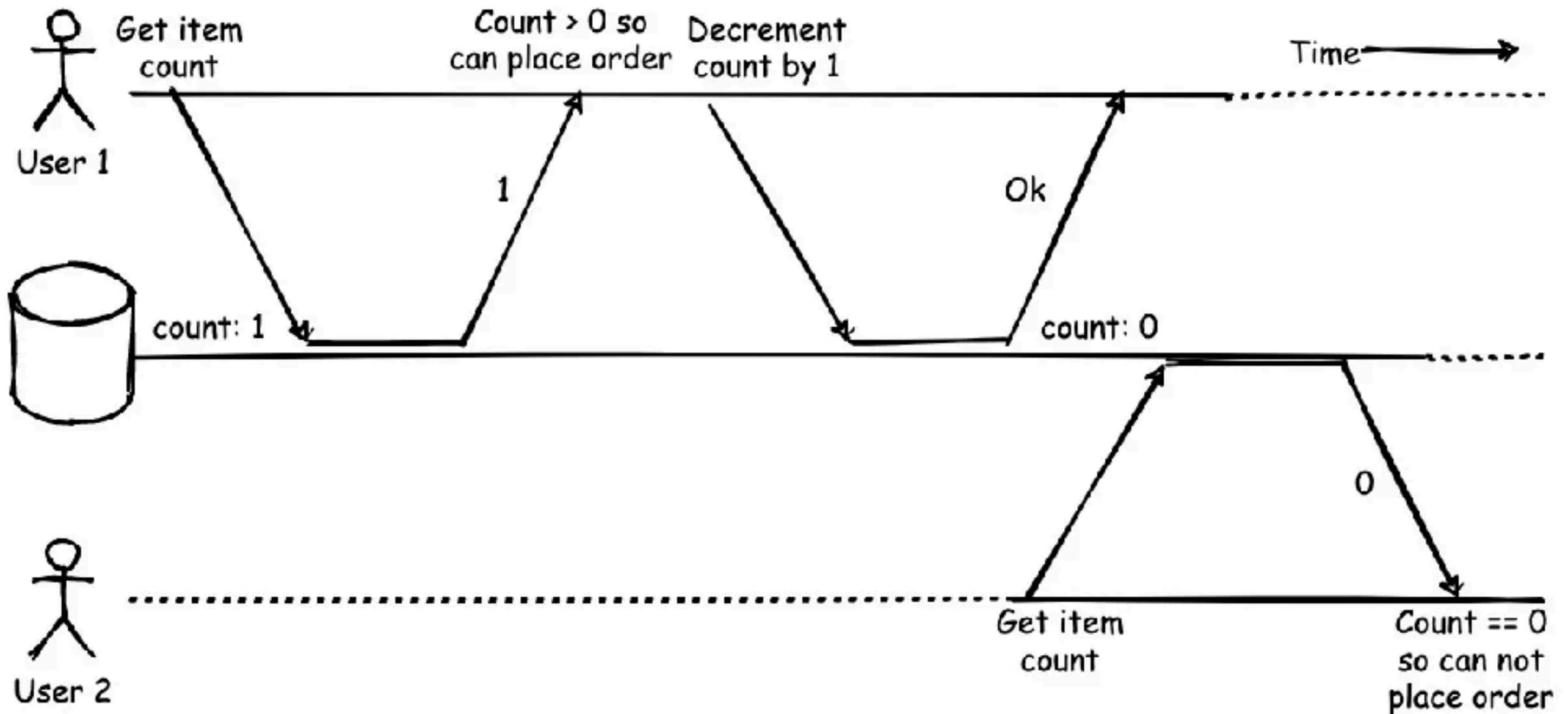
Race condition



https://en.wikipedia.org/wiki/Race_condition#Example



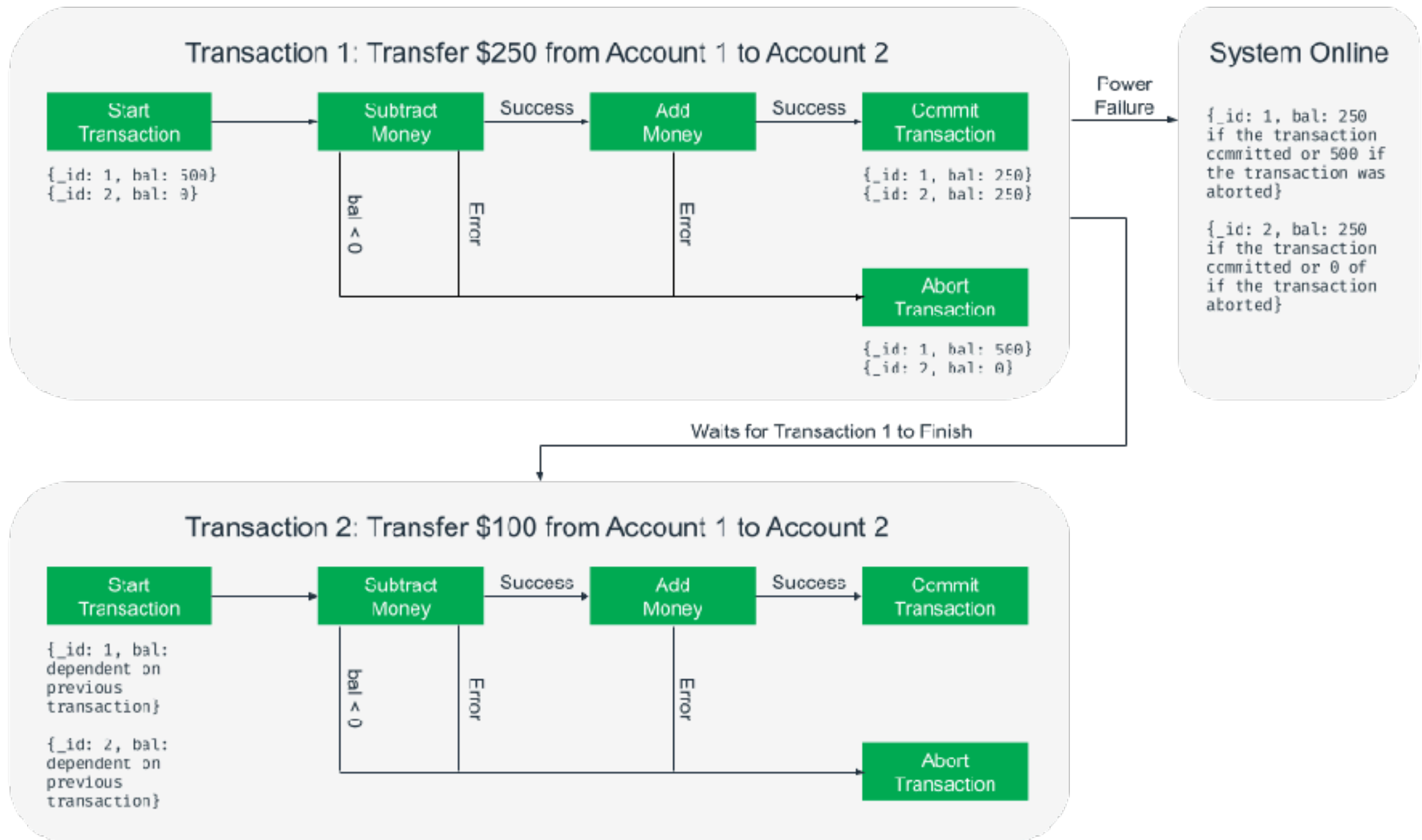
Sequential



<https://www.mongodb.com/basics/acid-transactions>



ACID in RDBMS



<https://www.mongodb.com/basics/acid-transactions>



Structure Query Language



SQL

Maintain data in **tables**

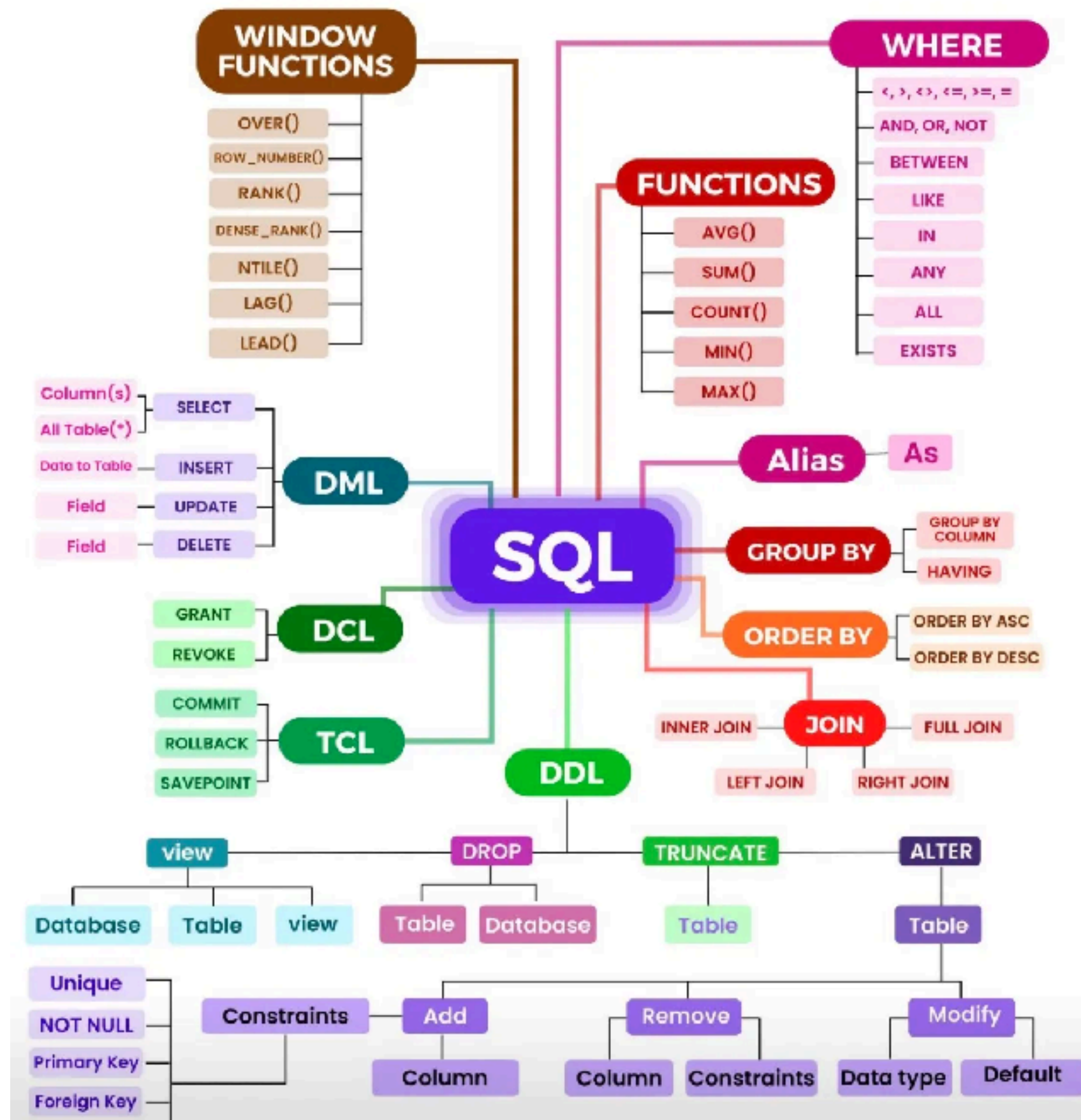
Pre-define structure

Each table has **Primary key**

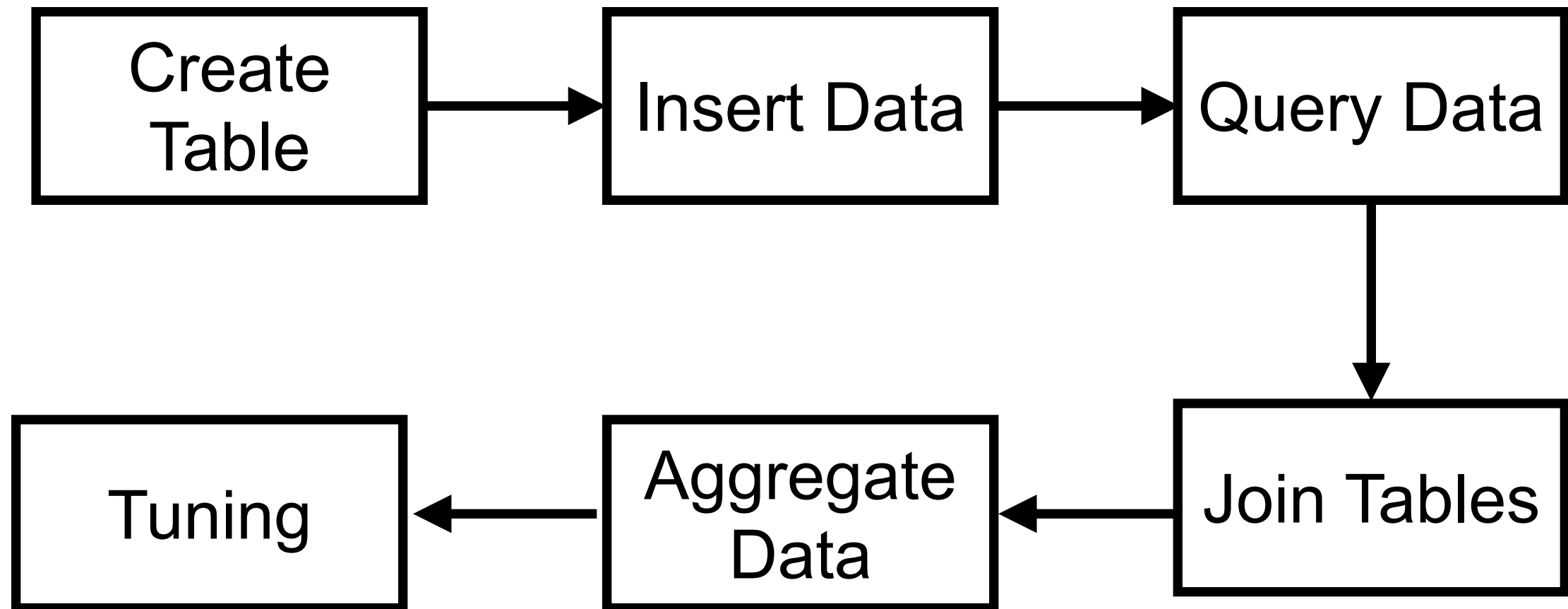
Many **columns**

Data is represented in term of **row**





Learning paths



Query Data



SELECT columns
FROM table
WHERE condition A
AND condition B
GROUP BY columns
ORDER BY columns **ASC|DESC**



Select Data process

Table A

Table B

Table C



Select Data process

Table A

Table B

Table C

FROM + JOIN
(merge)

Table A, B, C



Select Data process

Table A

Table B

Table C

FROM + JOIN
(merge)

Table A, B, C

WHERE
(filter)

Table A, B, C



Select Data process

Table A

Table B

Table C

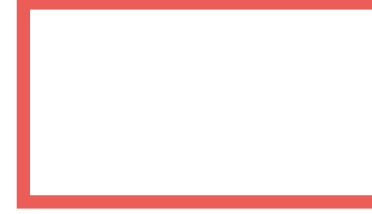
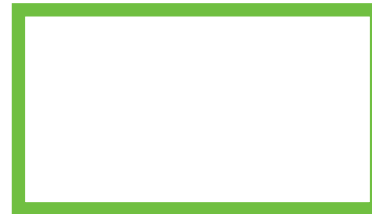
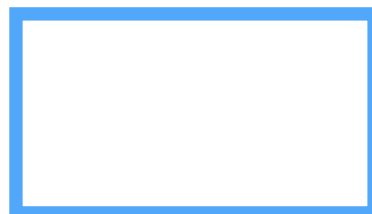
FROM + JOIN
(merge)

Table A, B, C

WHERE
(filter)

Table A, B, C

Group By
(Group)



Select Data process

Table A

Table B

Table C

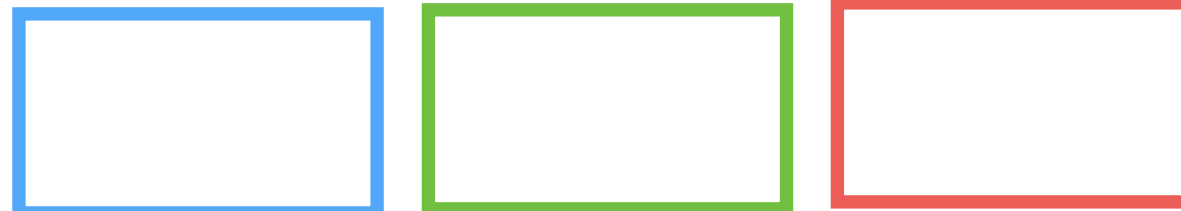
FROM + JOIN
(merge)

Table A, B, C

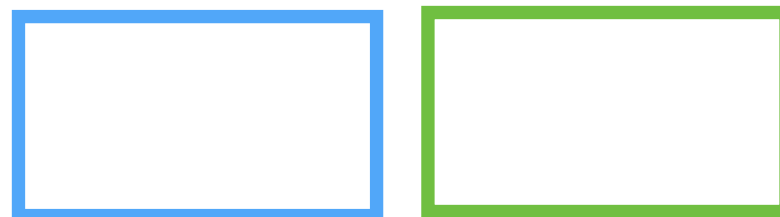
WHERE
(filter)

Table A, B, C

Group By
(Group)



Having
(filter)

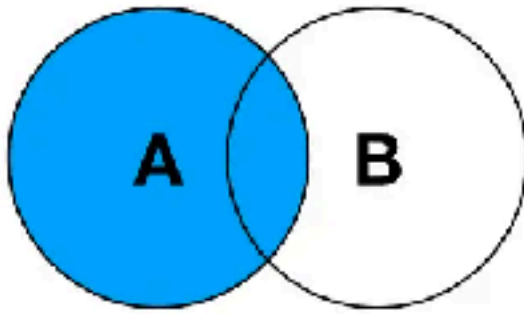


Joining

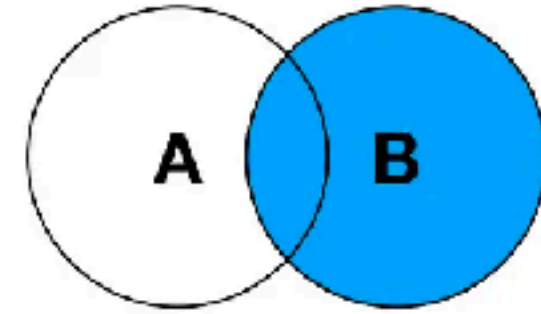
<https://www.postgresql.org/docs/current/tutorial-join.html>



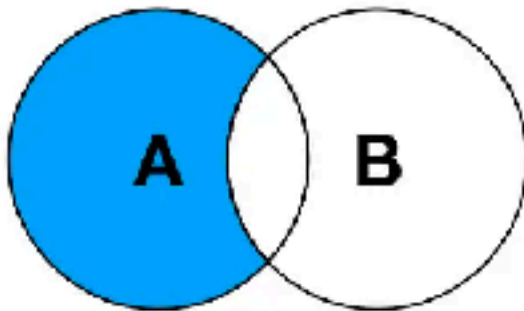
SQL JOINS



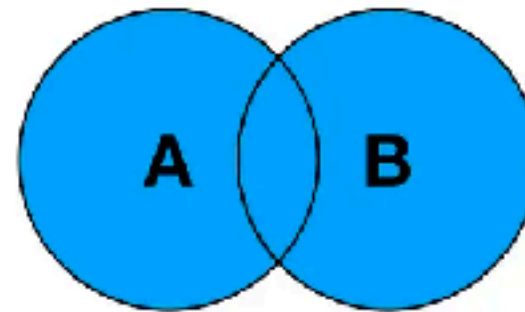
LEFT JOIN



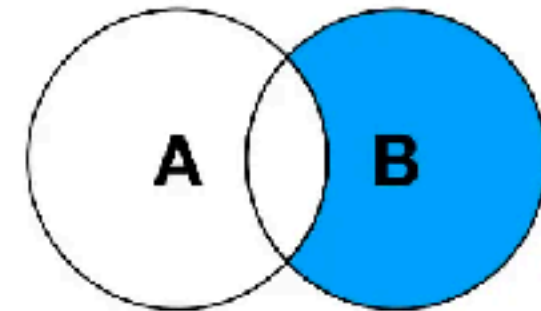
RIGHT JOIN



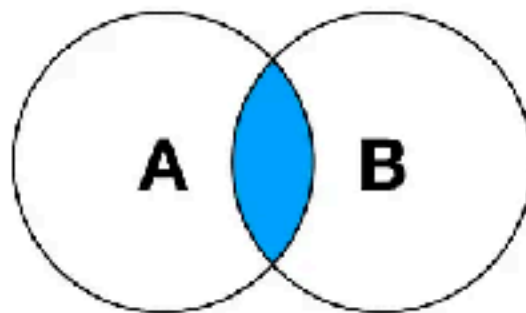
**LEFT JOIN EXCLUDING
INNER JOIN**



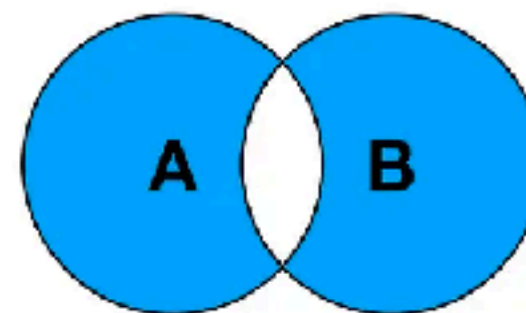
FULL OUTER JOIN



**RIGHT JOIN EXCLUDING
INNER JOIN**



INNER JOIN



**FULL OUTER JOIN EXCLUDING
INNER JOIN**



Self-join

Join in PostgreSQL

Left join

Inner join

Right join

Full outer
join



Workshop

Basket A

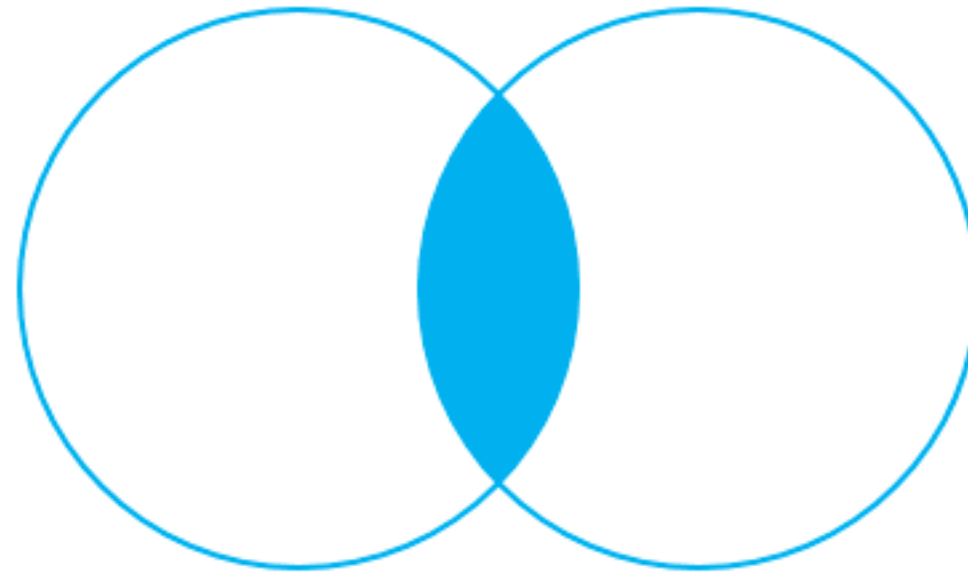
Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear



Inner join



INNER JOIN



Inner join

Basket A

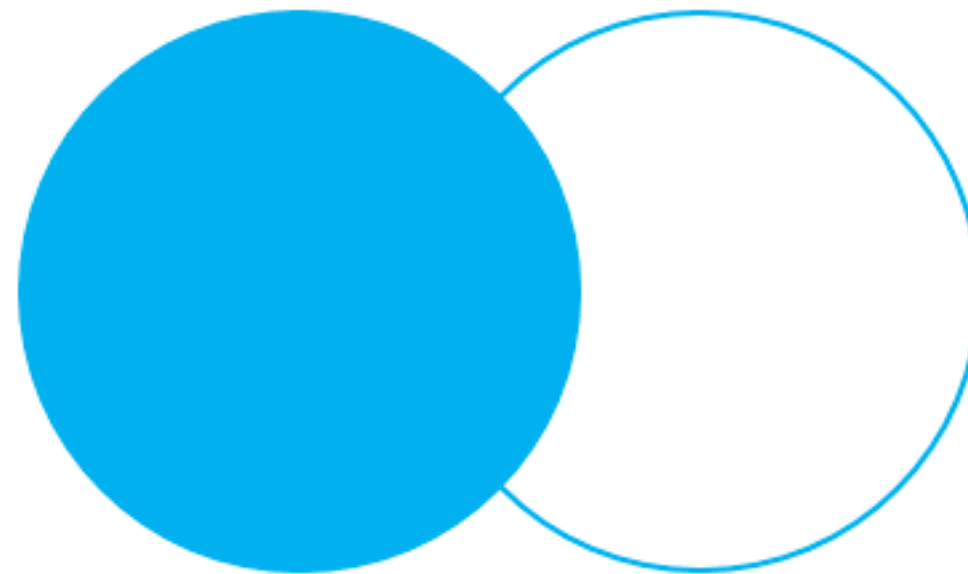
Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear



Left join (1)



LEFT OUTER JOIN



Left join (1)

Basket A

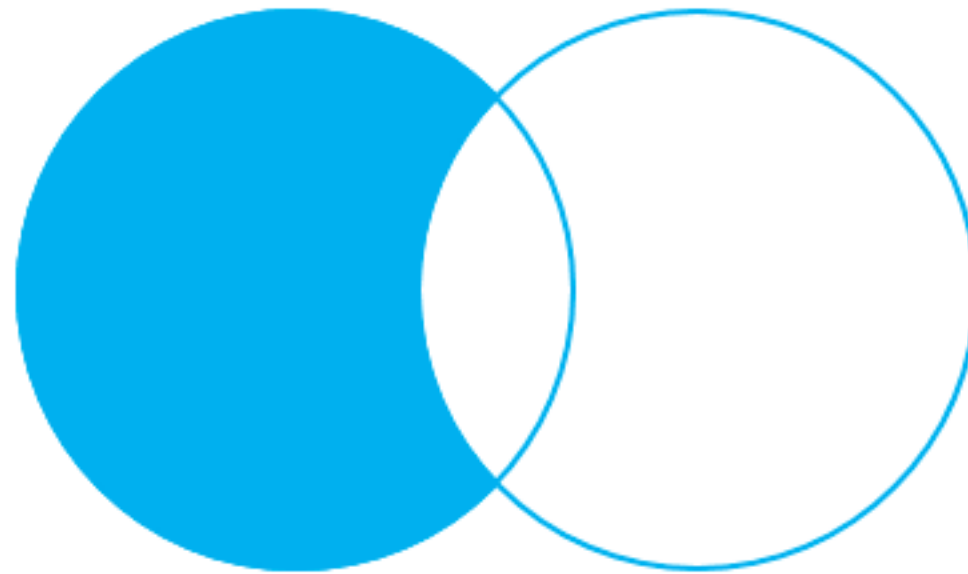
Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear



Left join (2)



LEFT OUTER JOIN – only
rows from the left table



Left join (2)

Basket A

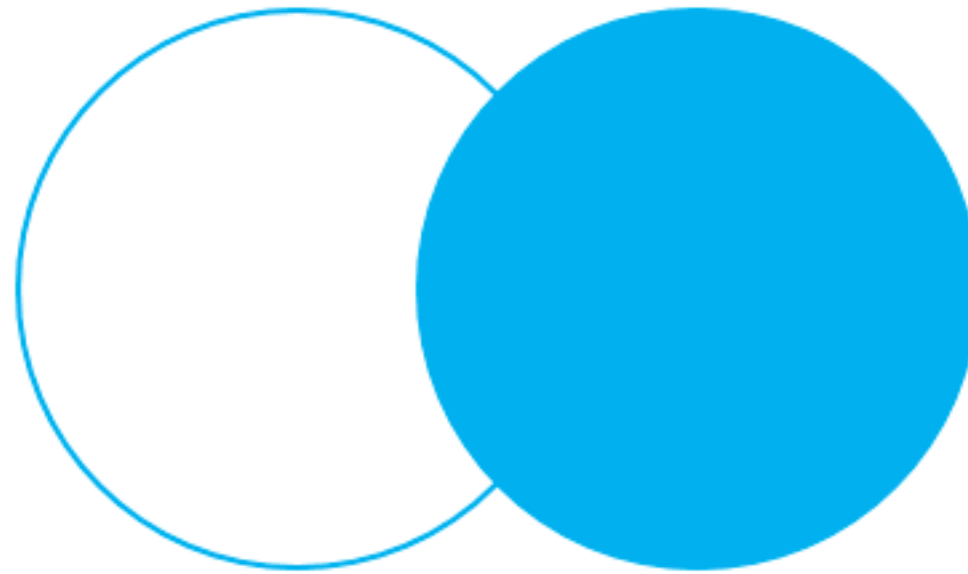
Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear



Right join (1)



RIGHT OUTER JOIN



Right join (1)

Basket A

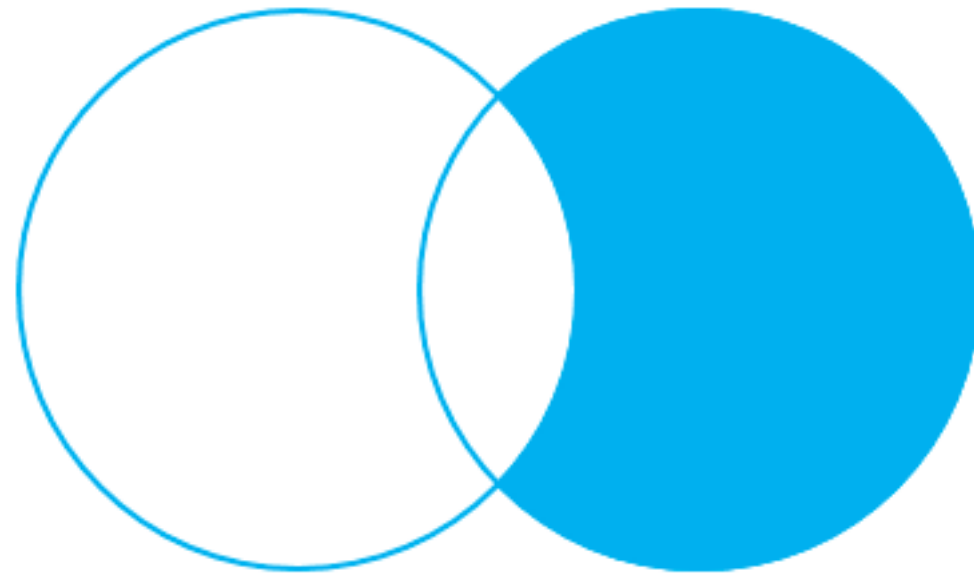
Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear



Right join (2)



RIGHT OUTER JOIN – only
rows from the right table



Right join (2)

Basket A

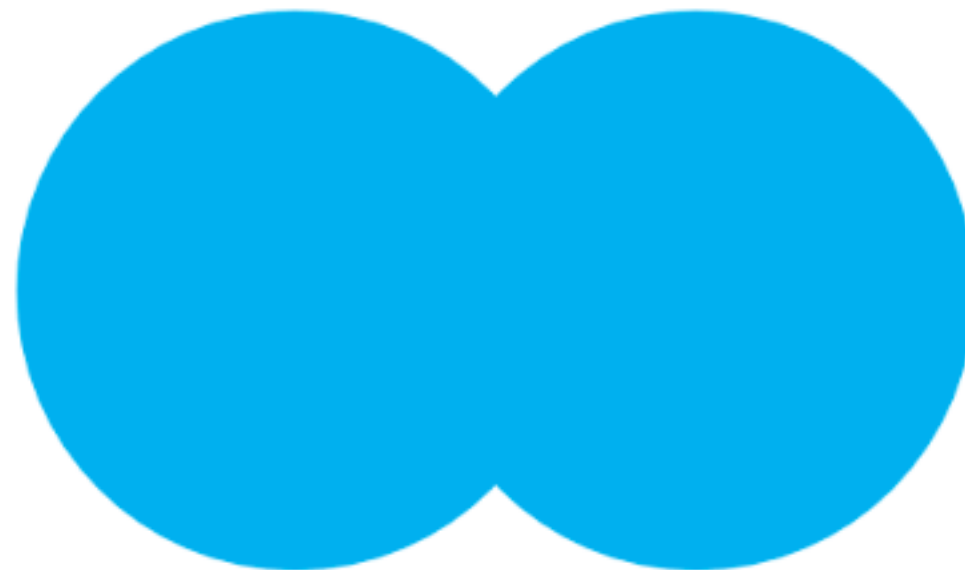
Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear



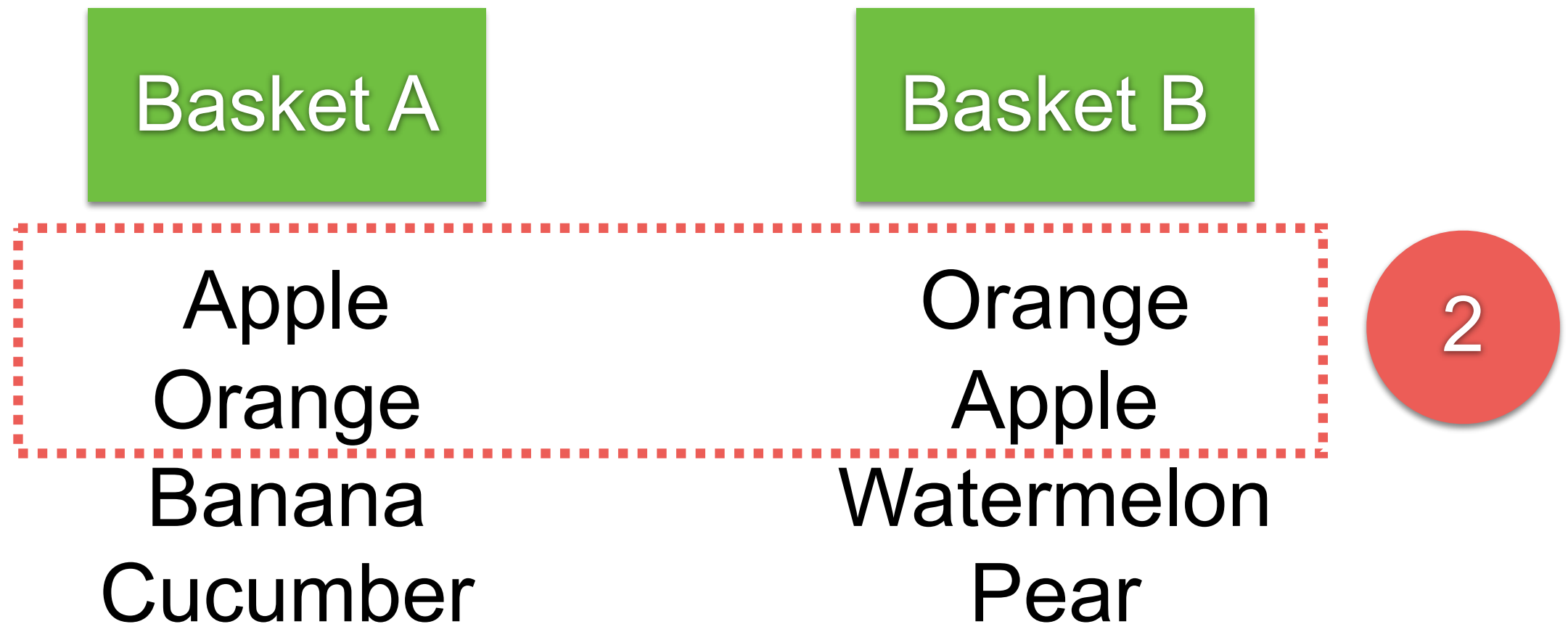
Full outer join (1)



FULL OUTER JOIN



Full outer join (1)



Full outer join (1)

Basket A

Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear

2



Full outer join (1)

Basket A

Apple
Orange
Banana
Cucumber

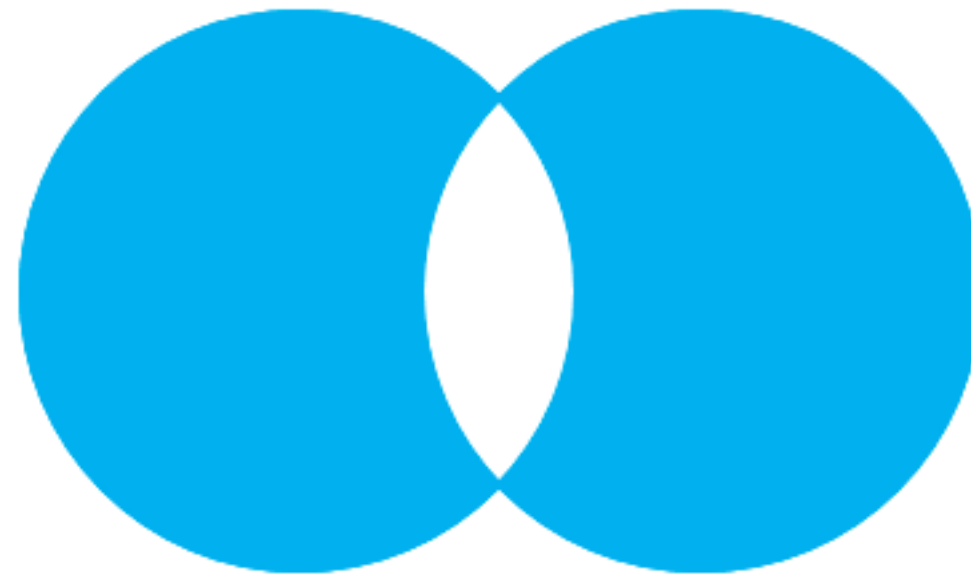
Basket B

Orange
Apple
Watermelon
Pear

2



Full outer join (2)



FULL OUTER JOIN – only
rows unique to both tables



Full outer join (2)

Basket A

Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear

2



Full outer join (2)

Basket A

Apple
Orange
Banana
Cucumber

Basket B

Orange
Apple
Watermelon
Pear

2



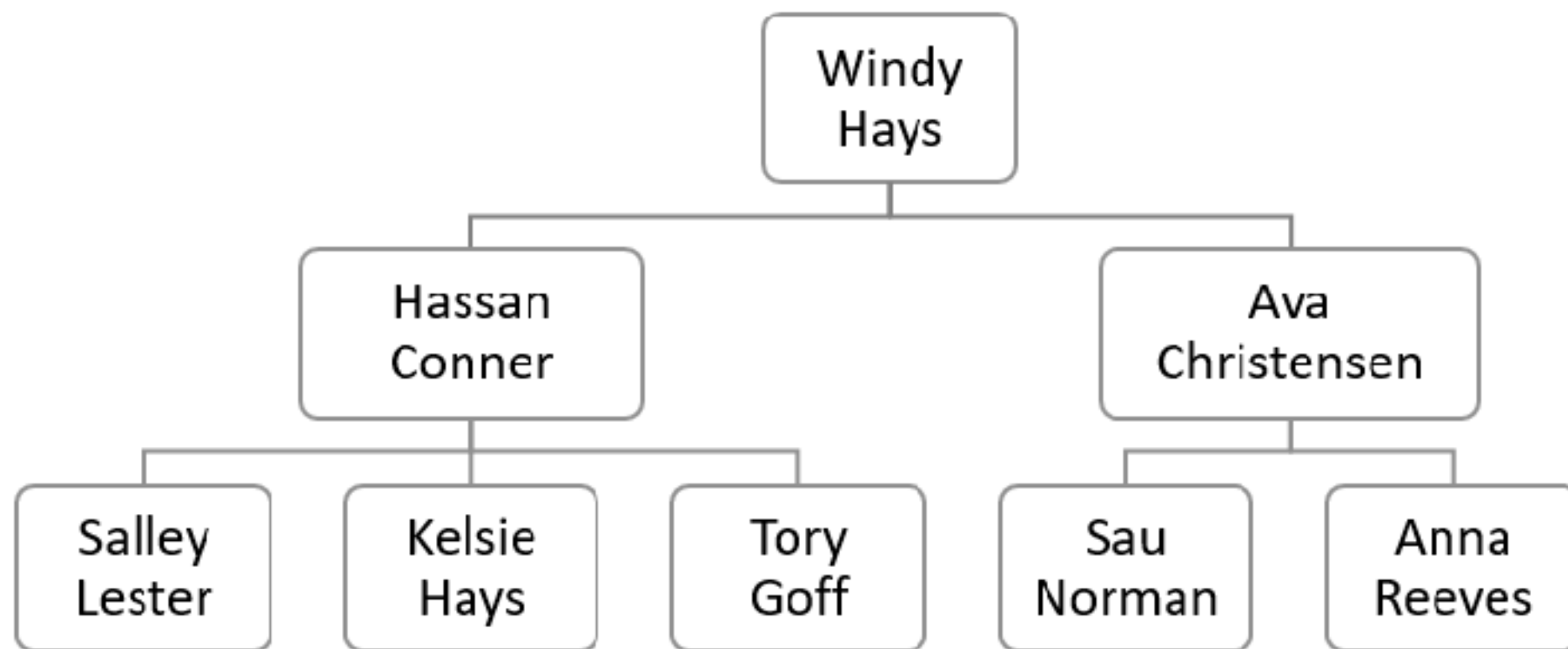
Workshop with Joins

<https://github.com/up1/workshop-database/blob/main/workshop/joins.md>



Self-join !!!

Regular join that joins a table to itself
Query hierarchical data



Workshop with Self-Join

<https://github.com/up1/workshop-database/blob/main/workshop/self-join.md>



Problems ?



Problems

Slow query

Accurate knowledge is required

Add complexity

Need denormalization ?

Rewrite query ?

Database vs application caching ?

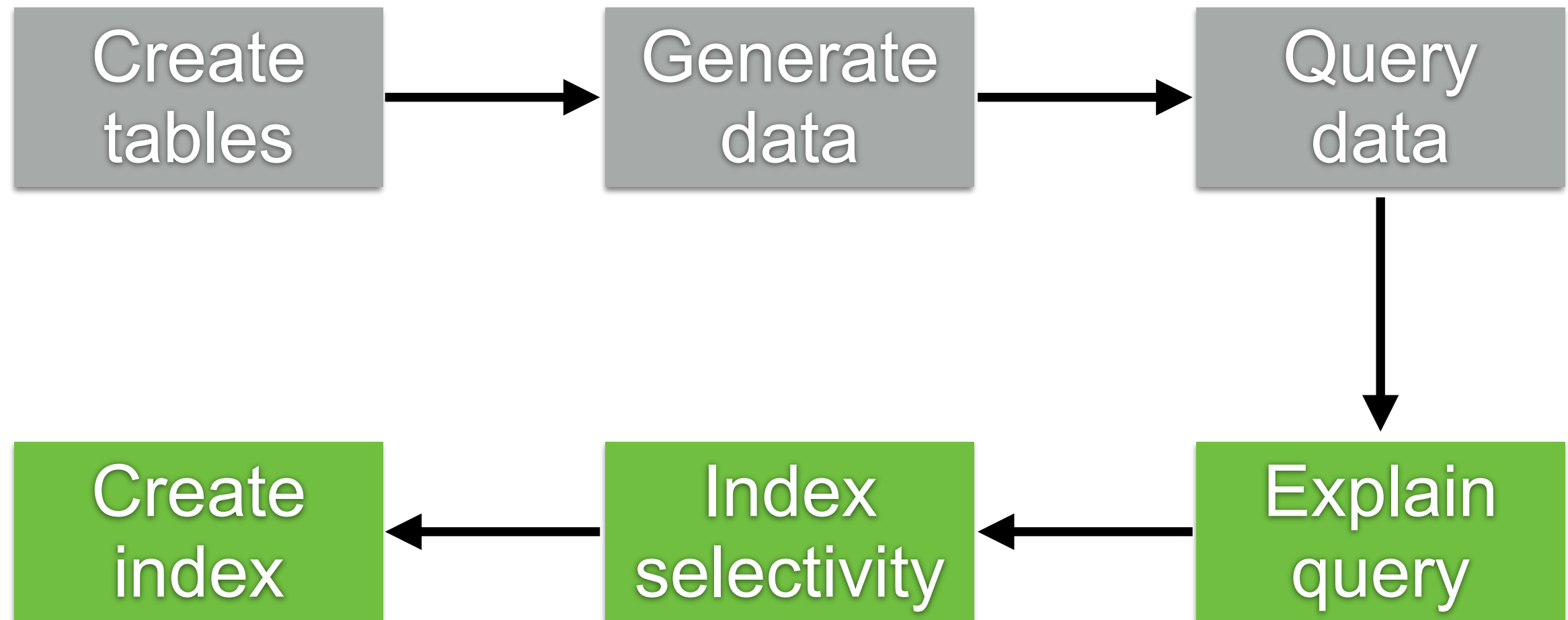


Improvement for query data

When ? Why ? How ? Who ?



Step in workshop



Tips for SQL Optimization

Bad	Good
SELECT *	SELECT column1, column2
SELECT ALL	SELECT with Limit, Offset, Top
Many joins !!	Use appropriate joins
SELECT distinct !! For large datasets	Use group by, union , exists, row number
%column%	Use full text search feature
count(*) > 0 or IN	exists (select 1 from table WHERE ...)
Don't use subquery	Use derived table or pre-joined tables

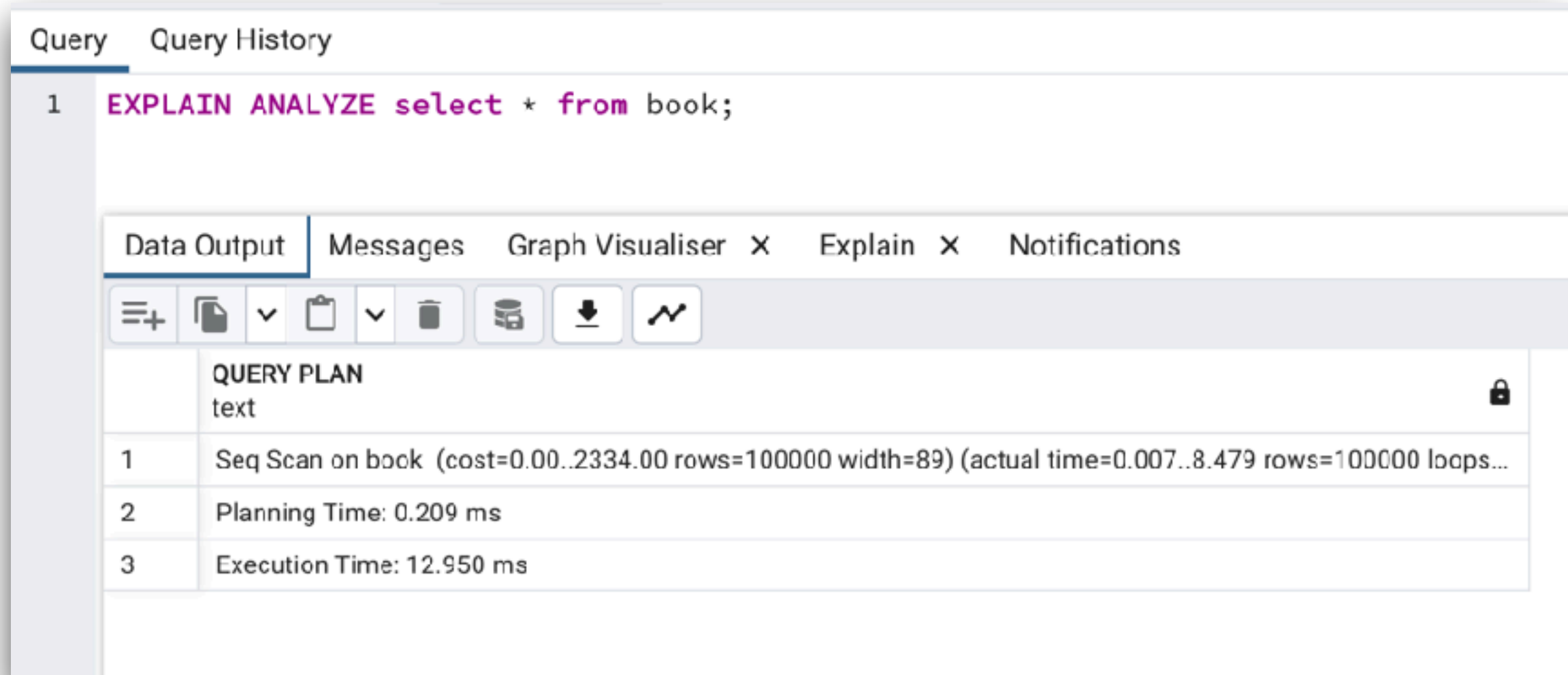


**Always
Explain Analyze Query !!**



Query plan in PostgreSQL

EXPLAIN ANALYZE <SQL>



The screenshot shows a PostgreSQL query interface with the following components:

- Query Tab:** Contains the SQL query: `1 EXPLAIN ANALYZE select * from book;`
- Query History Tab:** Empty.
- Data Output Tab:** Active, showing the query plan results.
- Messages Tab:** Empty.
- Graph Visualiser X:** Empty.
- Explain X:** Empty.
- Notifications:** Empty.

The query plan results are displayed in a table with the following structure:

	QUERY PLAN
1	Seq Scan on book (cost=0.00..2334.00 rows=100000 width=89) (actual time=0.007..8.479 rows=100000 loops...
2	Planning Time: 0.209 ms
3	Execution Time: 12.950 ms

<https://github.com/up1/workshop-database/tree/main/workshop#query-plan>



Indexing Strategies

<https://www.postgresql.org/docs/current/indexes-types.html>



Index

Symbols

2PC, 219, 346, 347, 349–351

A

A* pathfinding algorithms, 64, 84

ACID, 197, 208, 217, 219, 321

ActiveMQ, 92

adjacency lists, 77

Advanced Message Queuing Protocol,
125

Aeron, 402

aggregation window, 164, 176

Airbnb, 193, 199, 337

Amazon, 137, 199, 317

Amazon API Gateway, 303

Amazon Web Services, 251, 302

AML/CFT, 318

AMM, 409

AMQP, 125

Apache James, 231

append-only, 362, 365

Apple, 393

Apple Pay, 315

application loop, 398

ask price, 380

asynchronous, 328

At-least once, 122

at-least once, 93, 122

at-least-once, 331

at-most once, 93, 122

at-most-once, 331

atomic commit, 167

atomic operation, 218

audit, 360

Automatic Market Making, 409

Availability Zone, 253

Availability Zones, 268

availability zones, 27

AVRO, 165

AWS, 251, 302, 303

AWS Lambda, 303

AZ, 268

B

B+ tree, 267

Backblaze, 272

base32, 11

BEAM, 55

bid price, 380

Bigtable, 137, 235, 243

Blue/green deployment, 18

brokers, 95, 96, 98, 102, 105–107, 113,
118, 120, 122

buy order, 393

C

California Consumer Privacy Act,
2

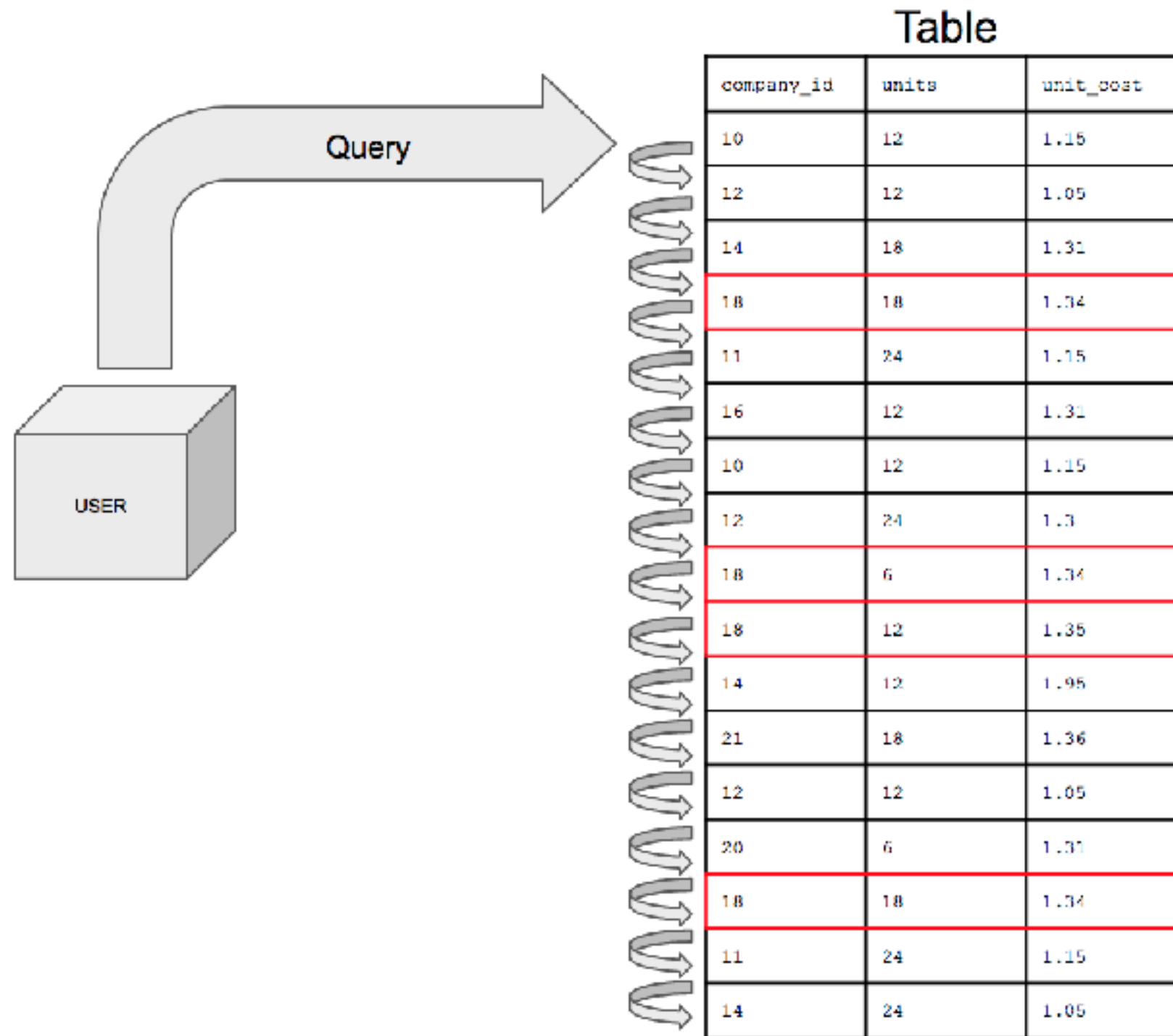
candlestick chart, 381

candlestick charts, 384, 387, 396,
407

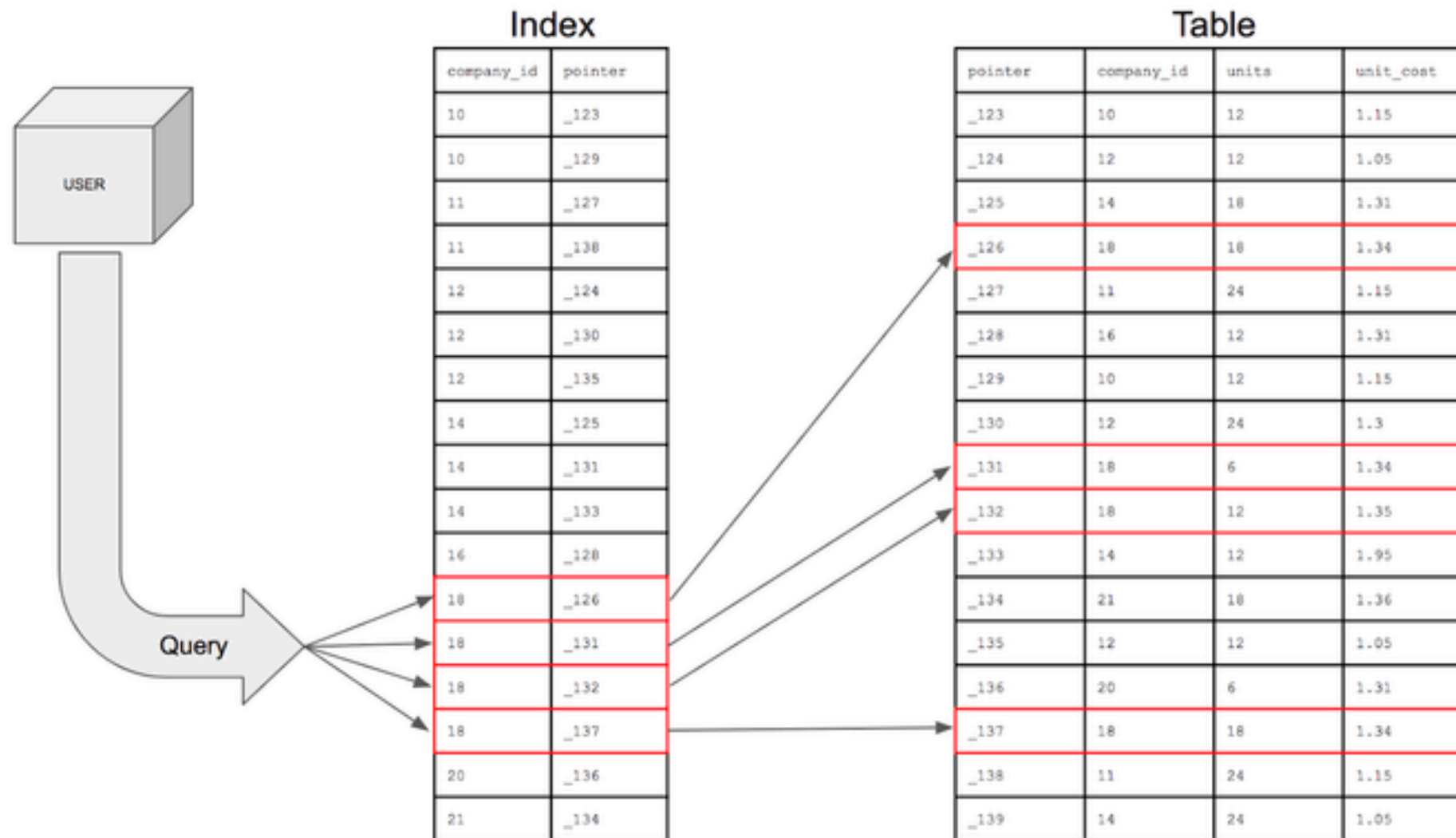
CAP theorem, 79



Without Index (full table scan)



With Index



With Index

Table

	Column A	Column B
RowId 1	Some value 1	Some another value 1
RowId 2	Some value 2	Some another value 2
RowId 3	Some value 3	Some another value 3
RowId 4	Some value 1	Some another value 4
RowId 5	Some value 5	Some another value 5
RowId 6	Some value 6	Some another value 6
RowId 7	Some value 1	Some another value 7
RowId 8	Some value 8	Some another value 8

Index

Some value 1	"RowId 1" "RowId 4" "RowId 7"
Some value 2	RowId 2
Some value 3	RowId 3
Some value 5	RowId 5
Some value 6	RowId 6
Some value 8	RowId 8

Index by column "Column A"



Main factors for good index ?

How much space does the index use ?

How fast does it read ?

How much overhead does it add to writes ?

How well does it handle concurrency ?

Quick lookup, fast insert, reliable delete



Index selectivity

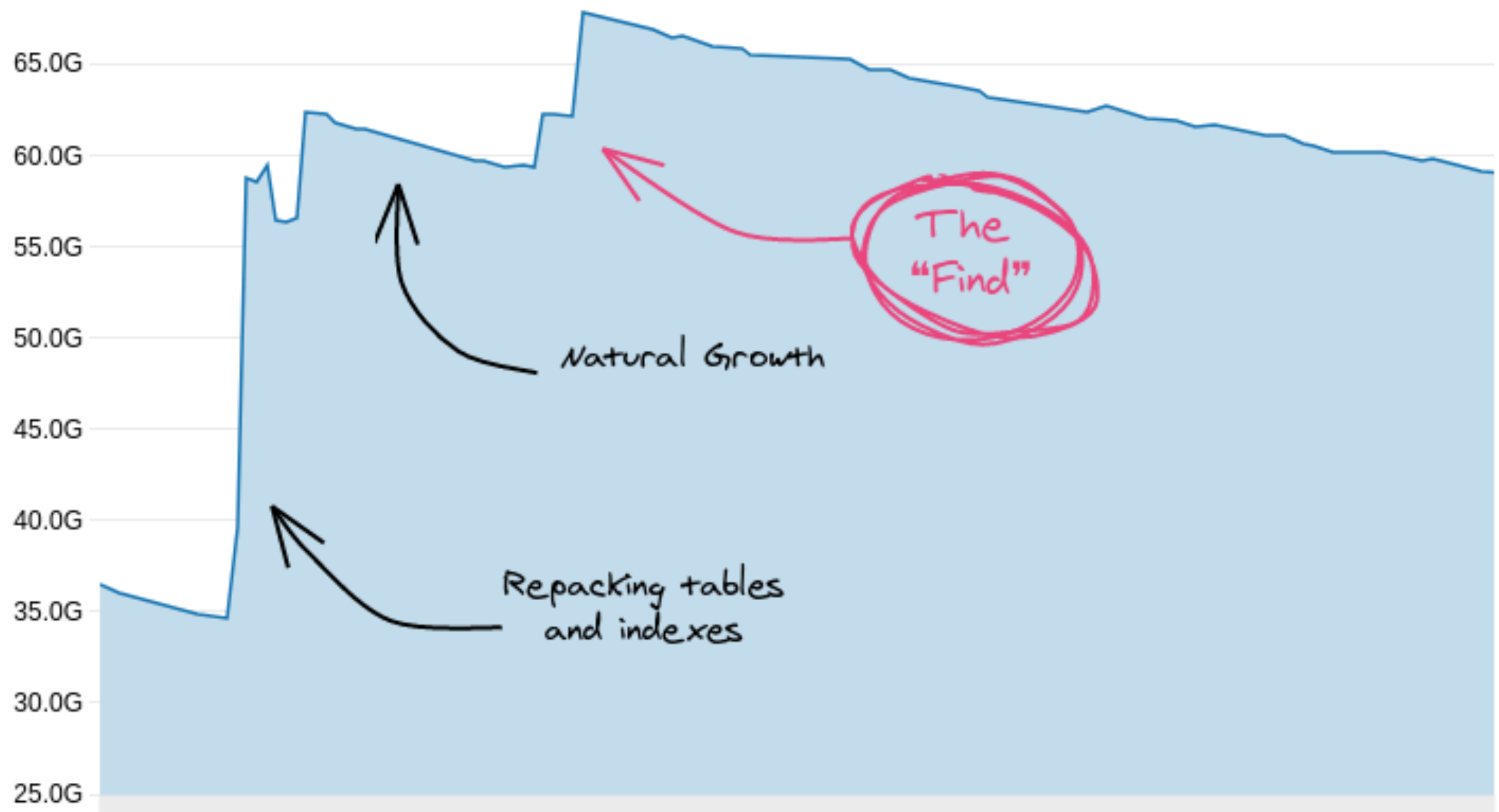
Factor to select a column to create an index

The number of **distinct values** in the indexed column divided by the number of records in the table is called a selectivity of an index

Prefer indexing columns with selectivity greater than > 0.85 (use Index scan)



Unused Index ?



<https://hakibenita.com/postgresql-unused-index-size>



Types Index in PostgreSQL

B-Tree

Hash

Block range indexes (BRIN)

Generalized inverted index (GIN)

Generalized inverted search tree index (GiST)

Space partitioned GiST (SP-GiST)

<https://www.postgresql.org/docs/current/indexes-types.html>

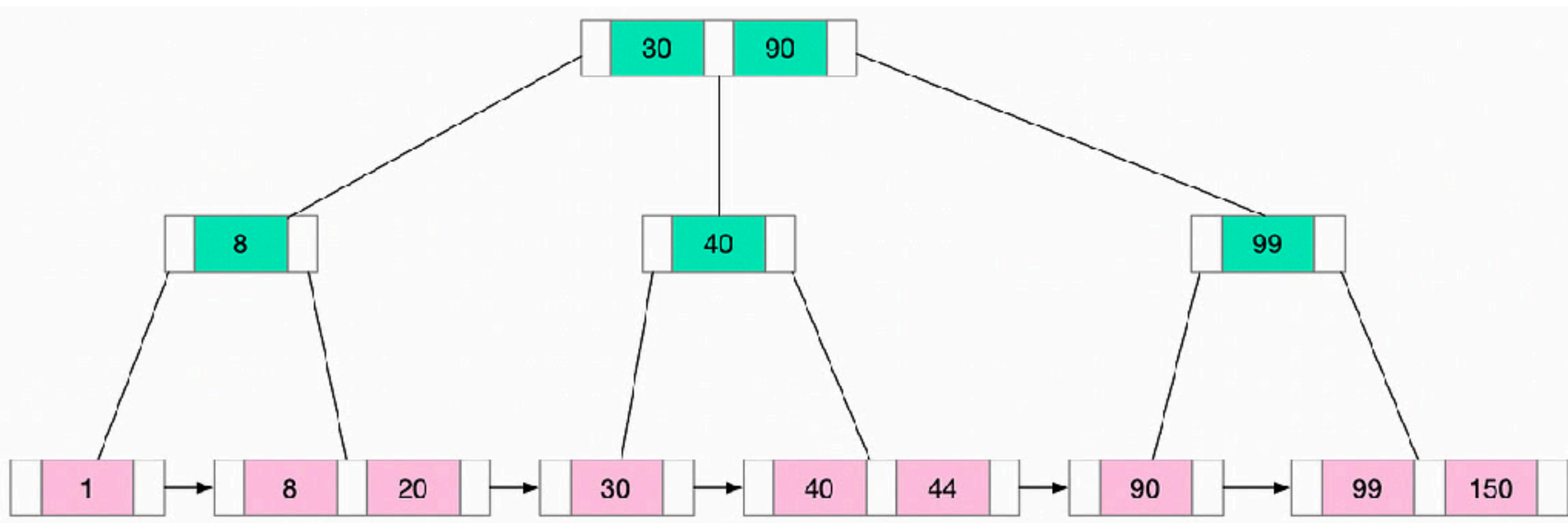


PostgreSQL Index Types

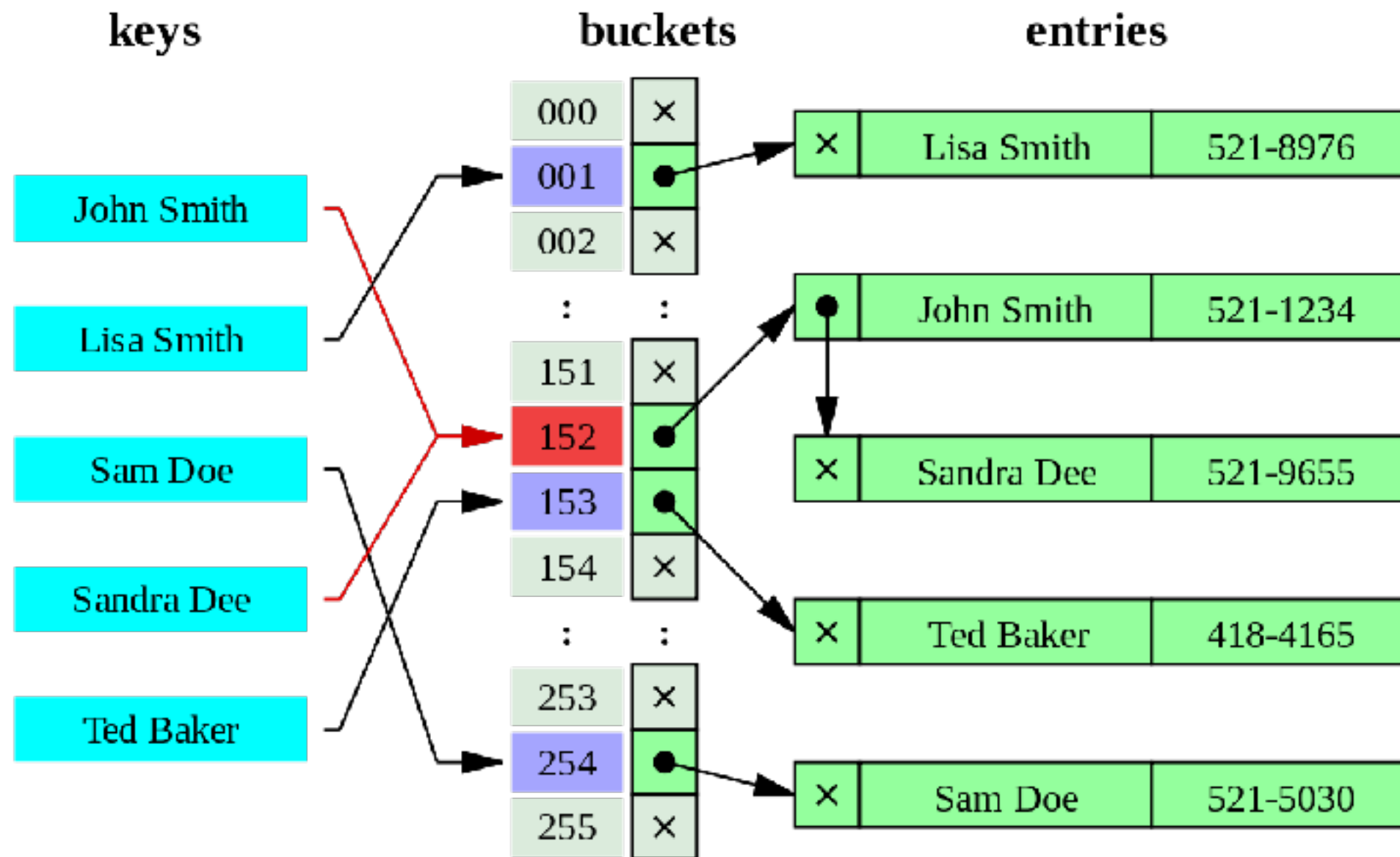
Index type	Data	Target	Use case
B-Tree	High-cardinality scalar columns	Fast equality, range, between operation	PK, FK, timestamp/range scan, sorting
GIN	Composite values (arrays, JSONB, tsvector)	Efficient contains, full-text, array membership	Full text search, JSONB containment, array lookup
GiST	Complex or non-scalar types	Flexible tree, Range and text	PostGIS spatial queries, trigram similarity, range types
BRIN	Very large data, ordered table	Lightweight index for append-only and range scan	Time-series data Log data
Bloom	Many low-cardinality columns	Compact multi-column equality filters	Filter data, check data existing



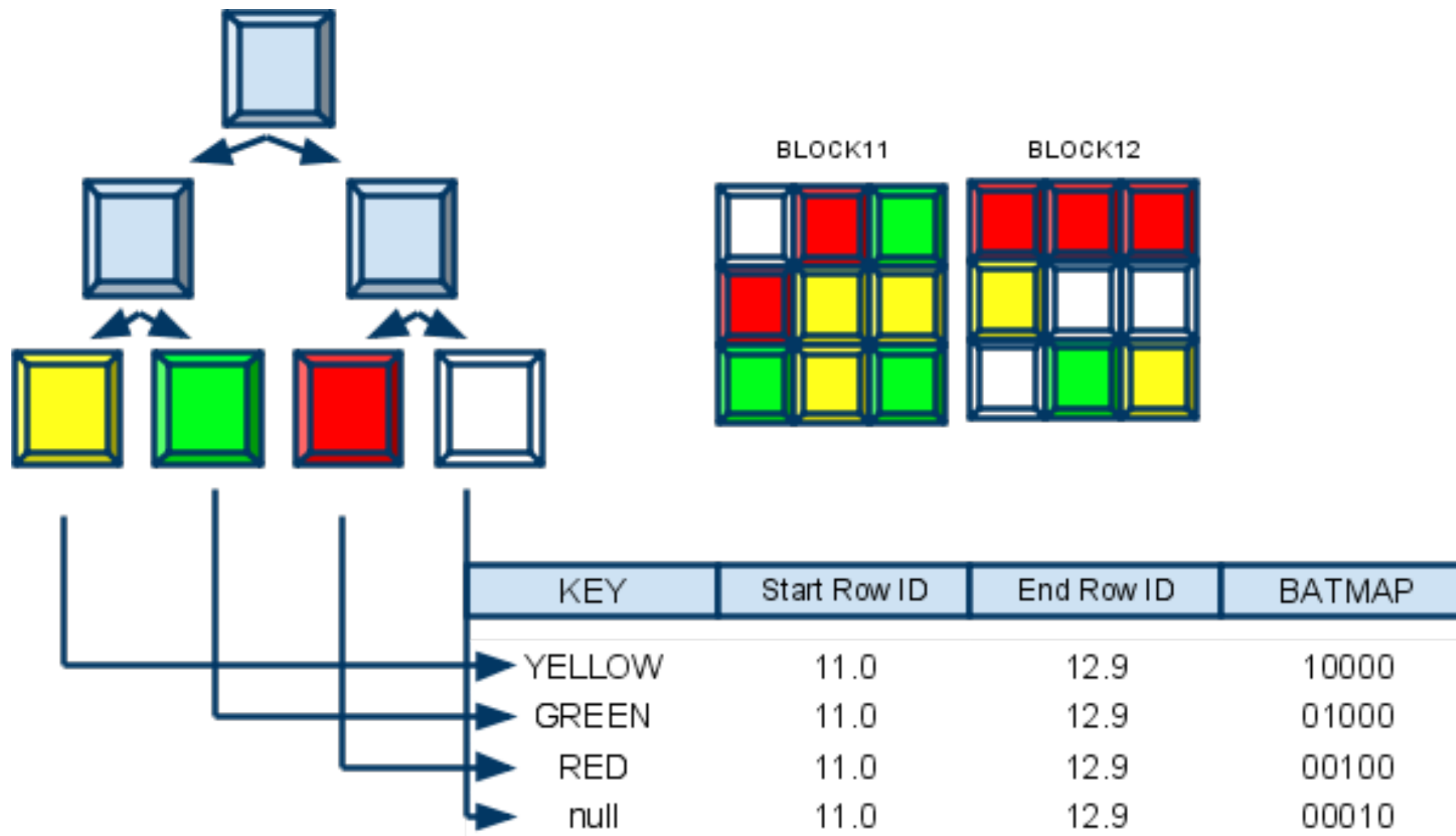
B-Tree +



Hash



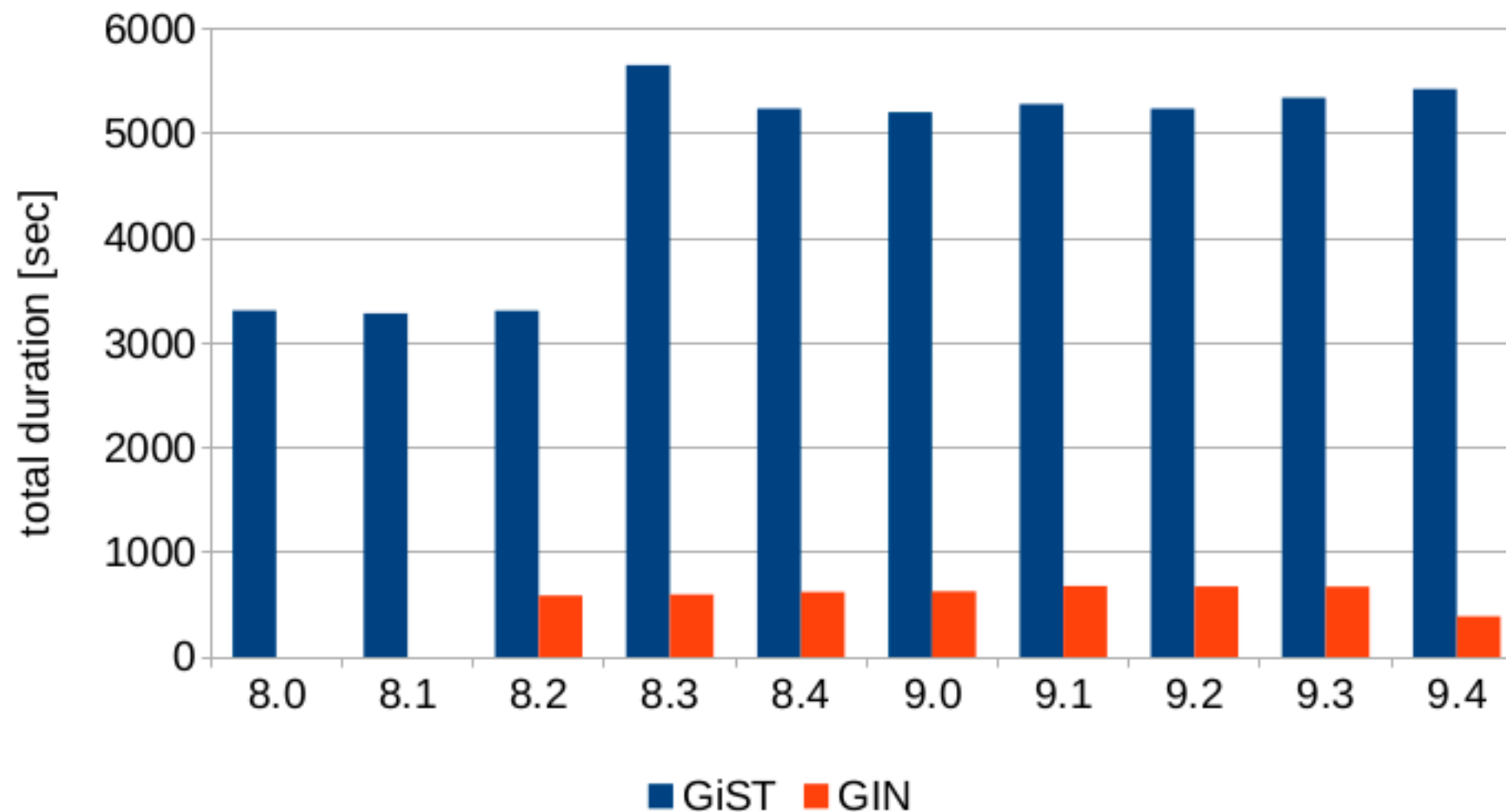
Bitmap



GIN vs GiST

Fulltext benchmark - GiST vs. GIN

33k queries from postgresql.org [TOP 100]



Tuning and Use Index ?

Size vs Speed

Extension setup

Combine indexes

Measure impact with **EXPLAIN** (before and after)



Don't over-index Unused index

Size of index ?



Workshop with Indexing

<https://github.com/up1/workshop-database/tree/main/workshop/basic-workshop>



Tuning Performance



Tuning performance

Normalization vs Denormalization

Partition

Sharding

Database architecture

Database parameteres



Partition

<https://hakibenita.com/postgresql-unused-index-size>



Partition

Table partitioning refers to **splitting** what is logically one large table into **smaller physical pieces**

Reduce index size



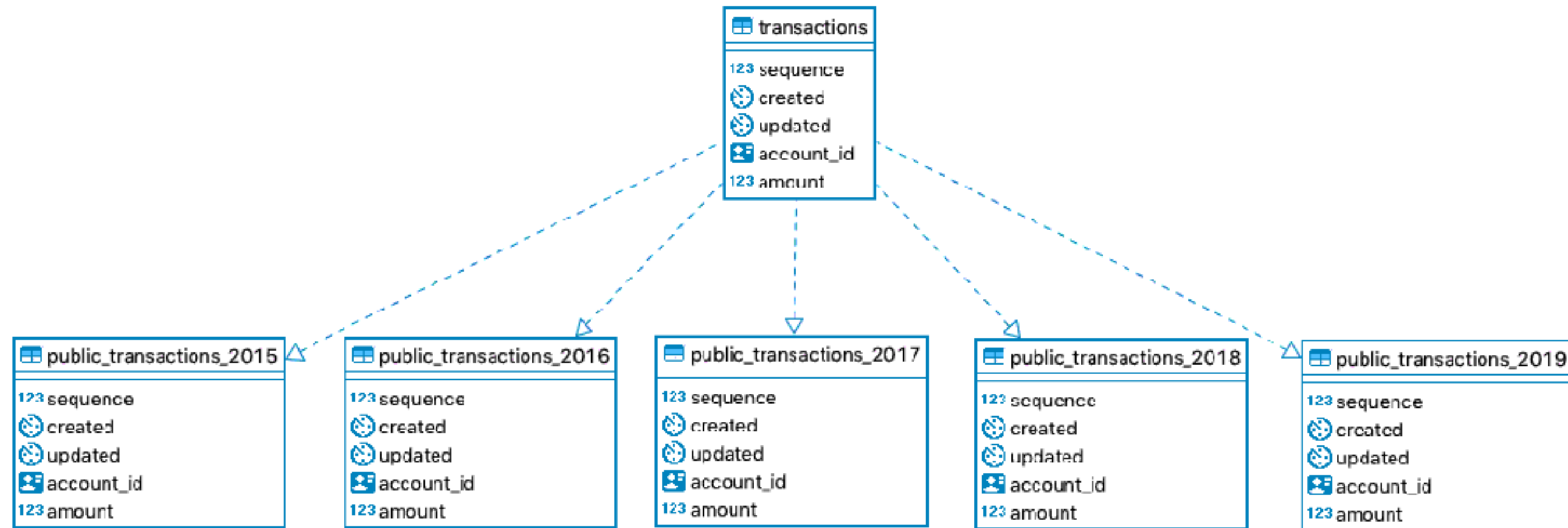
Partition

Horizontal partitioning
Vertical partitioning



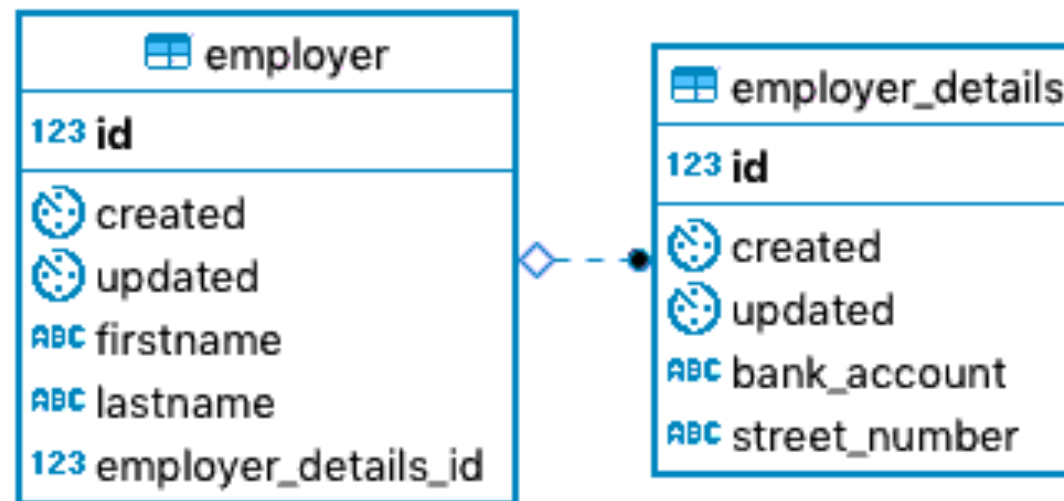
Horizontal partitioning

Putting different rows into different table



Vertical partitioning

Horizontal partitioning



Types Horizontal partitioning

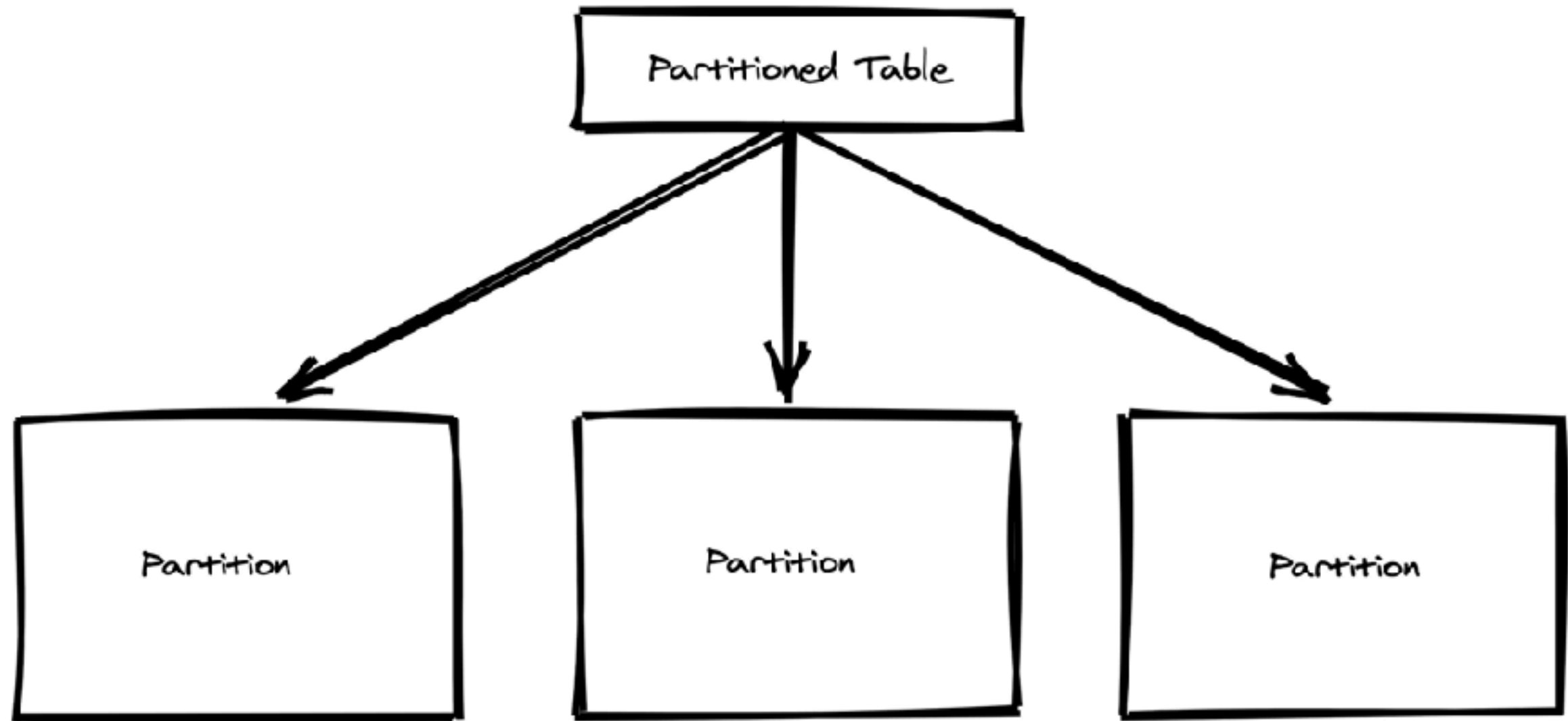
Range
List



Example



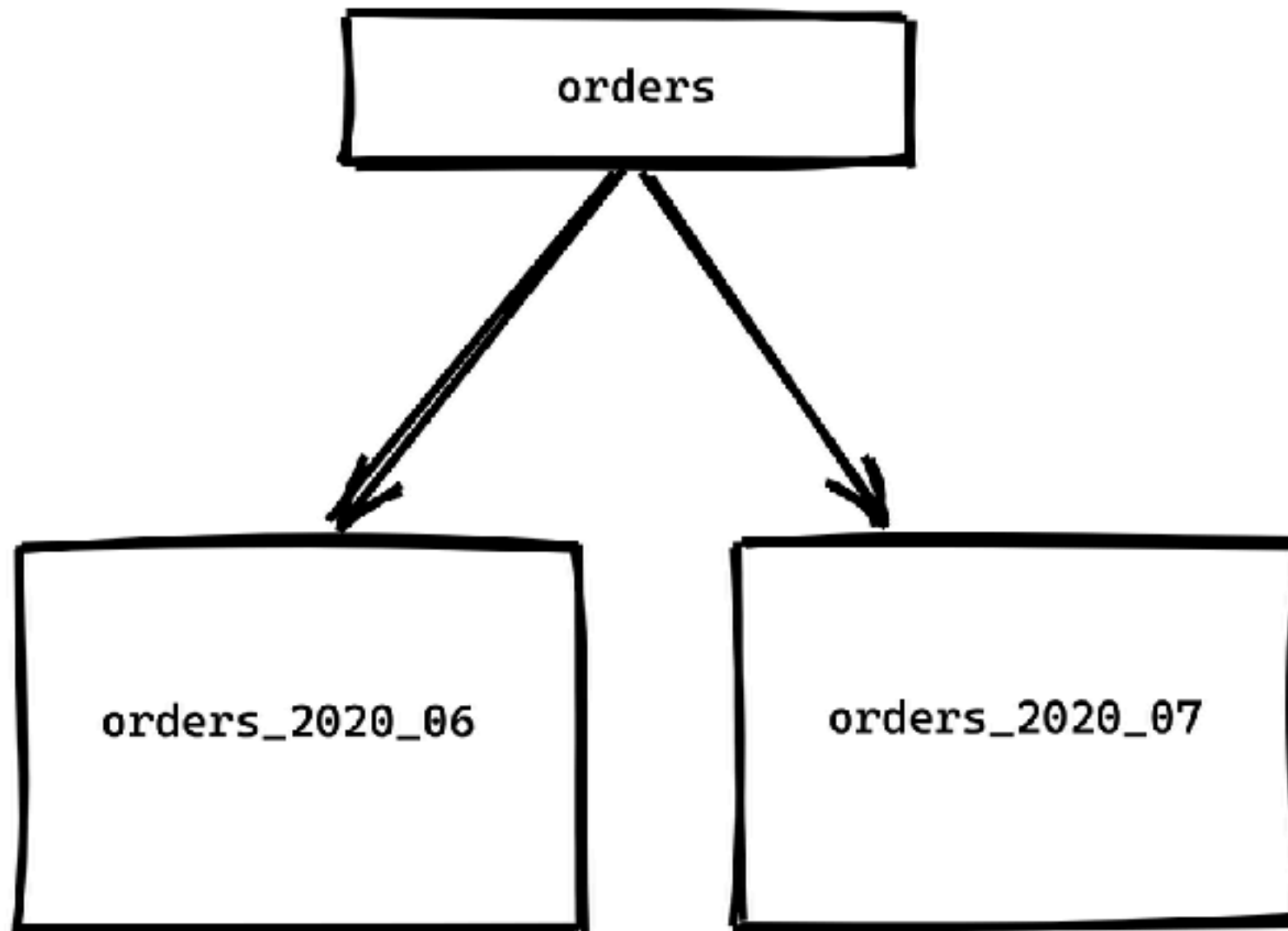
Partition tables



<https://chrisherwin.com/table-partitioning/>



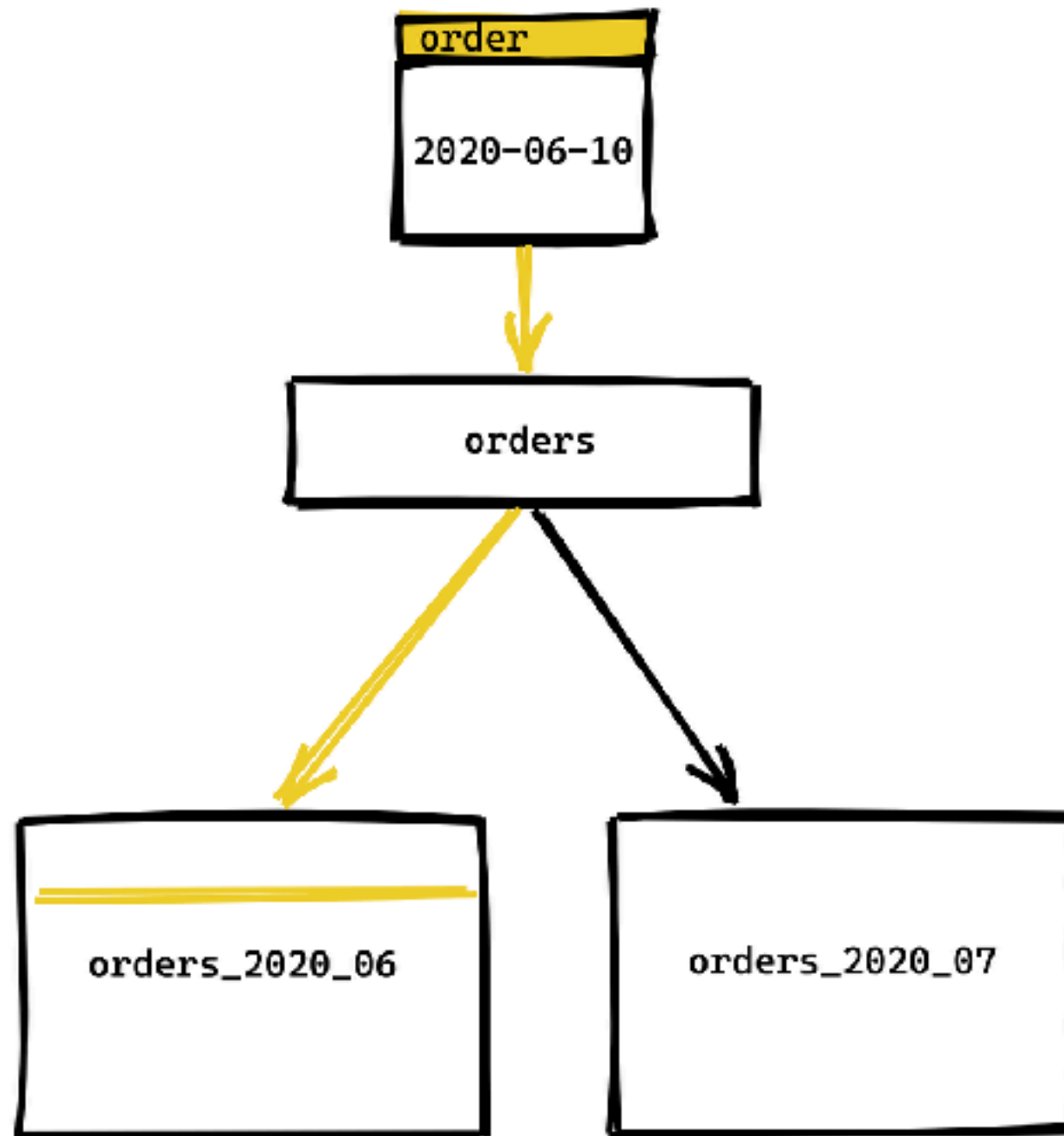
Orders table



<https://chriserwin.com/table-partitioning/>



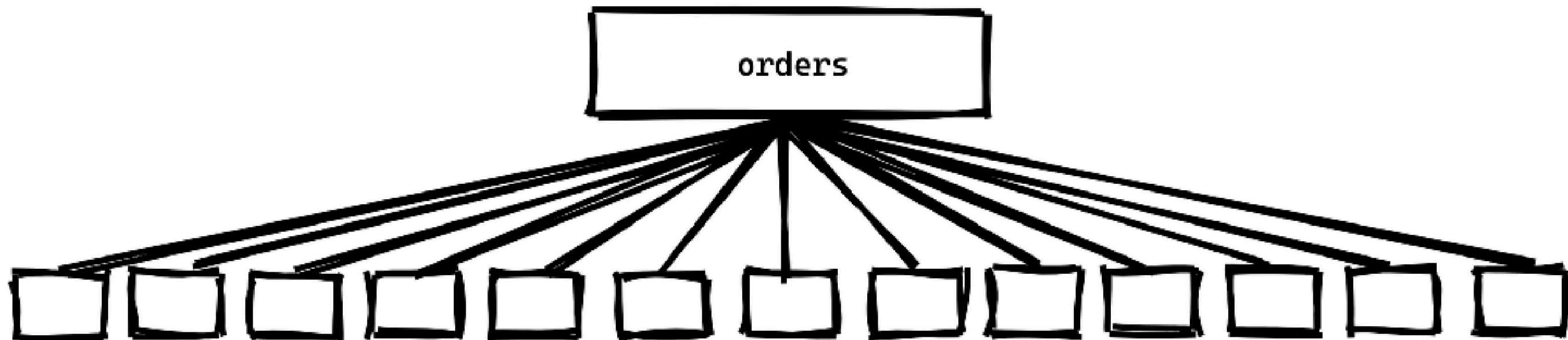
Insert data into orders



<https://chriserwin.com/table-partitioning/>



...



<https://chrisherwin.com/table-partitioning/>



Workshop with Partition

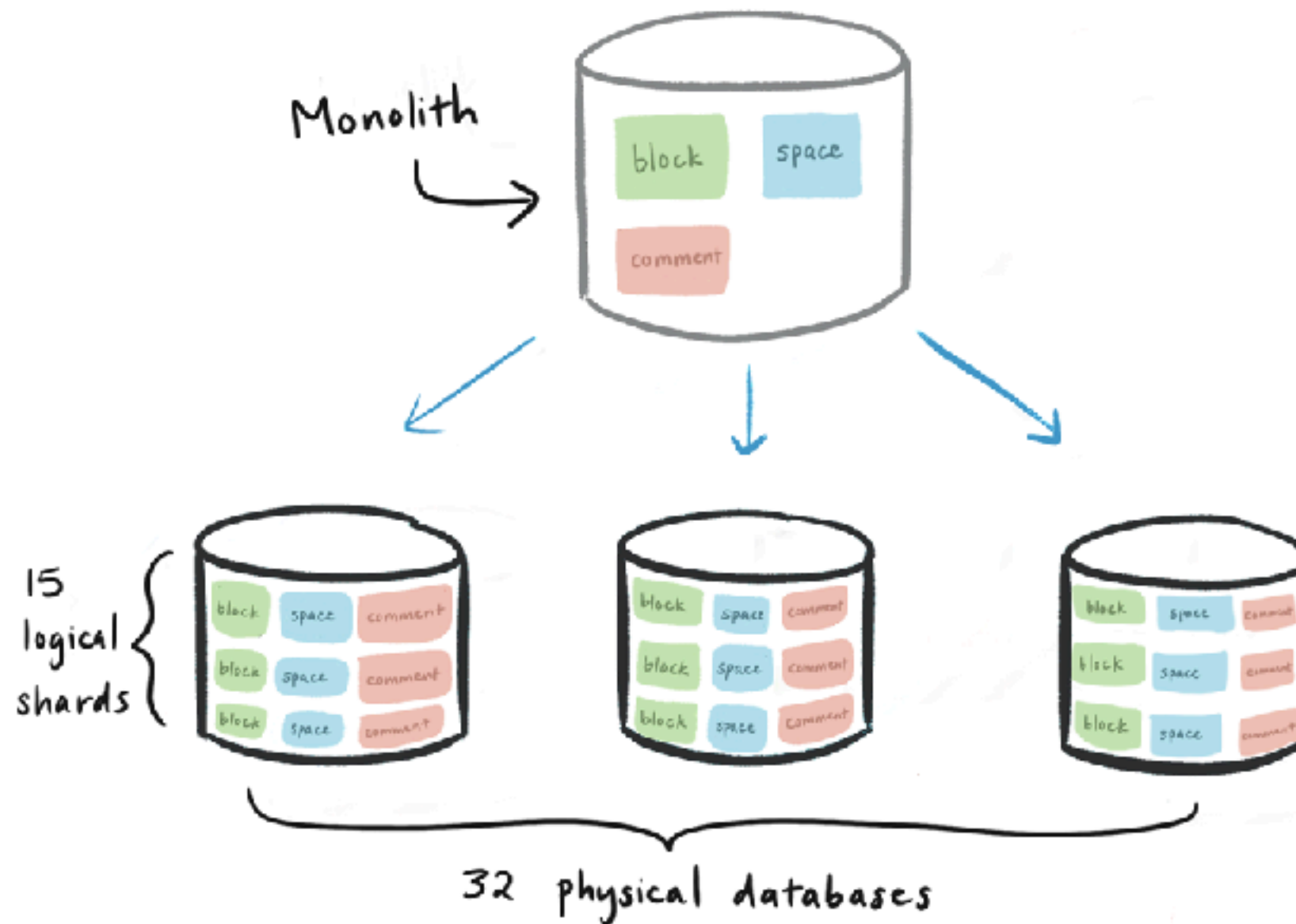
<https://github.com/up1/workshop-database/blob/main/workshop/basic-workshop/partition.md>



Sharding



Sharding



<https://www.notion.so/blog/sharding-postgres-at-notion>



Sharding techniques

Hash

Range

Location

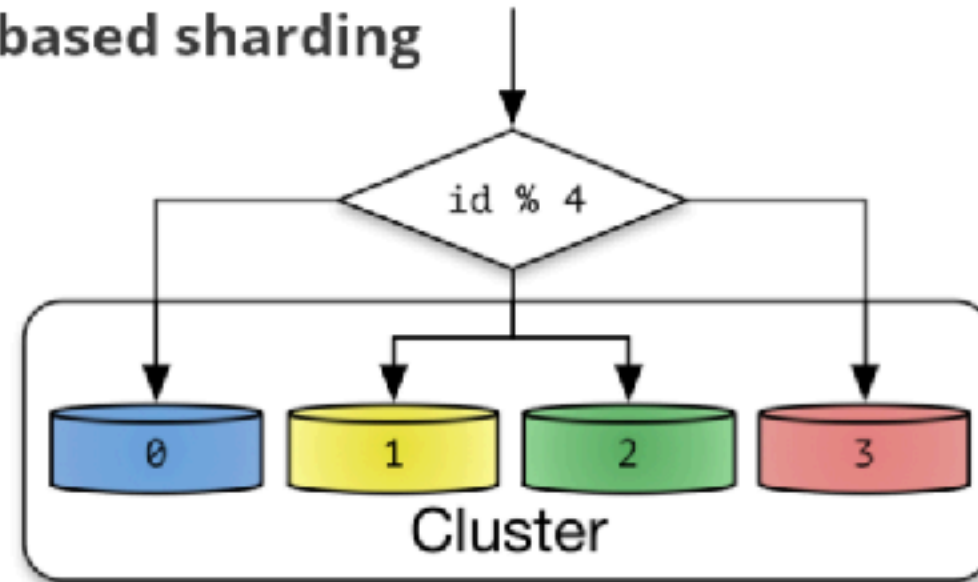
Key/Directory

<https://www.somkiat.cc/introduction-database-sharding/>

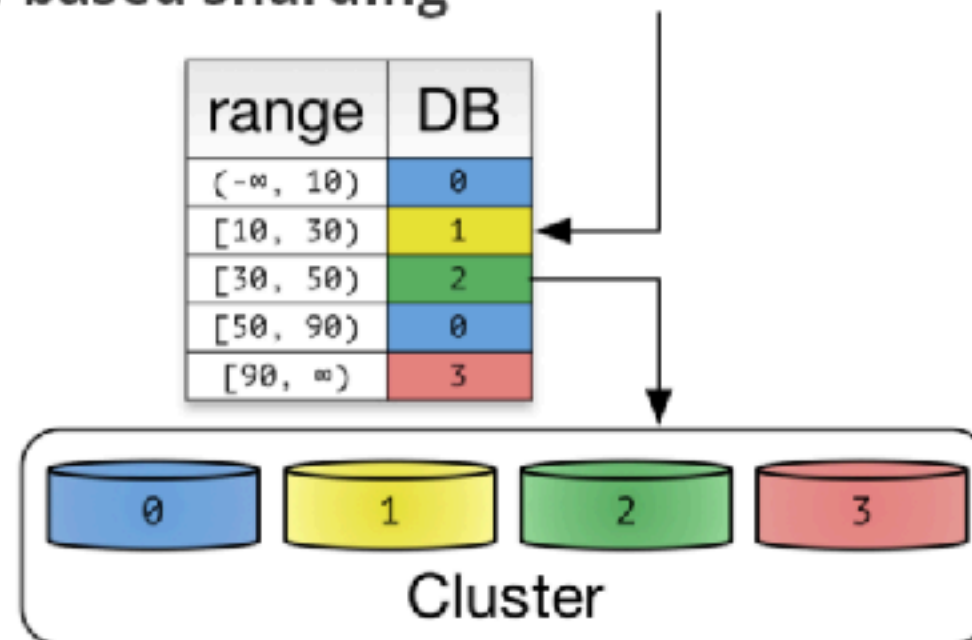


Sharding

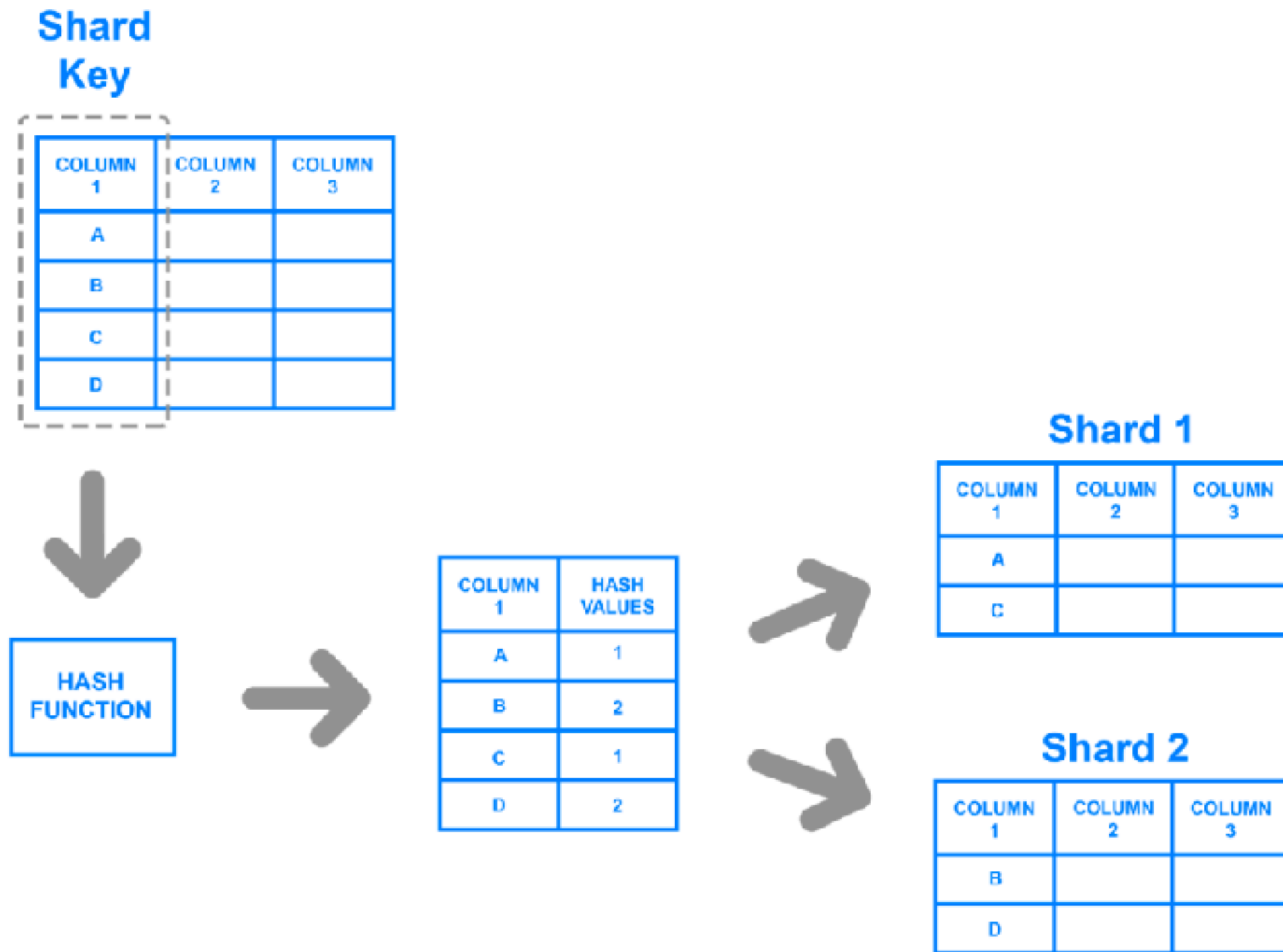
(a) Hash-based sharding



(b) Range-based sharding



Shard Key



Range

PRODUCT	PRICE
WIDGET	\$118
GIZMO	\$88
TRINKET	\$37
THINGAMAJIG	\$18
DOODAD	\$60
TCHOTCHKE	\$999



(\$0-\$49.99)

PRODUCT	PRICE
TRINKET	\$37
THINGAMAJIG	\$18

(\$50-\$99.99)

PRODUCT	PRICE
GIZMO	\$88
DOODAD	\$60

(\$100+)

PRODUCT	PRICE
WIDGET	\$118
TCHOTCHKE	\$999



Directory

Pre-Sharded Table

Shard Key

DELIVERY ZONE	FIRST NAME	LAST NAME
3	DARCY	CLAY
1	DENISE	LASALLE
2	HIROSHI	YOSHIMURA
4	KIRSTY	MACCOLL



Shard Key

DELIVERY ZONE	SHARD ID
1	S1
2	S2
3	S3
4	S4



S1

1	DENISE	LASALLE
---	--------	---------

S2

2	HIROSHI	YOSHIMURA
---	---------	-----------

S3

3	DARCY	CLAY
---	-------	------

S4

4	KIRSTY	MACCOLL
---	--------	---------



Sharding

Shard 1

Stores	
id	name
1	my book store
5	my other store

Products		
id	name	store_id
1	foo	1
2	bar	1
3	baz	1

Purchases			
id	product_id	store_id	price
1	2	1	1000
2	1	1	1200
3	3	1	1199

Shard 2

Stores	
id	name
2	my sock store
6	old things

Products		
id	name	store_id
33	new socks	2
34	old socks	6
35	old tie	6

Purchases			
id	product_id	store_id	price
102	35	6	600
43	33	2	800

<https://www.citusdata.com/blog/2016/08/10/sharding-for-a-multi-tenant-app-with-postgres/>



Denormalization



Denormalization techniques

Redundant column/Pre-joining tables

Table splitting (horizontal/vertical)

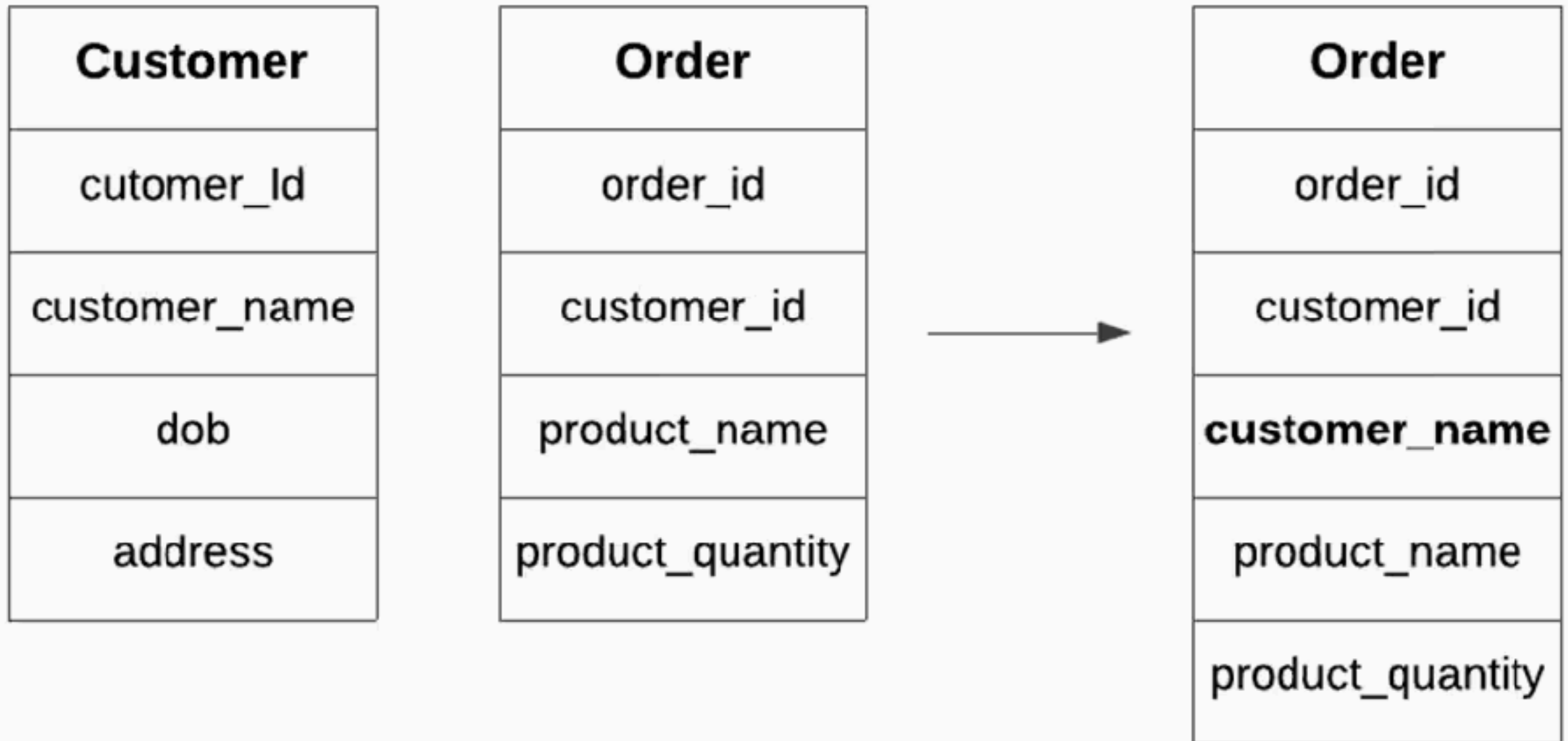
Add derived columns

Mirrored tables

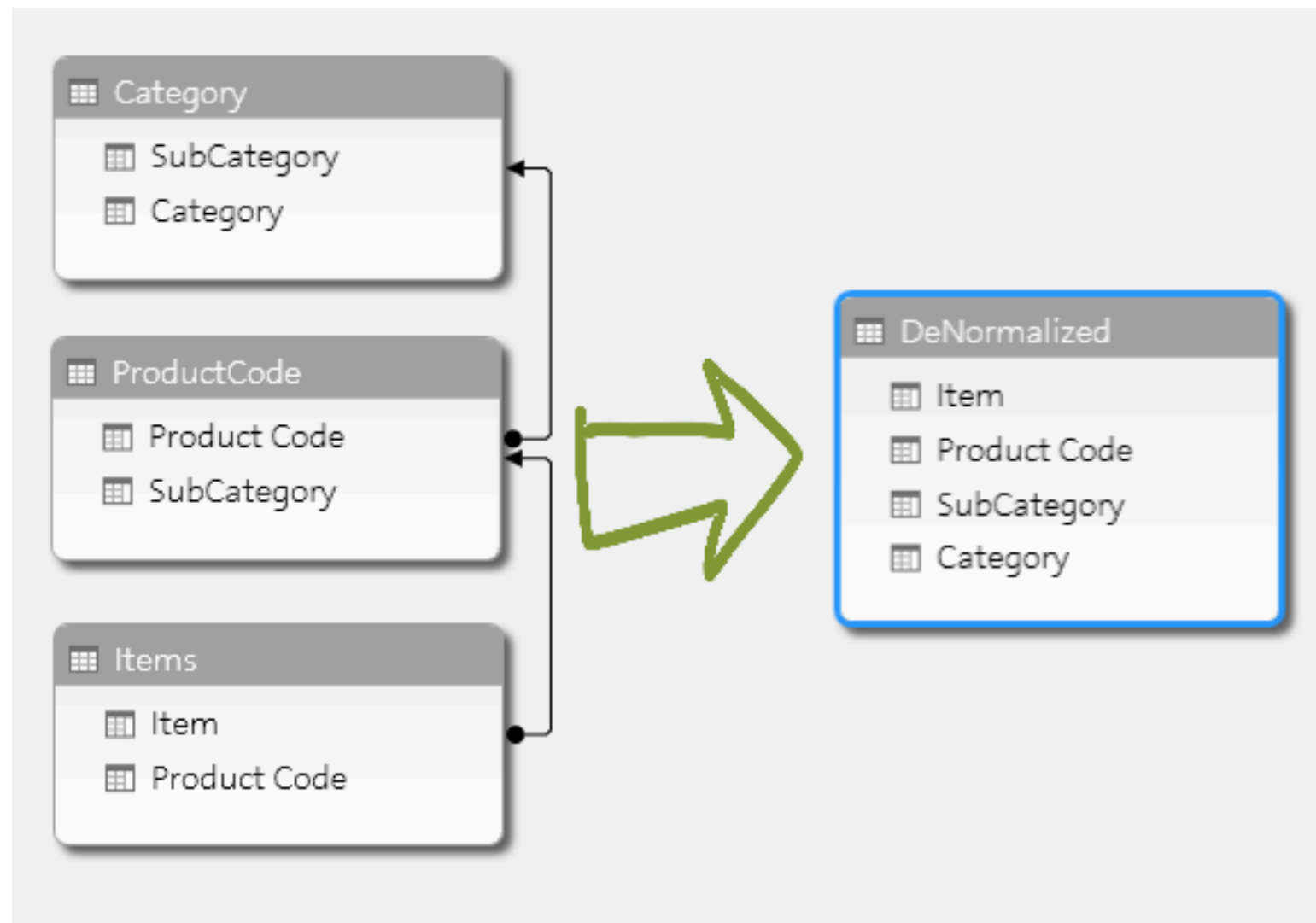
Materialized views



Redundant column/Pre-joining tables



Redundant column/Pre-joining tables



Horizontal splitting

Student_ID	Department_ID	Student_Name
1	1	Alex
2	2	Marie
3	1	Sam
4	3	Sara



Computer Science

Student_ID	Department_ID	Student_Name
1	1	Alex
3	1	Sam

Chemistry

Student_ID	Department_ID	Student_Name
2	2	Marie

Botany

Student_ID	Department_ID	Student_Name
4	3	Sara



Vertical splitting



Add derived columns

Student_ID	Name
1	Alex
2	Marie

Student_ID	Assignment_ID	Mark
1	1	20
1	2	35.50
2	1	45
2	2	45



Student_ID	Name	Total_Marks
1	Alex	55.5
2	Marie	90



Pros of data denormalization

Improve query performance (UX)

Reduce complexity of data model

Improve application scalability

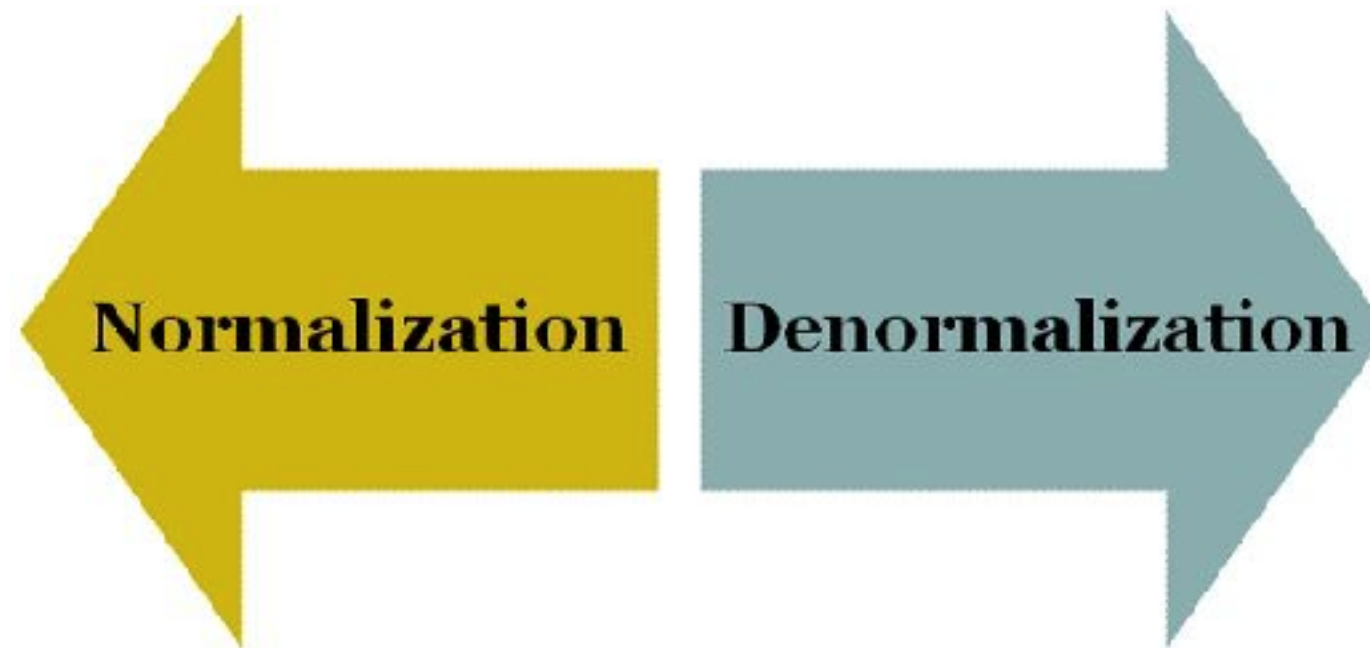
Generate data report faster



Cons of data denormalization

- Increase data redundancy
- Inconsistency between data sets
- Require more disk space (split tables)
- More data model (hard to maintain)
- Difficult to insert and update data
- Maintenance costs





Setup Postgres



Database Topologies

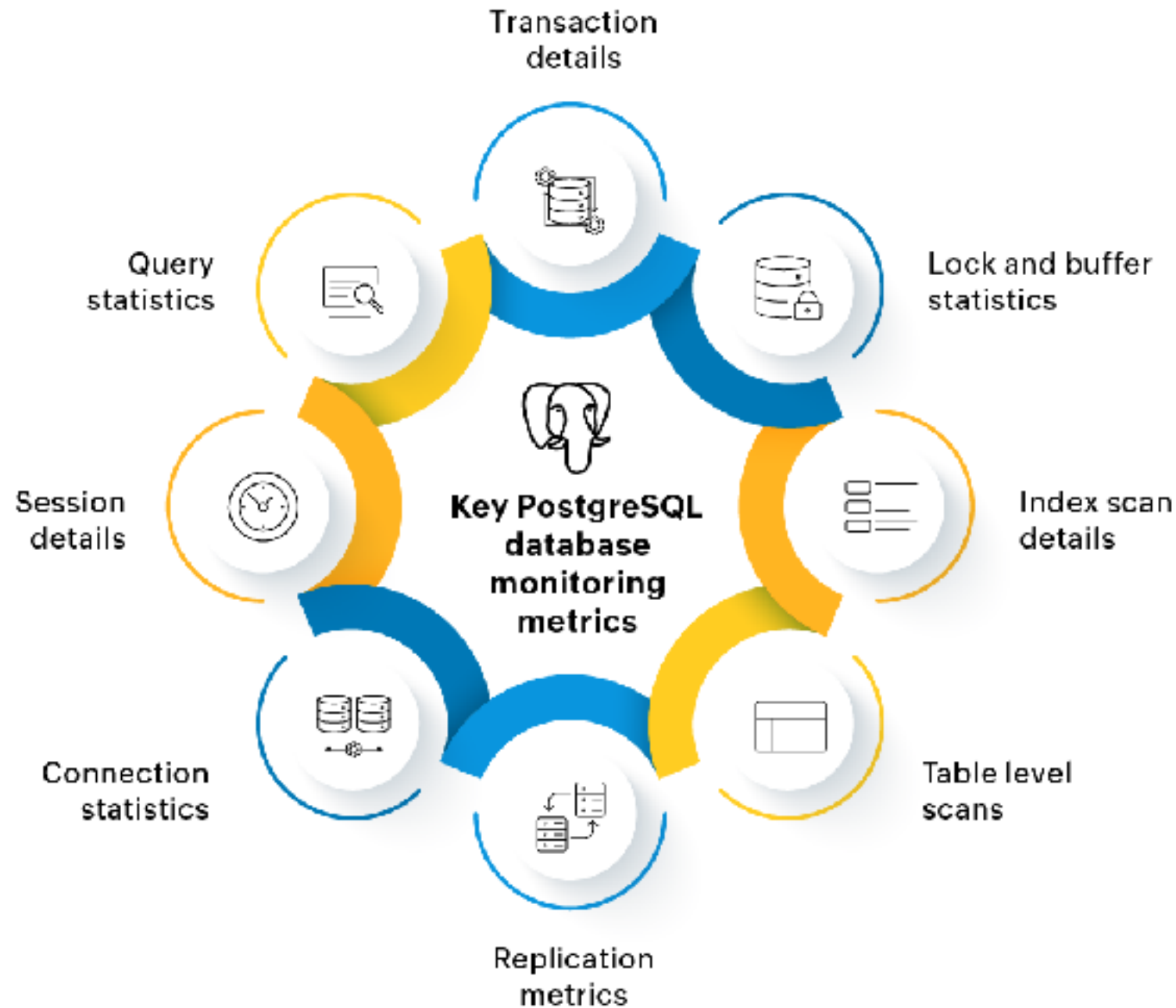
Standalone
Master-slave (scale-read)
Cluster and sharding



Monitoring



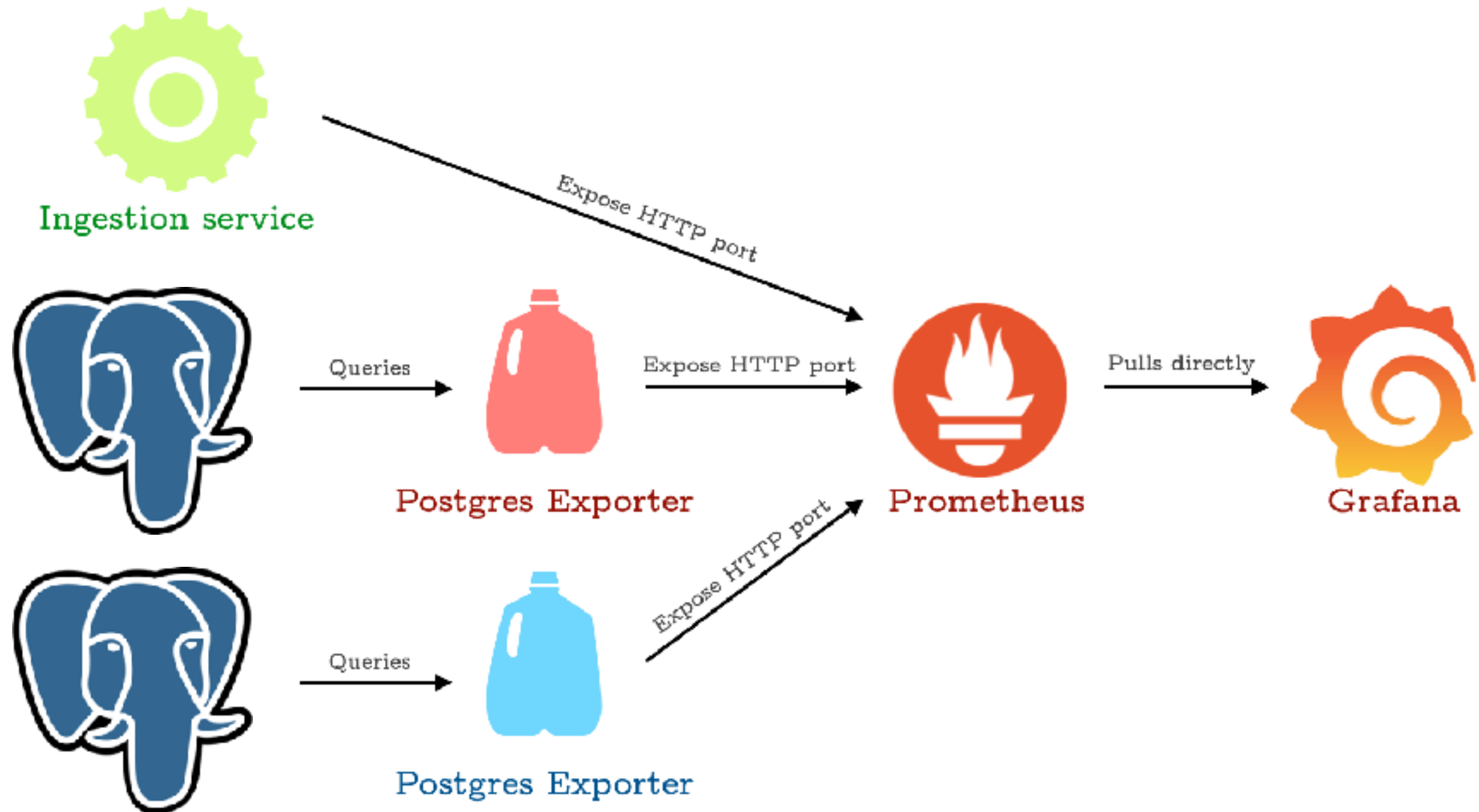
Basic Monitoring



<https://blogs.manageengine.com/application-performance-2/appmanager/2024/04/19/postgres-performance-monitoring-best-practices-and-key-metrics.html>



Basic Monitoring



<https://github.com/up1/workshop-database/tree/main/workshop/monitoring-testing>



Essential configuration parameters



Max connections

Purpose

Maximum number of concurrent connections the server will accept

Impact

Too low: connection refused

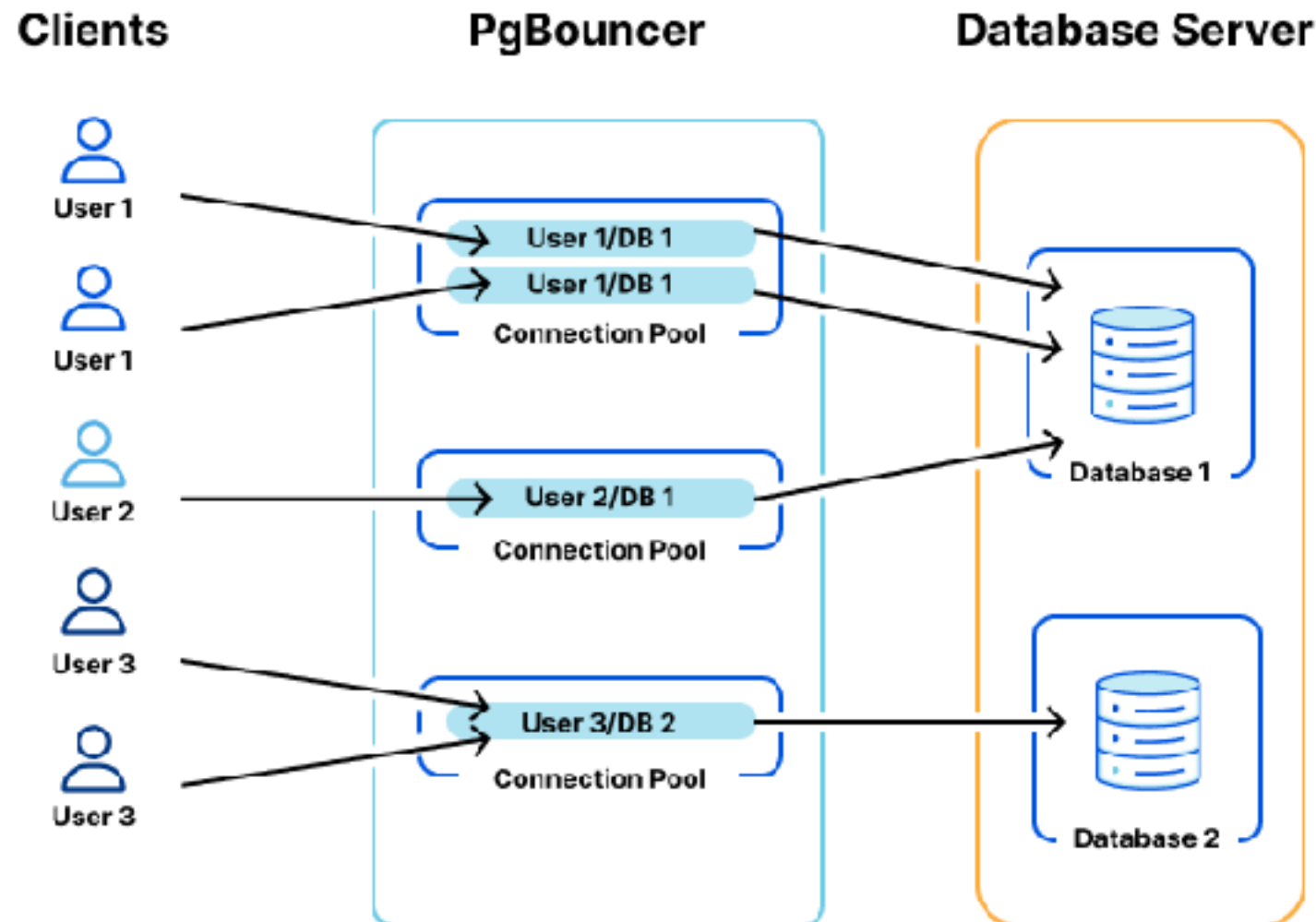
Too high: excessive memory consumption, thrashing

Recommendation

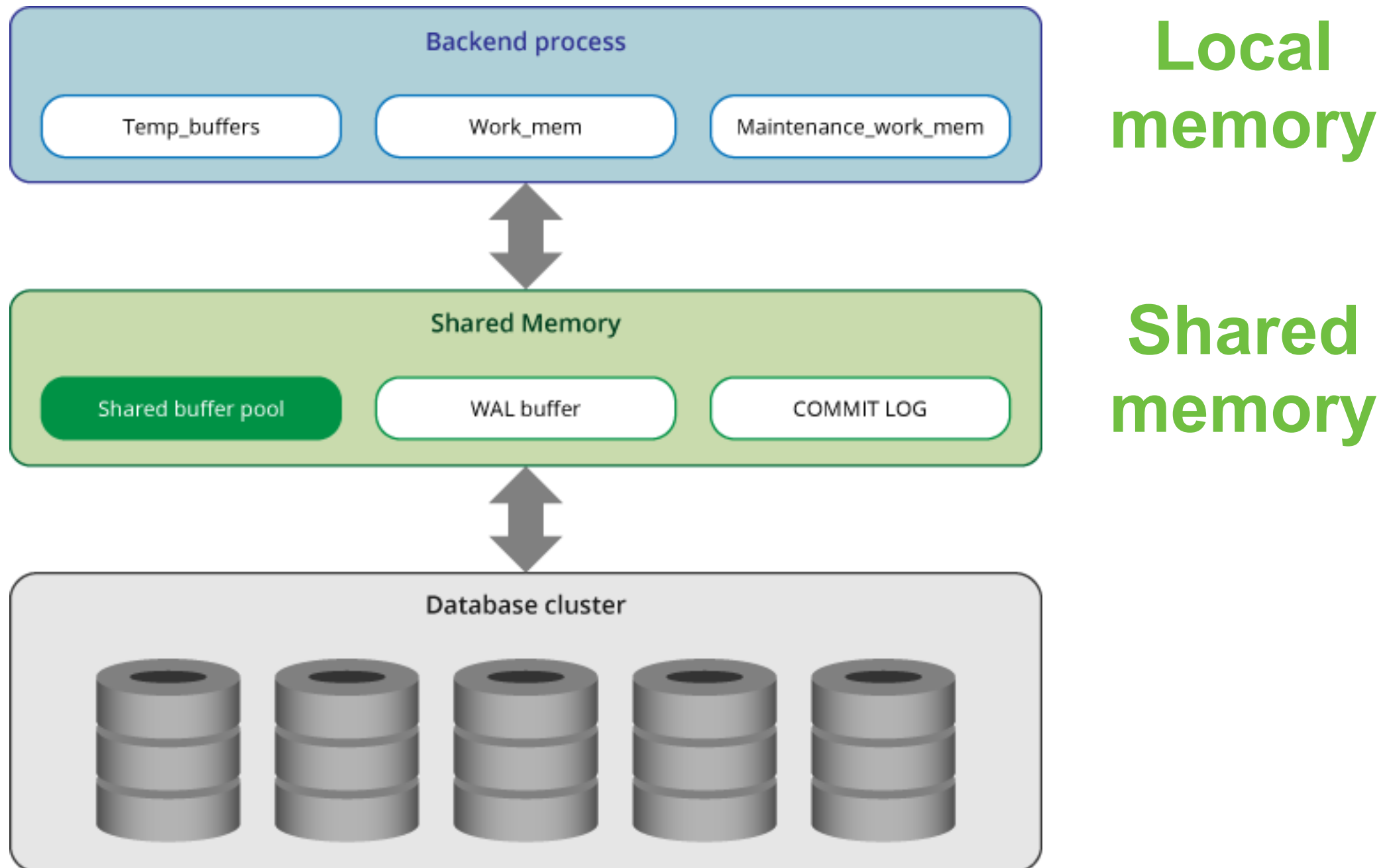
Start with a reasonable number based on application needs



Manage connection pooling ?



Memory architecture



Local memory

Each backend process allocate local memory
Query processing

work_mem

temp_buffer

maintenance_work_mem



Shared memory

Allocate by server when it start up

shared_buffer

wal_buffer

Commit log (CLOG)



Work mem (not max memory)

Purpose

Memory used by internal sort operations and hash tables before spilling to disk

Impact

Crucial for query performance, especially with large sorts
Too low leads to temp file creation on disk

Recommendation

Start with a reasonable value (e.g., 4-64MB)
Tune based on EXPLAIN ANALYZE



Shared buffers

Purpose

Amount of memory dedicated to shared buffer pool for caching data pages

Impact

Directly affects read performance by reducing disk I/O

Recommendation

Typically 25% of total RAM, but can vary



Effective cache size

Purpose

Inform the query planner about the total amount of memory available for caching data (pg shared buffer + OS file cache)

Impact

Doesn't allocate memory

Help the planner make better decision (index vs sequential)

Recommendation

50-75% of total RAM



Write-Ahead Log (WAL)

The foundation of Durability



Write-Ahead Log (WAL) ?

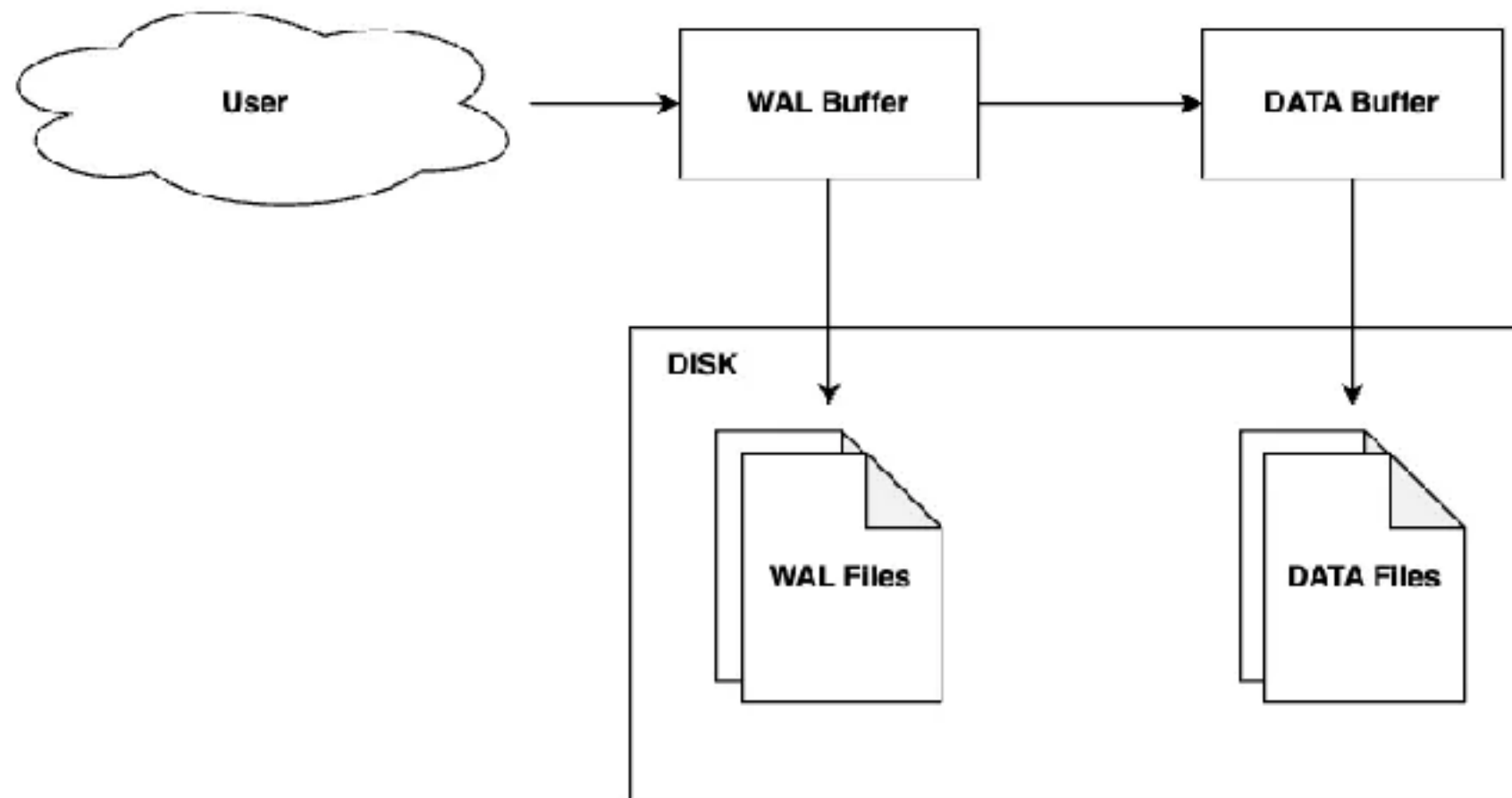
Transaction log or XLOG

Use to ensure data durability and consistency

All changes to data must be written to the WAL before applied to the actual data files on disk



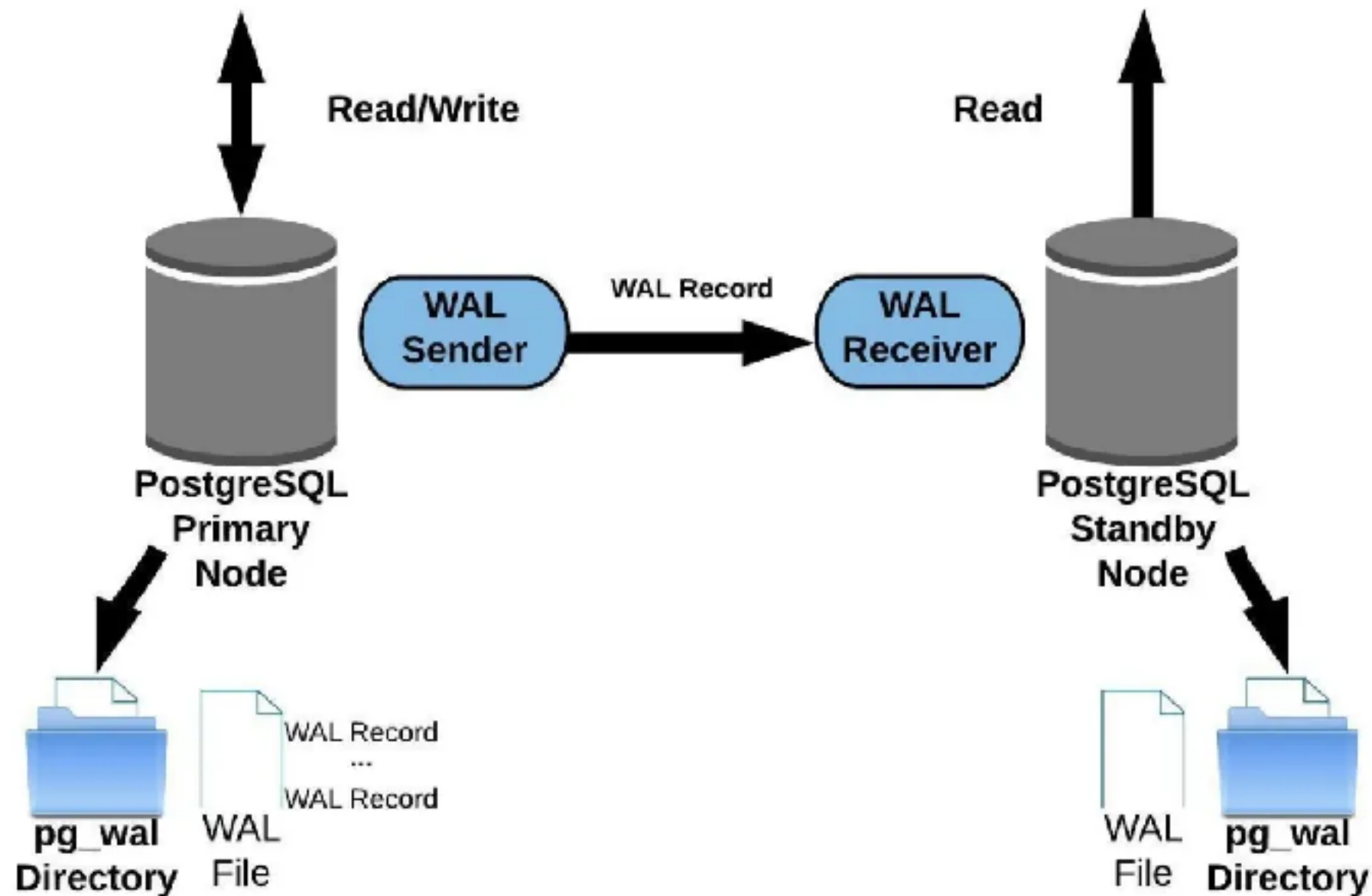
Write-Ahead Log (WAL) ?



<https://levelup.gitconnected.com/when-databases-crash-wal-saves-the-day-write-ahead-log-wal-710e31aa9cdd>



Async Replication



<https://levelup.gitconnected.com/when-databases-crash-wal-saves-the-day-write-ahead-log-wal-710e31aa9cdd>



Checkpoint timeout

Purpose

Controls how often checkpoints occur

Checkpoints flush dirty shared buffers to disk

Impact

Too frequent: I/O spikes during checkpoints

Too infrequent: Longer recovery time, large pg_wal directory

Recommendation

5-10 minutes



Max wal size

Purpose

Controls how often checkpoints occur

Checkpoints flush dirty shared buffers to disk

Impact

The maximum size of the WAL directory that triggers a checkpoint

Recommendation

Based on I/O capacity and recovery time tolerance
(e.g., 4GB-16GB+)



High volume tuning ?

Read heavy

Write heavy

Many
concurrent user

Complex query



Principles for high volume tuning

Allocate more memory
Balance I/O (optimize WAL and checkpoint)
Parallelism
Monitoring (pg_stat_statement, EXPLAIN)
Hardware (Fast disk, RAM, CPU)
Connection pooling



Slow Query !!



Autovacuum

Prevent table bloat and ensure query performance

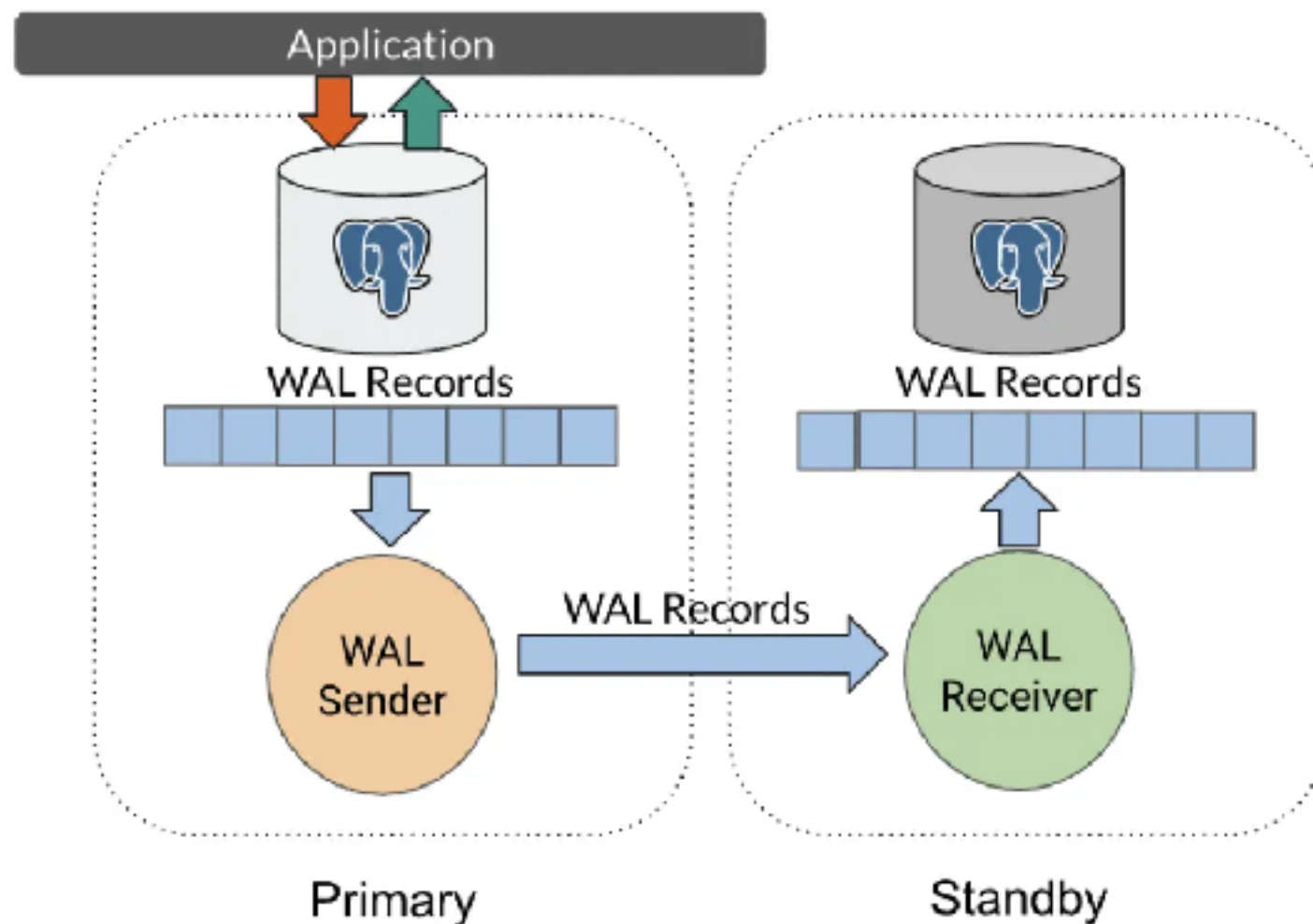


PostgreSQL Cluster



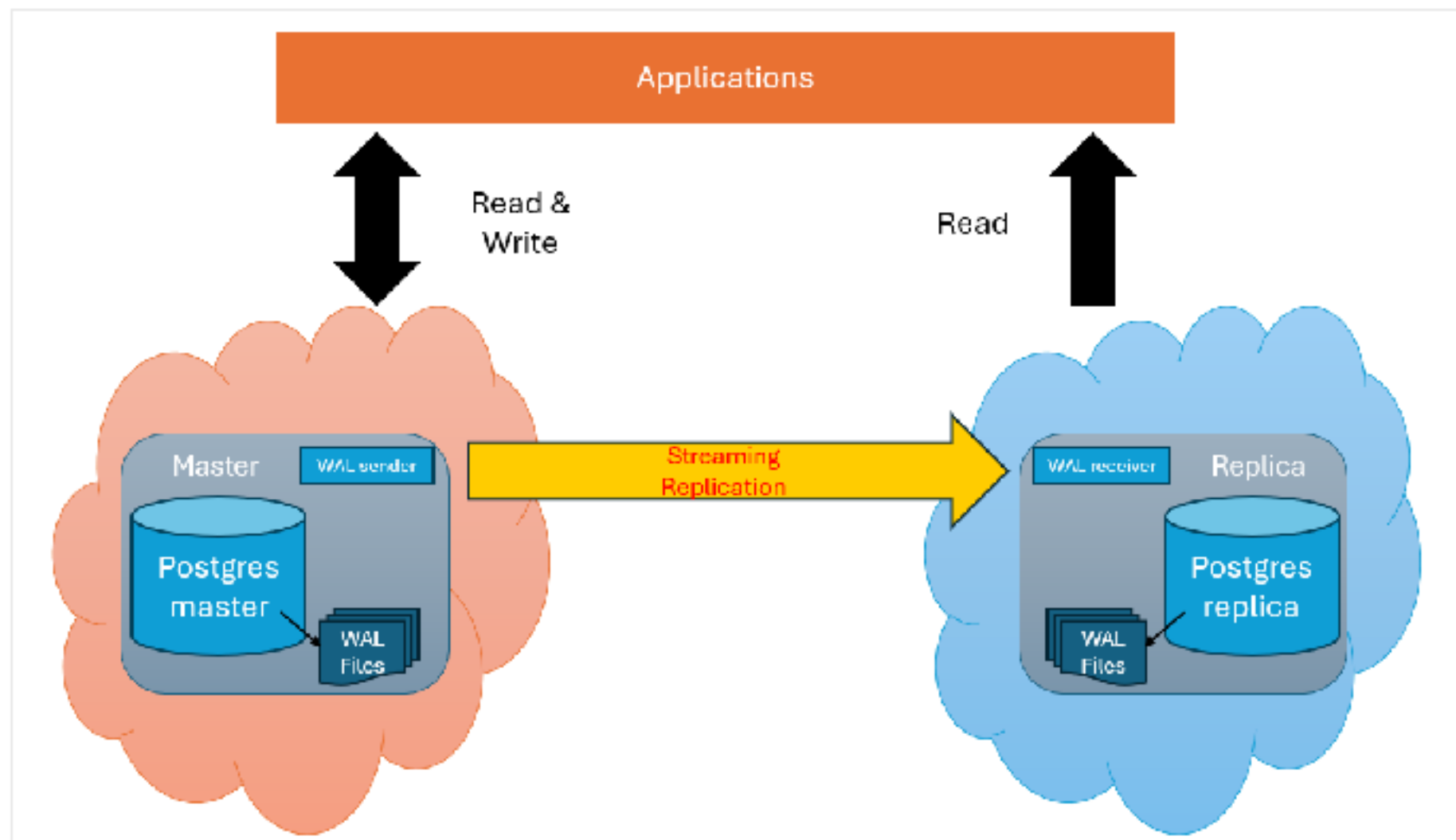
Stream Replication

Build-in sync or async replication
Simple DR site and backup cluster



Stream Replication

PostgreSQL stream replication (build-in)

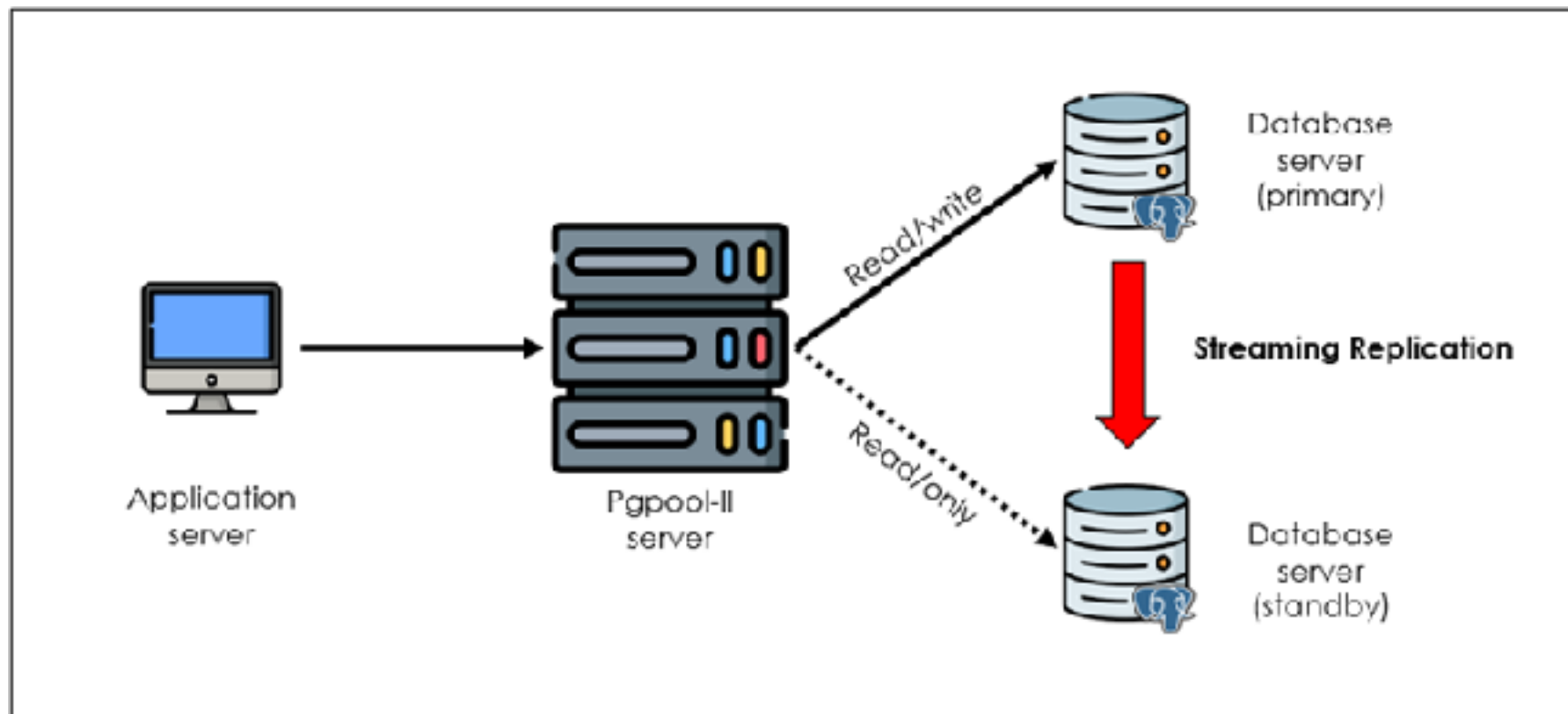


Scale with PgPool II

Connection pooling

Load balancing

In-memory query caching

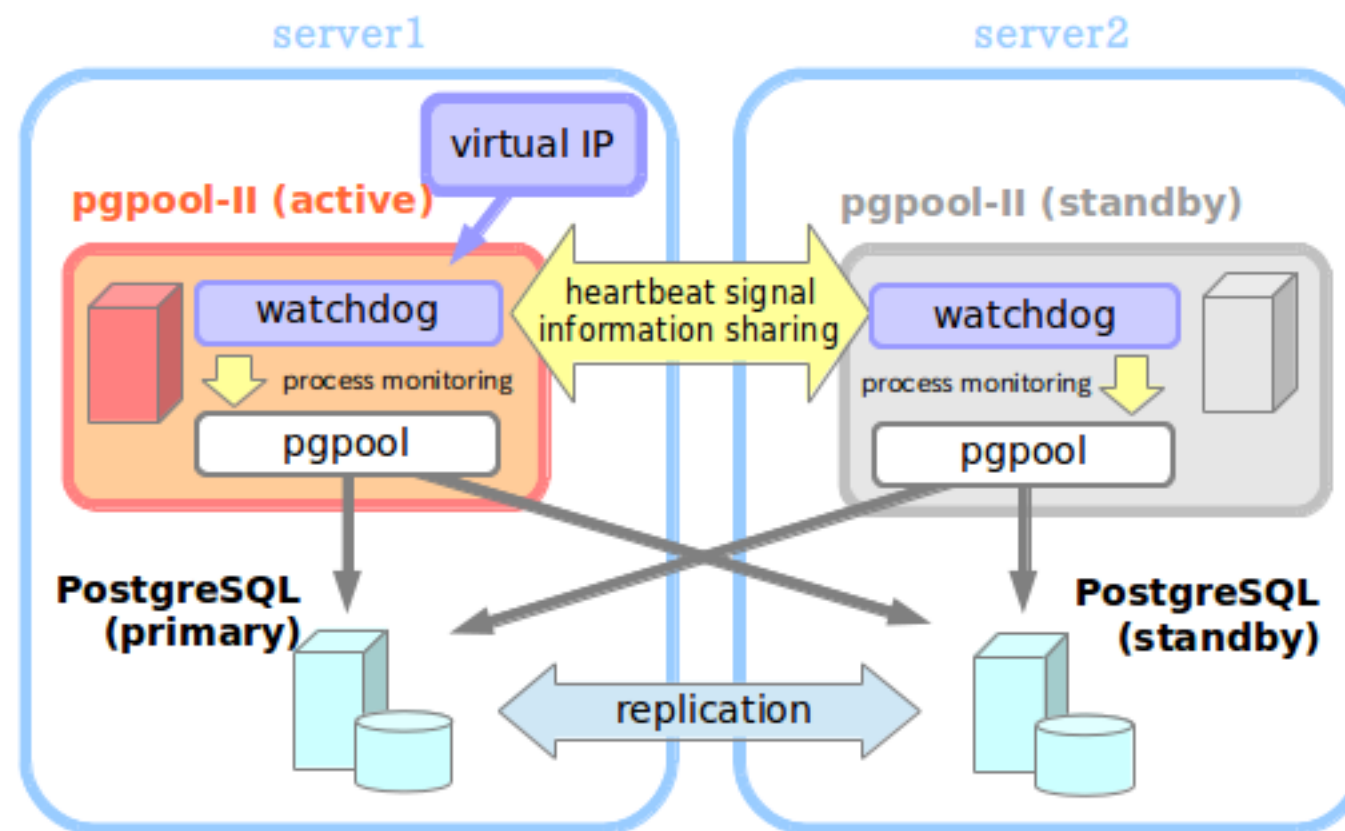


https://www.pgpool.net/mediawiki/index.php/Main_Page



Scale with PgPool II

Master-slave mode with watchdog Failure management



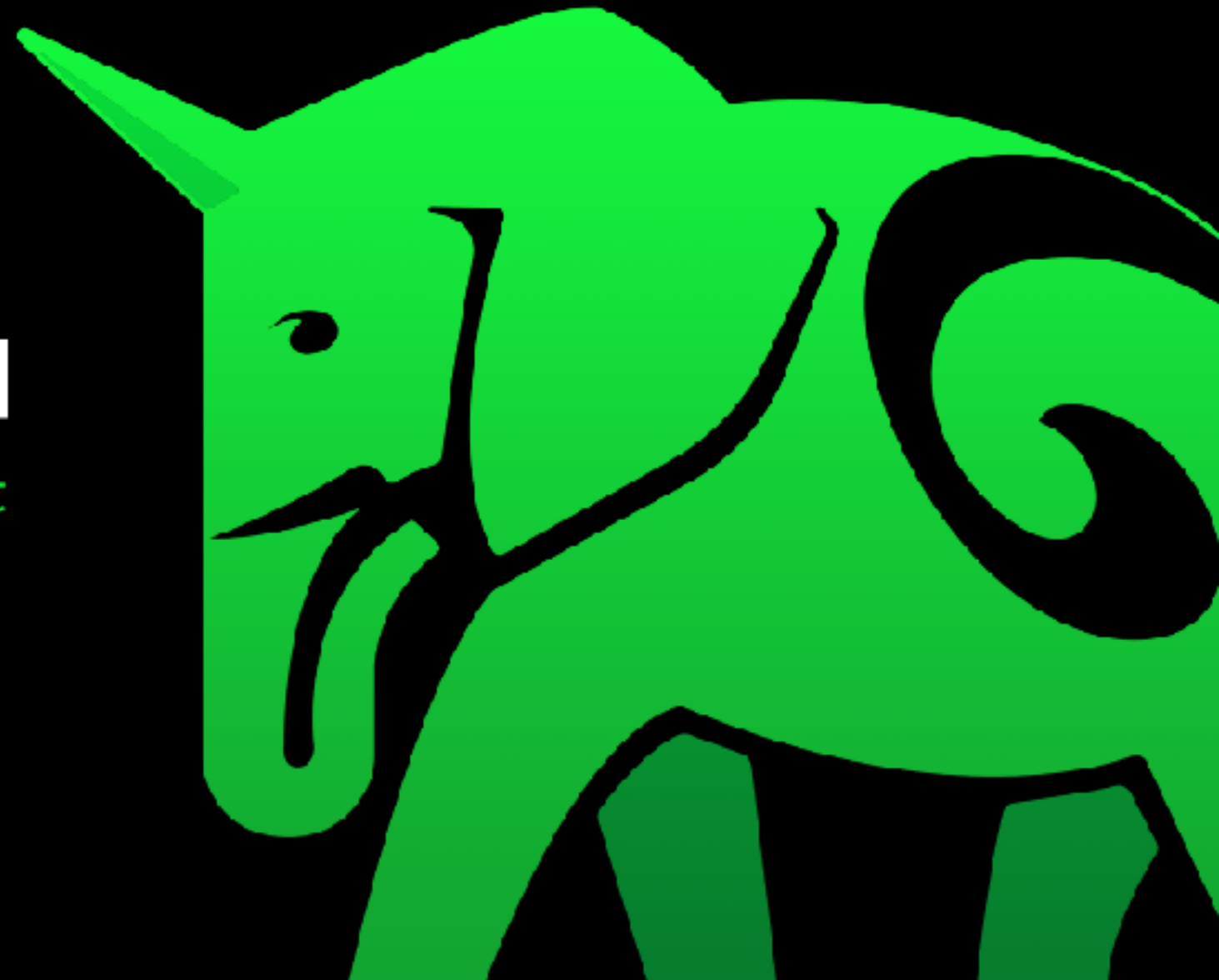
https://www.pgpool.net/mediawiki/index.php/Main_Page



Citus = Distributed PostgreSQL



[POSTGRES AT ANY SCALE]

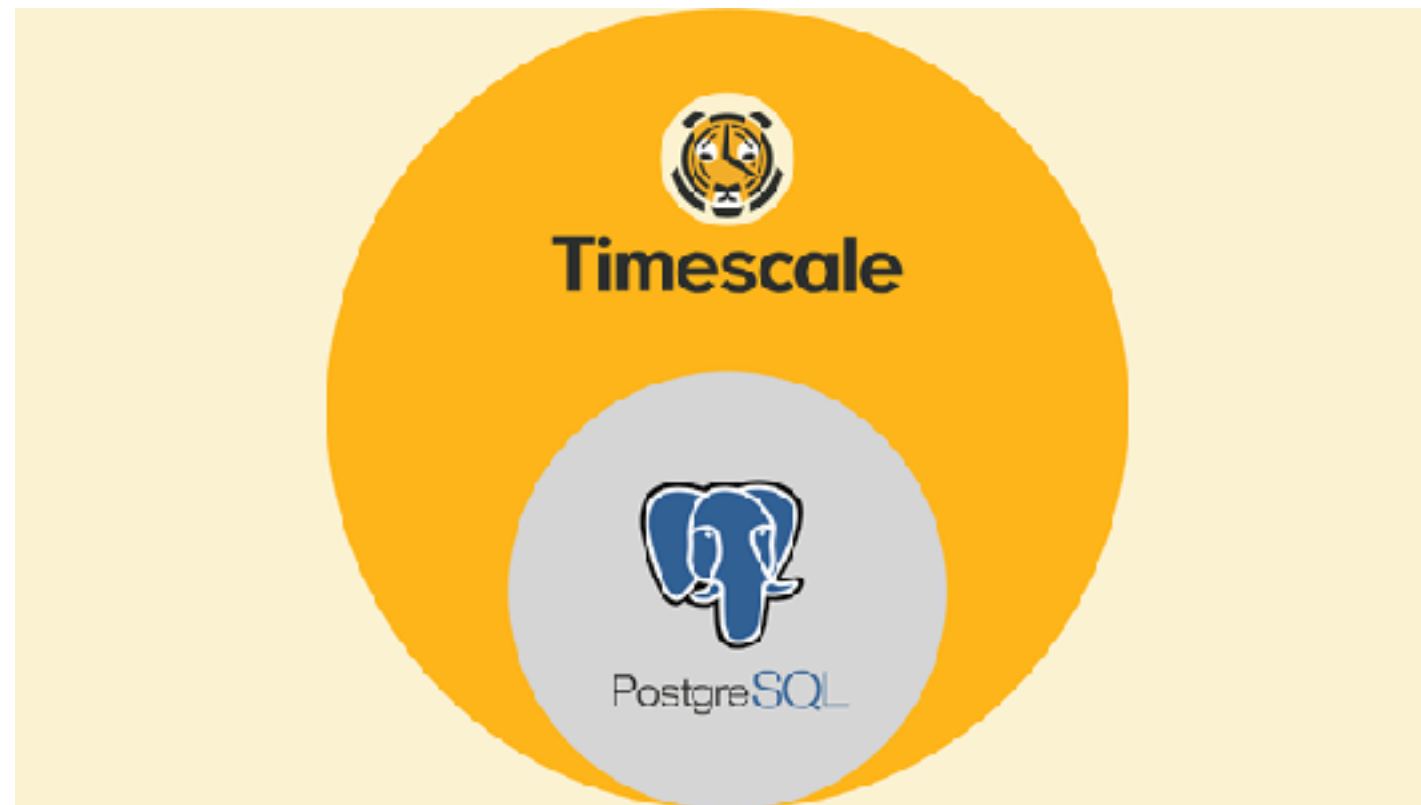


<https://github.com/citusdata/citus>



TimeScaleDB

Time-series database for realtime analytic
PostgreSQL extension

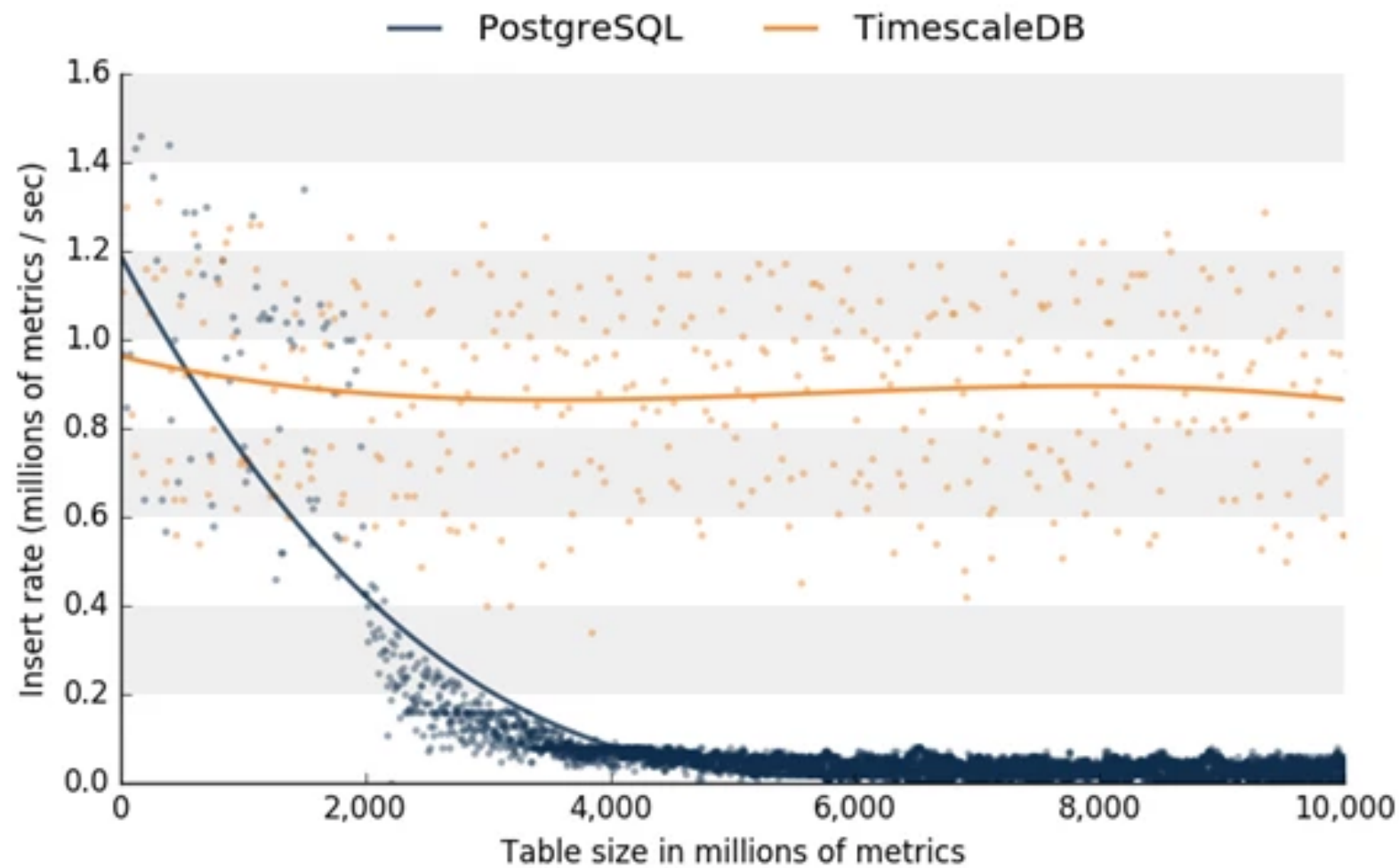


<https://github.com/timescale/timescaledb>



TimeScaleDB

Insert operation

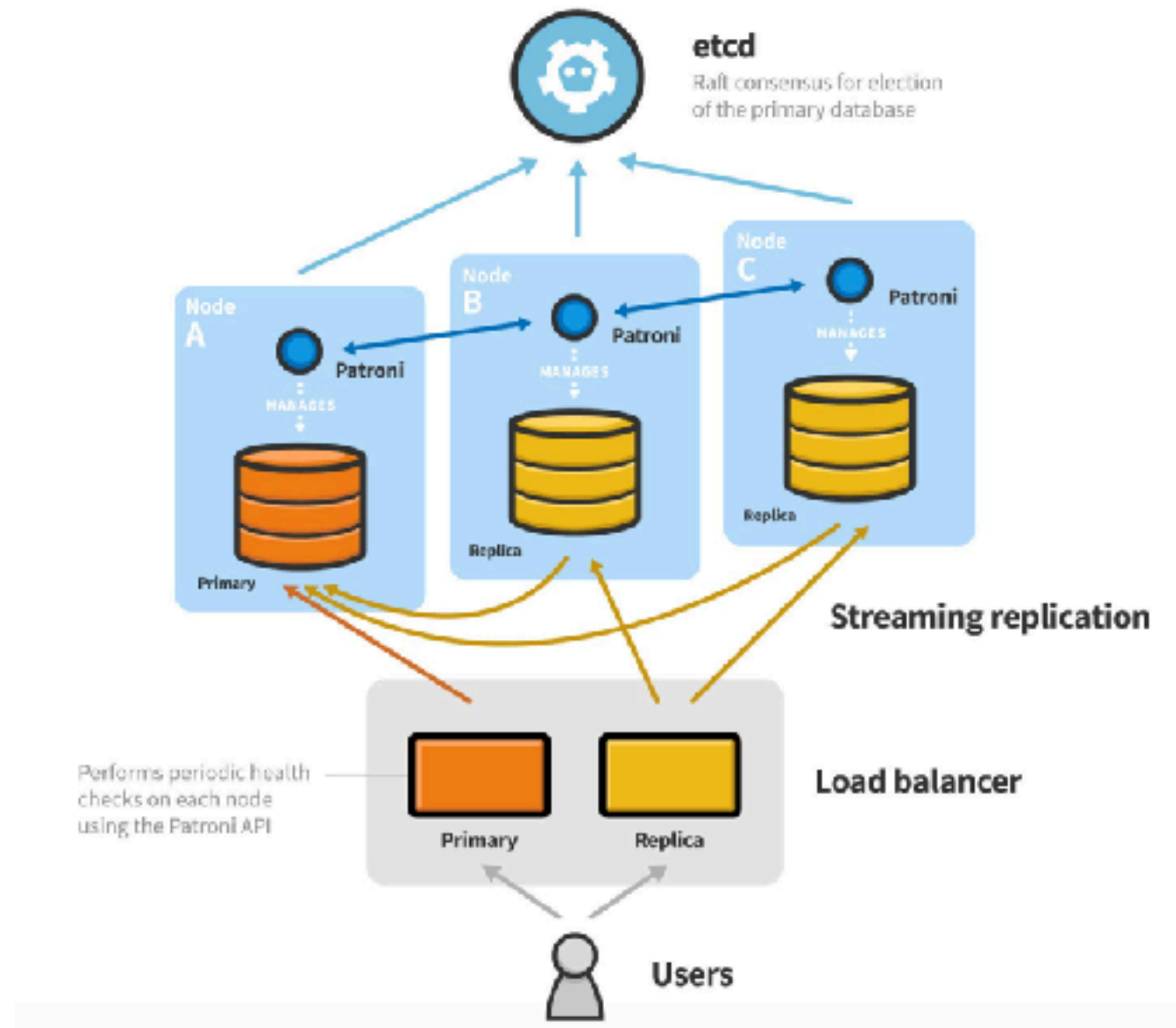


<https://azure.microsoft.com/en-us/blog/power-iot-and-time-series-workloads-with-timescaledb-for-azure-database-for-postgresql/>



Patroni (HA for PostgreSQL)

With Etcd, Consul and Zookeeper



<https://github.com/patroni/patroni>



Challenge of Large datasets



Challenge of Large datasets

Disk I/O, CPU, Memory pressure

Index size and maintenance

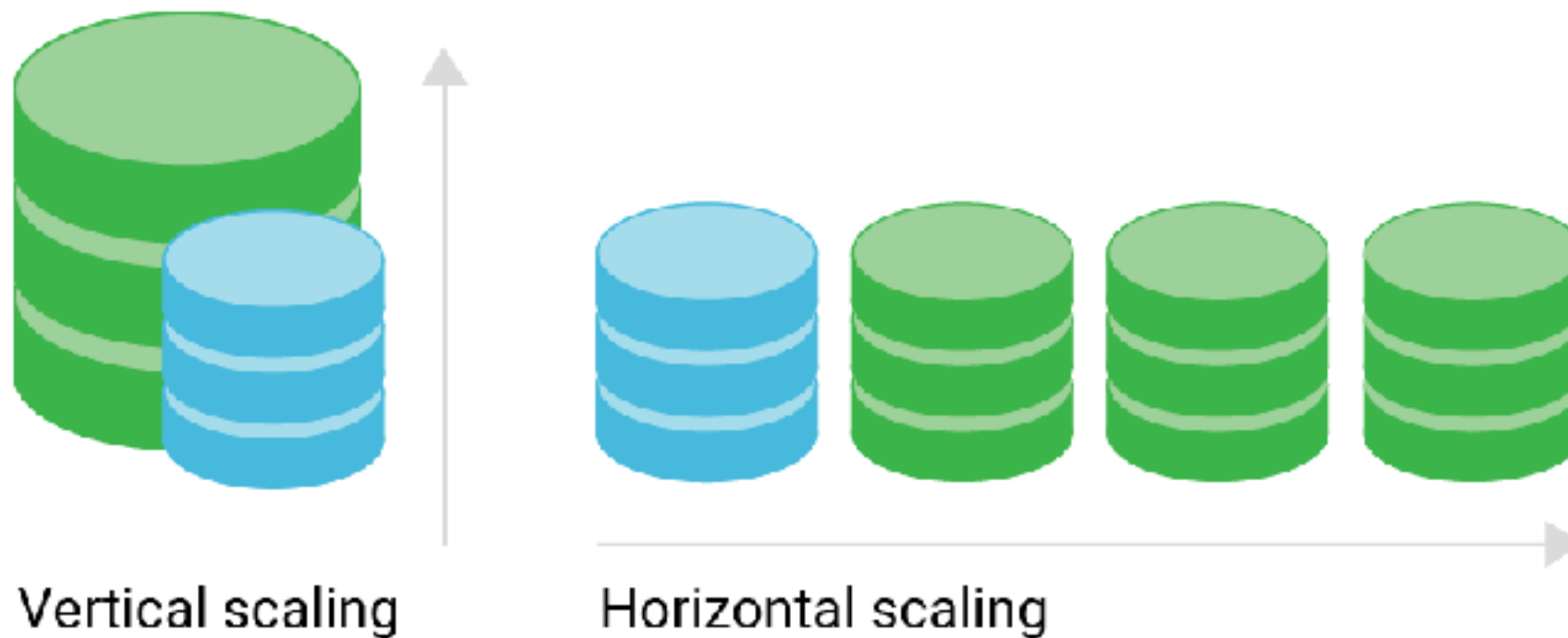
Locking and concurrency

Why database slow down with more data ?



Scaling Database ?

Vertical vs Horizontal



Scalability metrics

Throughput (transactions per second)

Latency (response time)

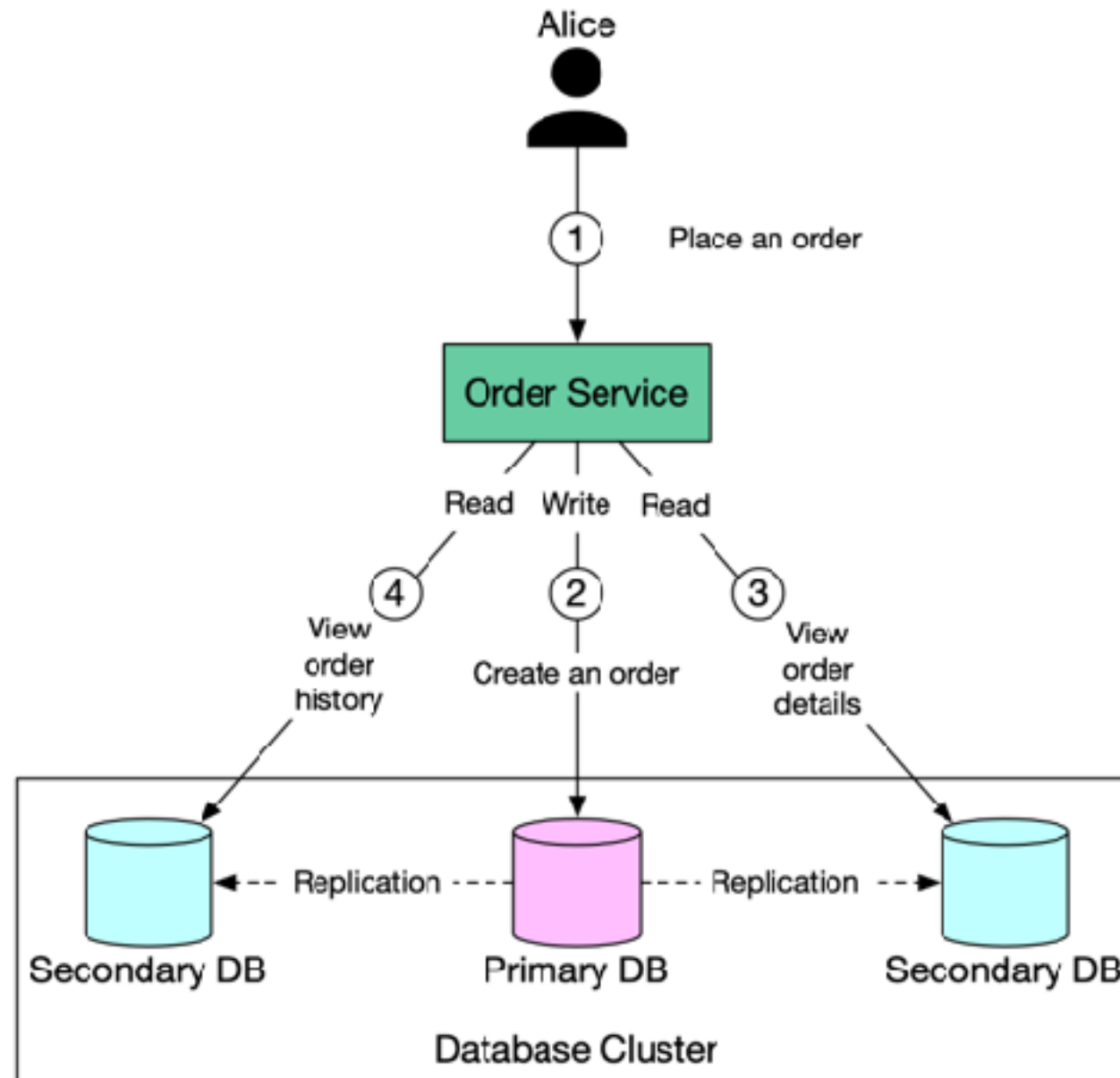
Availability (uptime)

Durability (data safety)



Read replicas

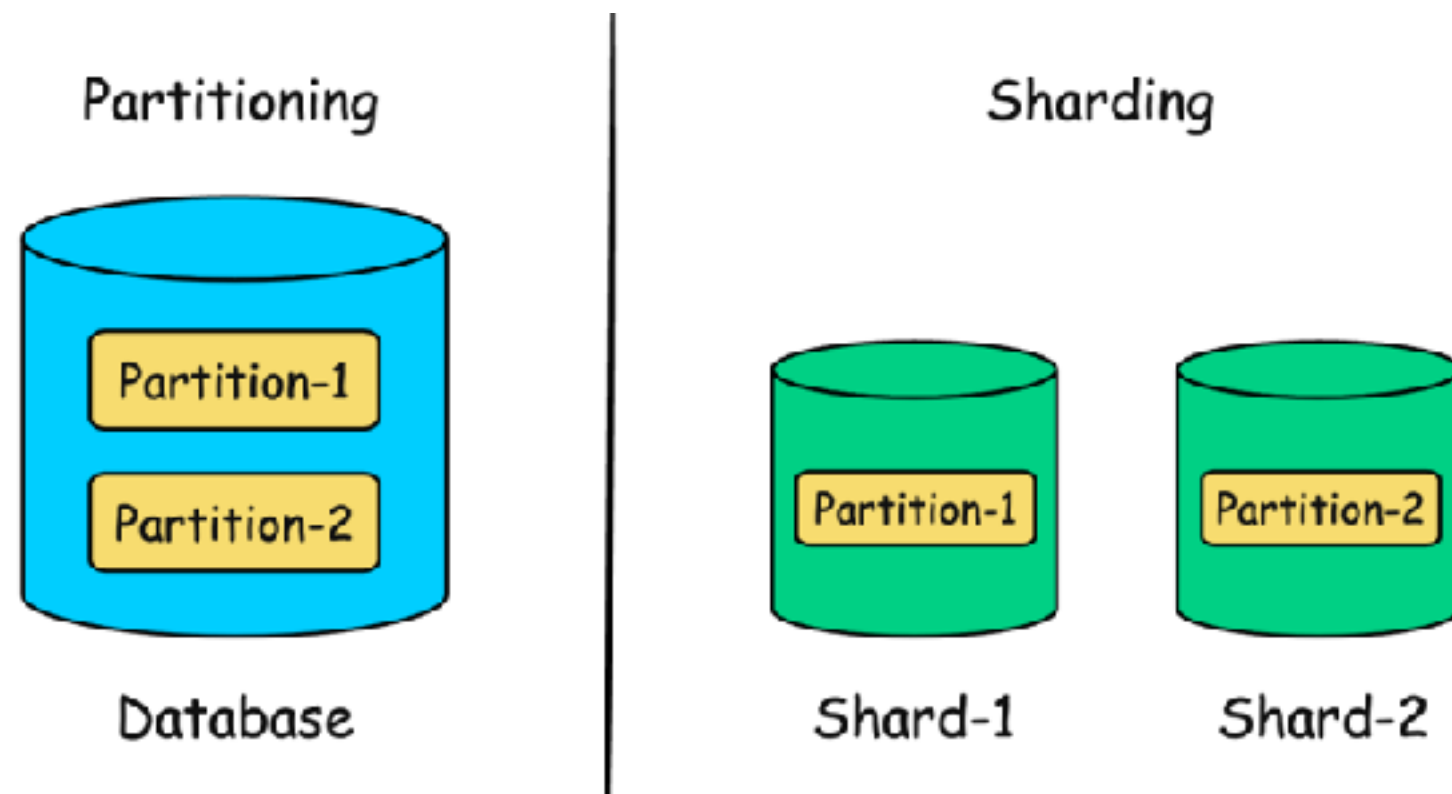
Read-scaling solution



Data distribution strategies

Partitioning (within a single instance)

Sharding (across multiple instances)



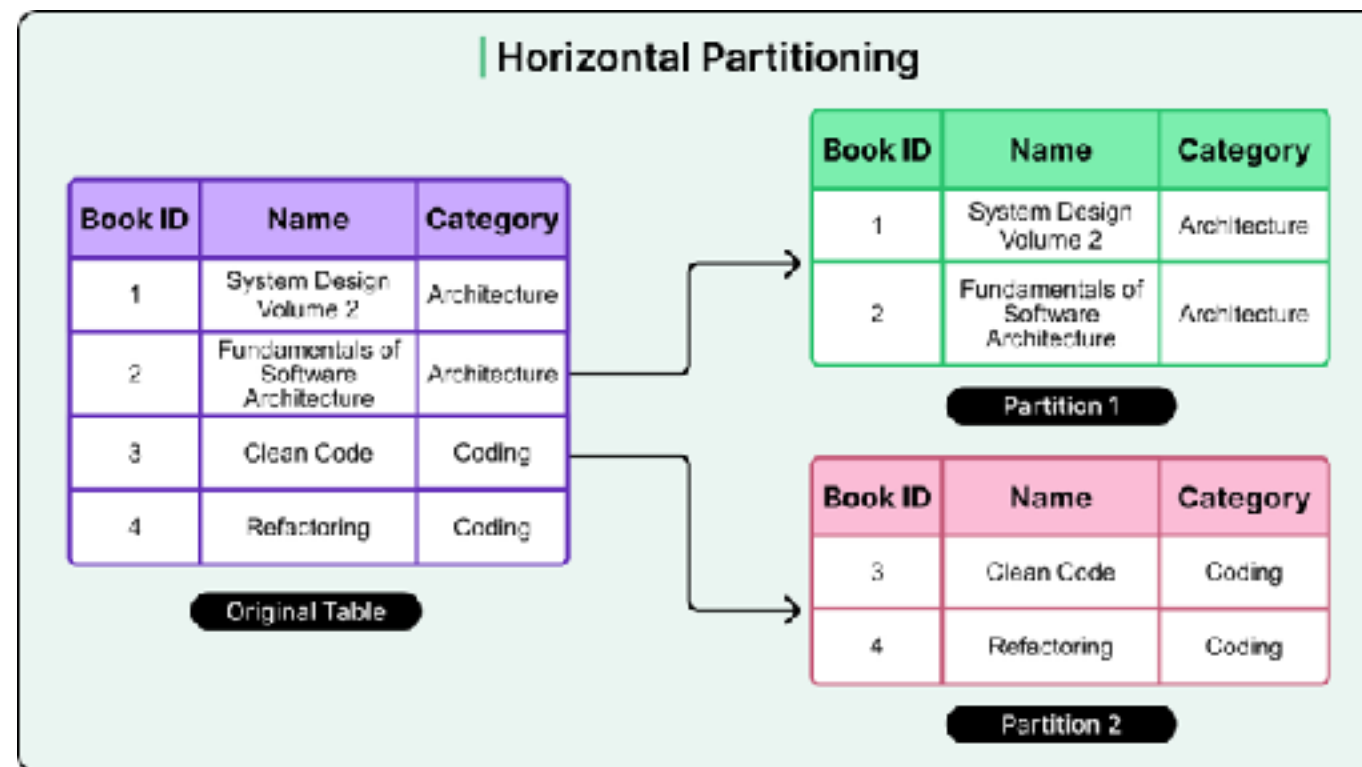
Database sharding

Design for large dataset by split into smaller

Divided across multiple instance

Based-on horizontal partitioning

Use when to overcome single instance limitations



Sharding techniques

Range

Hash or key-based

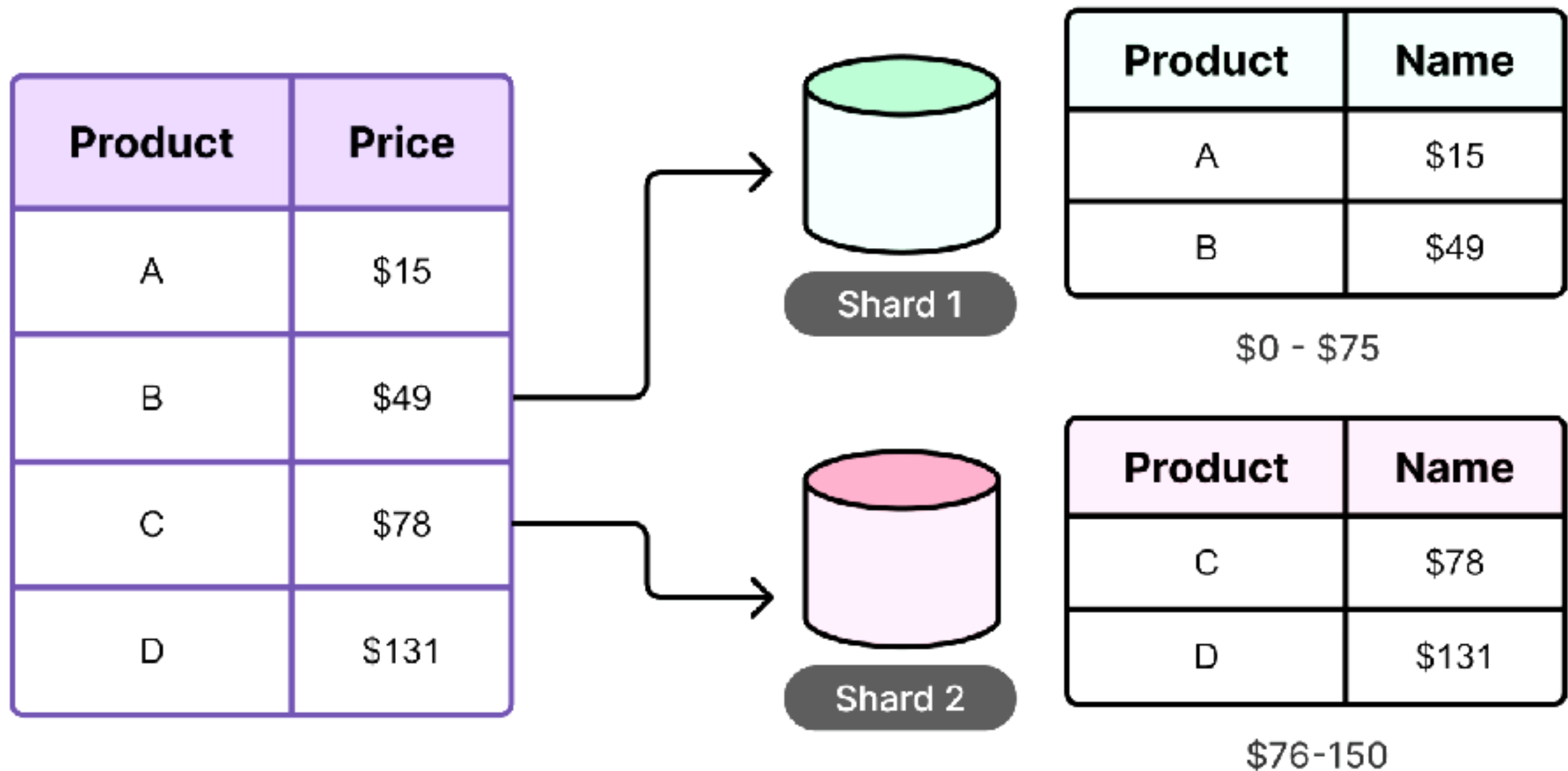
Directory-based

Request routing

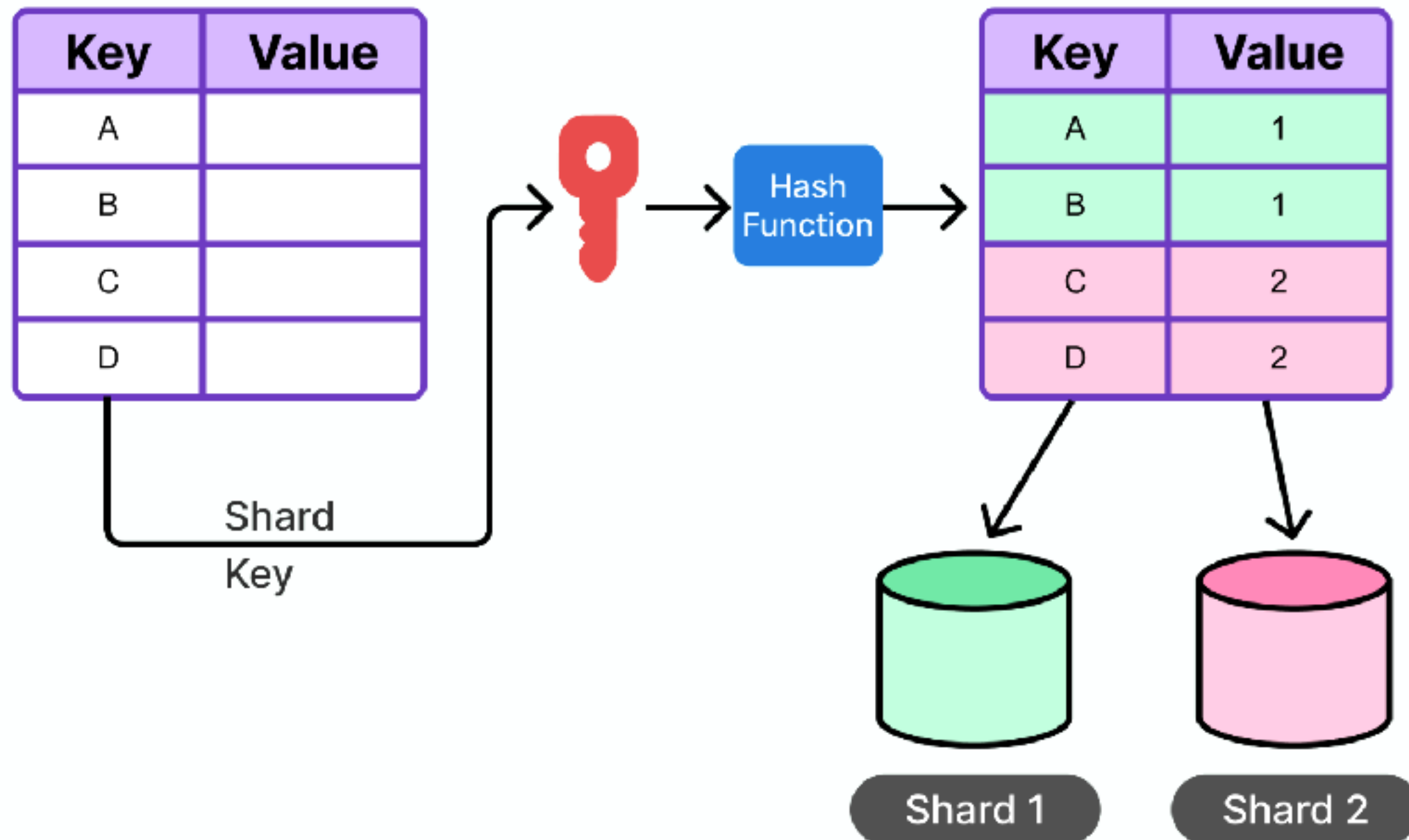
<https://blog.bytebytego.com/p/a-guide-to-database-sharding-key>



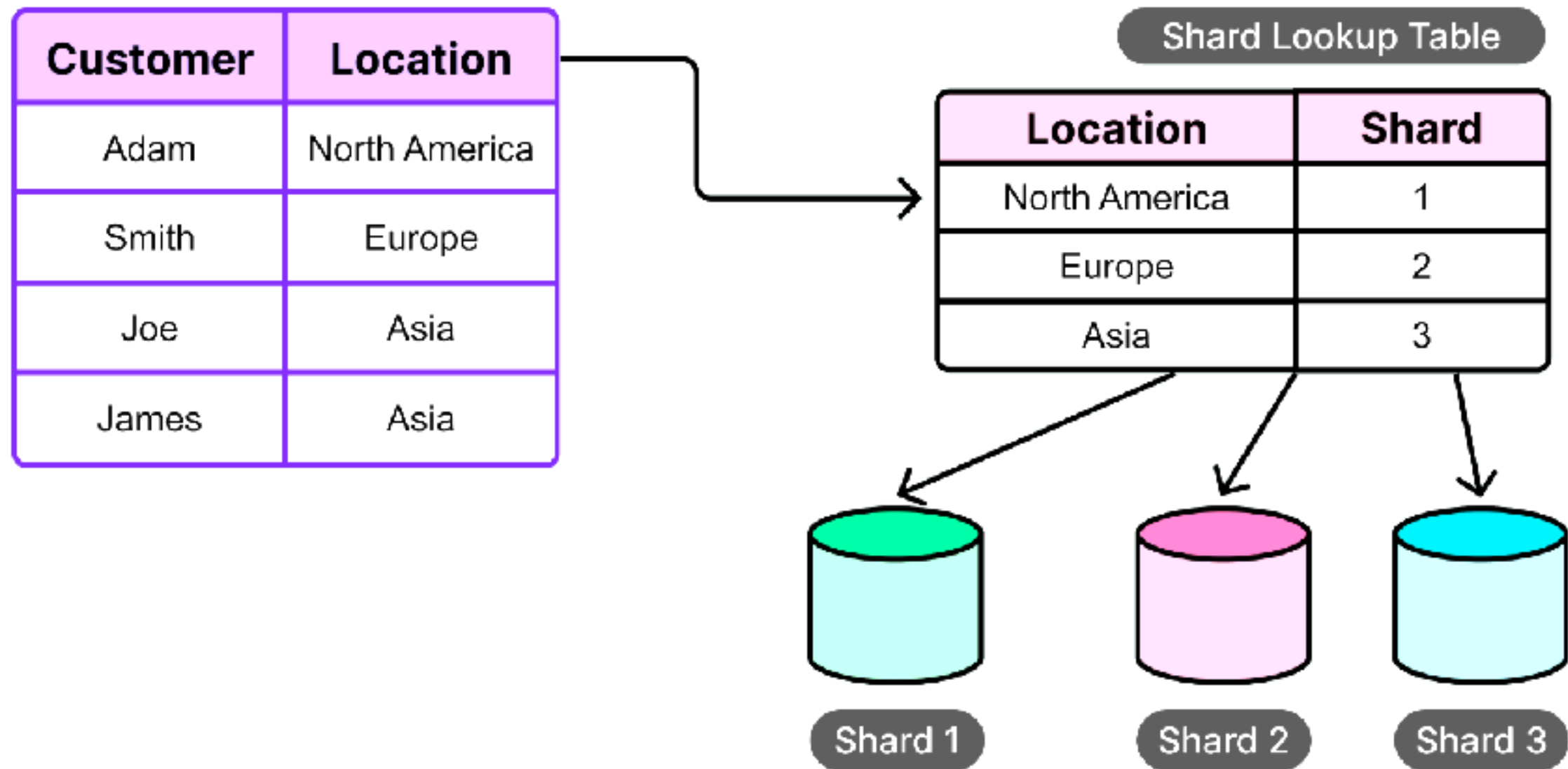
Range-based sharding



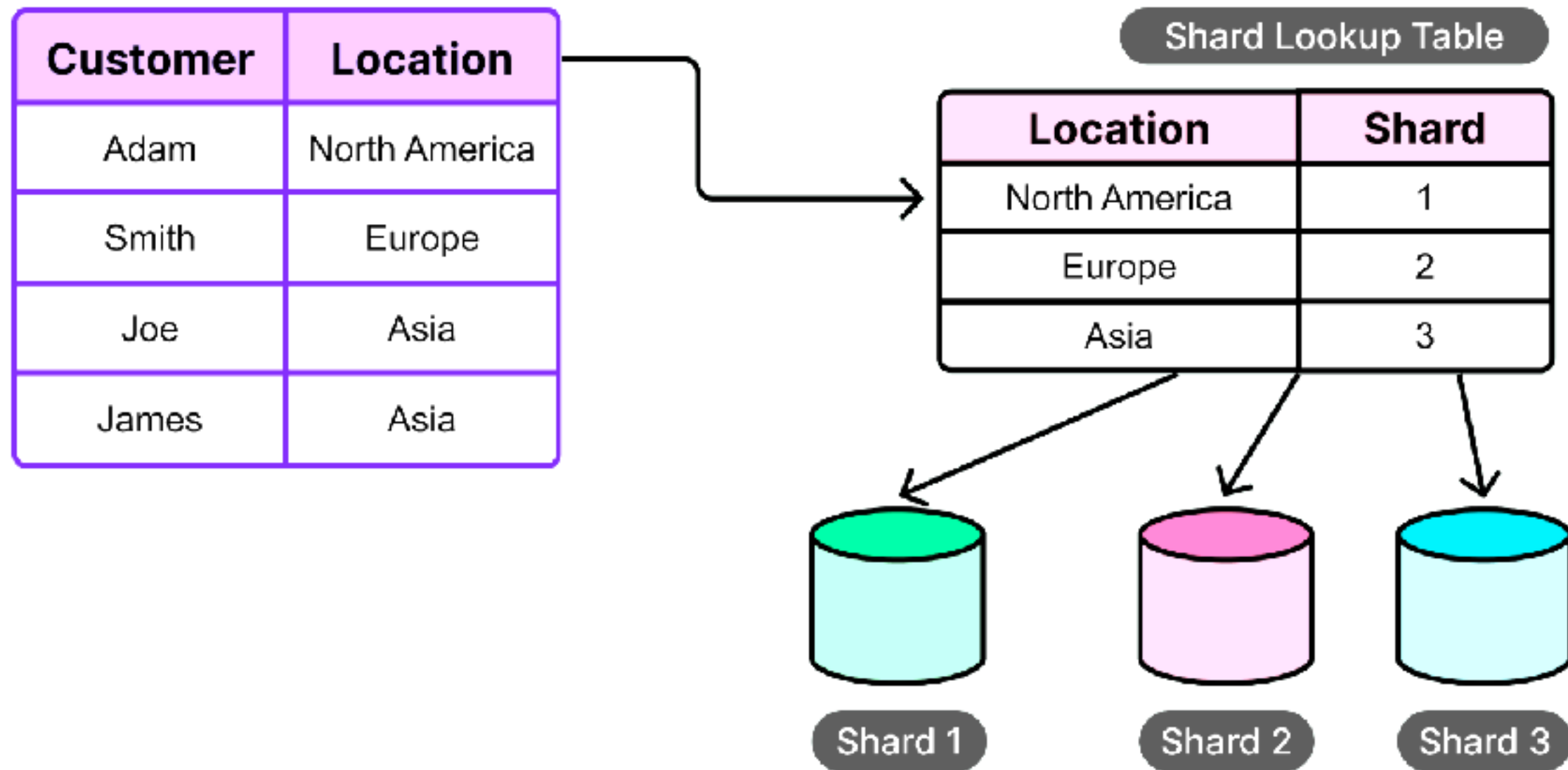
Key-based sharding



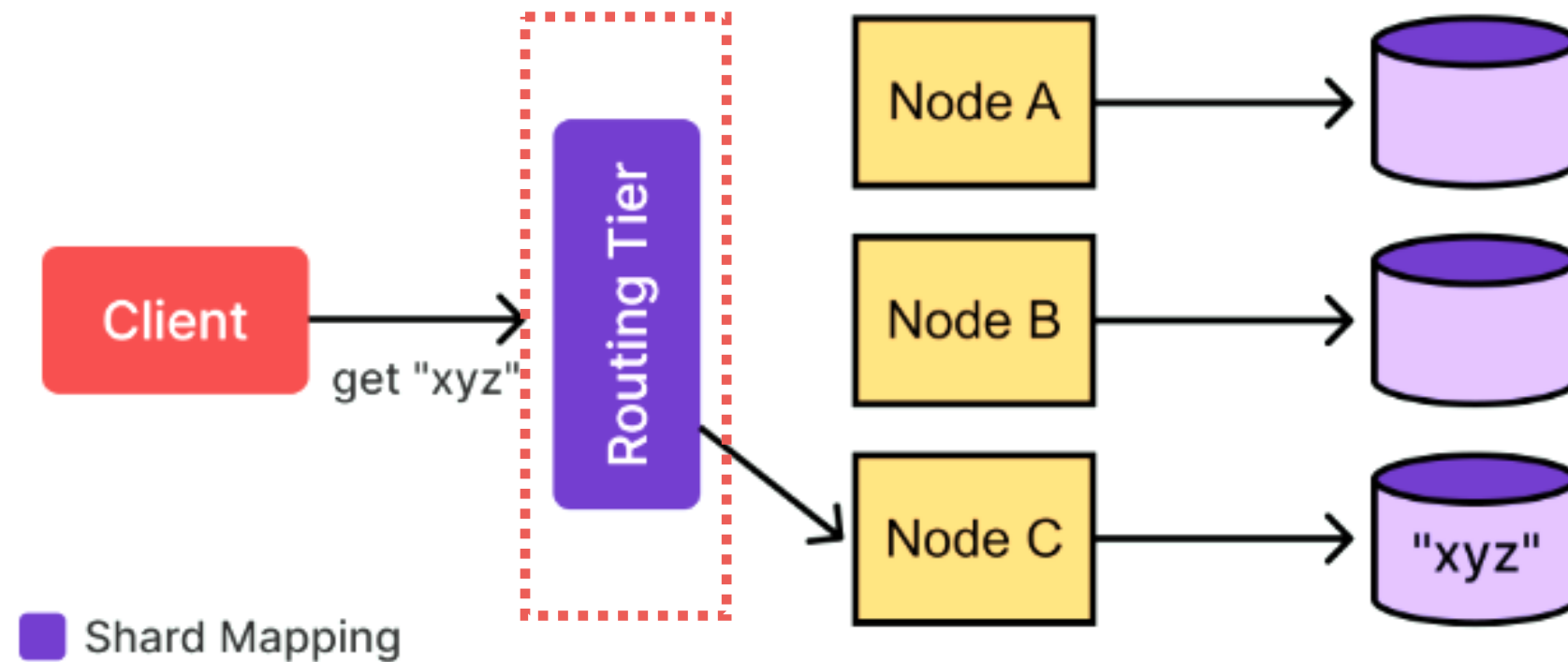
Directory-based sharding



Directory-based sharding



Request routing



Workshop with Sharding

<https://github.com/up1/workshop-database/blob/main/workshop/basic-workshop/partition.md>

